

16374-Cooperative Multirobot Systems, Winter Term 11-12

Lecture 1 – Thursday March 29, 2012

Introduction to Multirobot Systems

Outline

- Seminar Description
- Introduction to Multirobot Systems
- Benchmark Problems of Multirobot Systems
- Organizational paradigms of Multirobot Systems
- Challenging Problems of Multirobot Systems
- Summary

Outline

- **Seminar Description**
- Introduction to Multirobot Systems
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Seminar Description

This seminar provides well-grounded and informative answers to questions like:

- What are the main features of multirobot systems?
- What are the challenging problems of multirobot systems?
- What is meant by cooperation?
- Why cooperation is important?
- Why it is difficult to achieve robust and effective cooperation in multirobot systems? and
- How to achieve cooperative behaviors in these systems?

Seminar Description

Day	Date	Duration	Lectures
1	Thursday March 29, 2012	5:00- 9:00PM	• Introduction to Multirobot Systems • Benchmark Problems of Multirobot Systems • Organizational paradigms of Multirobot Systems • Challenging Problems of Multirobot Systems
2	Friday March 30, 2012	5:00- 9:00PM	• What is Cooperation? • Cooperation in Multirobot Systems • Generic Typology of Cooperation • Agent as Cooperative Entity • Technology Enablers

Seminar Description

• Textbooks

- E. Bonabeau, M. Dorigo, and G. Theraulaz. *Swarm Intelligence: From Natural to Artificial Systems*. Sante Fe Institute Studies in the Sciences of Complexity, 1999.
- J. Kennedy and R. Eberhart. *Swarm Intelligence*. Morgan Kaufman Publishers, 2000. - R. Tuomela. *Cooperation: A Philosophical Study*. Philosophical Studies Series, Kluwer Academic Publishers, 2000.
- L. Parker, “Multiple Mobile Robot Systems,” Ch. 40 in *Springer Handbook of Robotics*, 2008.
- F. Dressler. *Self-Organization in Sensor and Actor Networks*. Wiley, 2007.
- Alaa Khamis. *Conceptual Foundations of Cooperation in Distributed Surveillance*. Technical Reports, Thales Canada, Naval Division, 2010.

Seminar Description

- **Website**

<http://www.alaakhamis.org/teaching/UC3M/coop/index.html>

- **Instructor**

Dr. Alaa Khamis

Email: a[dot]khamis[at]pami[dot]uwaterloo[dot]ca

Office: 1.3B12

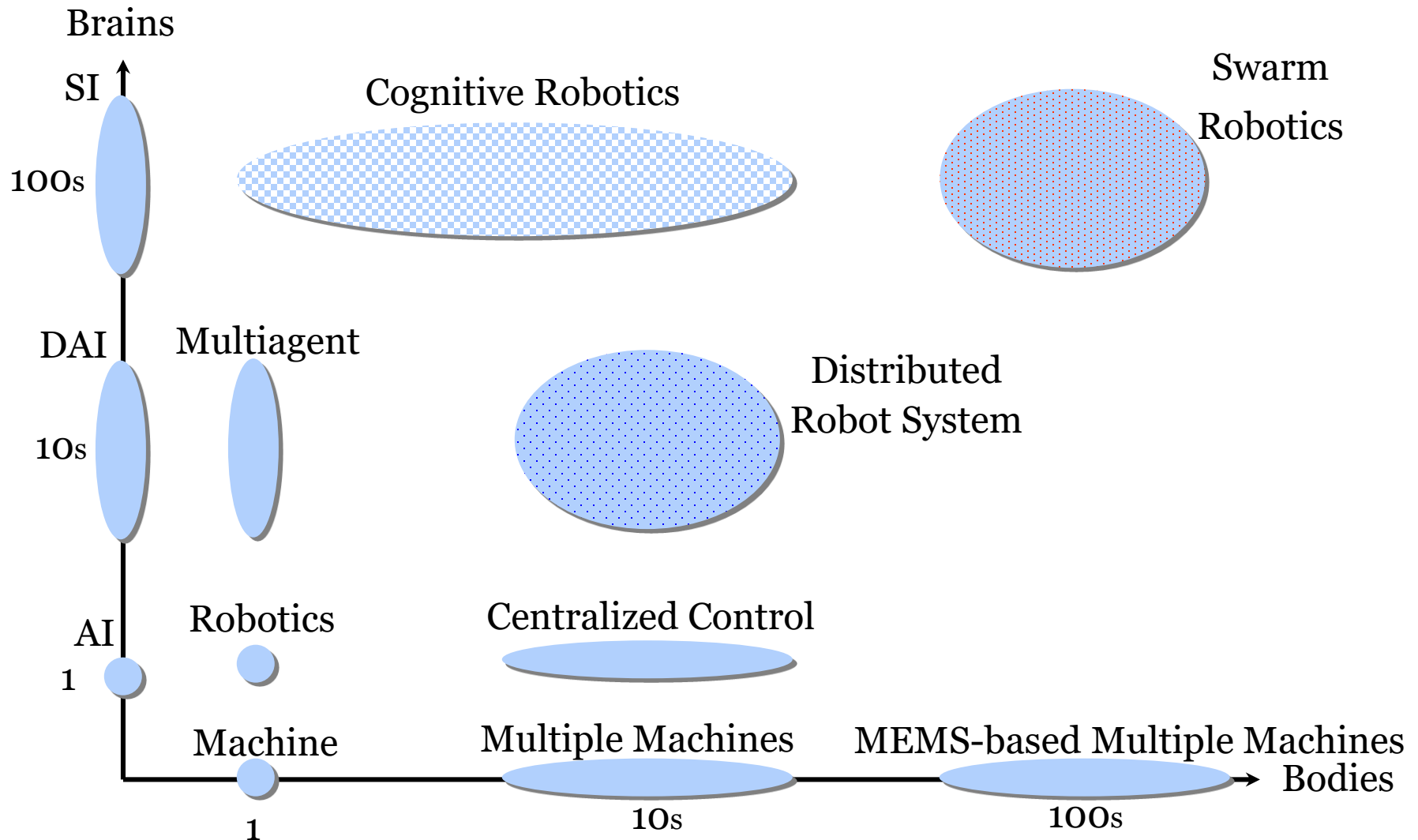
URL: <http://www.alaakhamis.org/>

Outline

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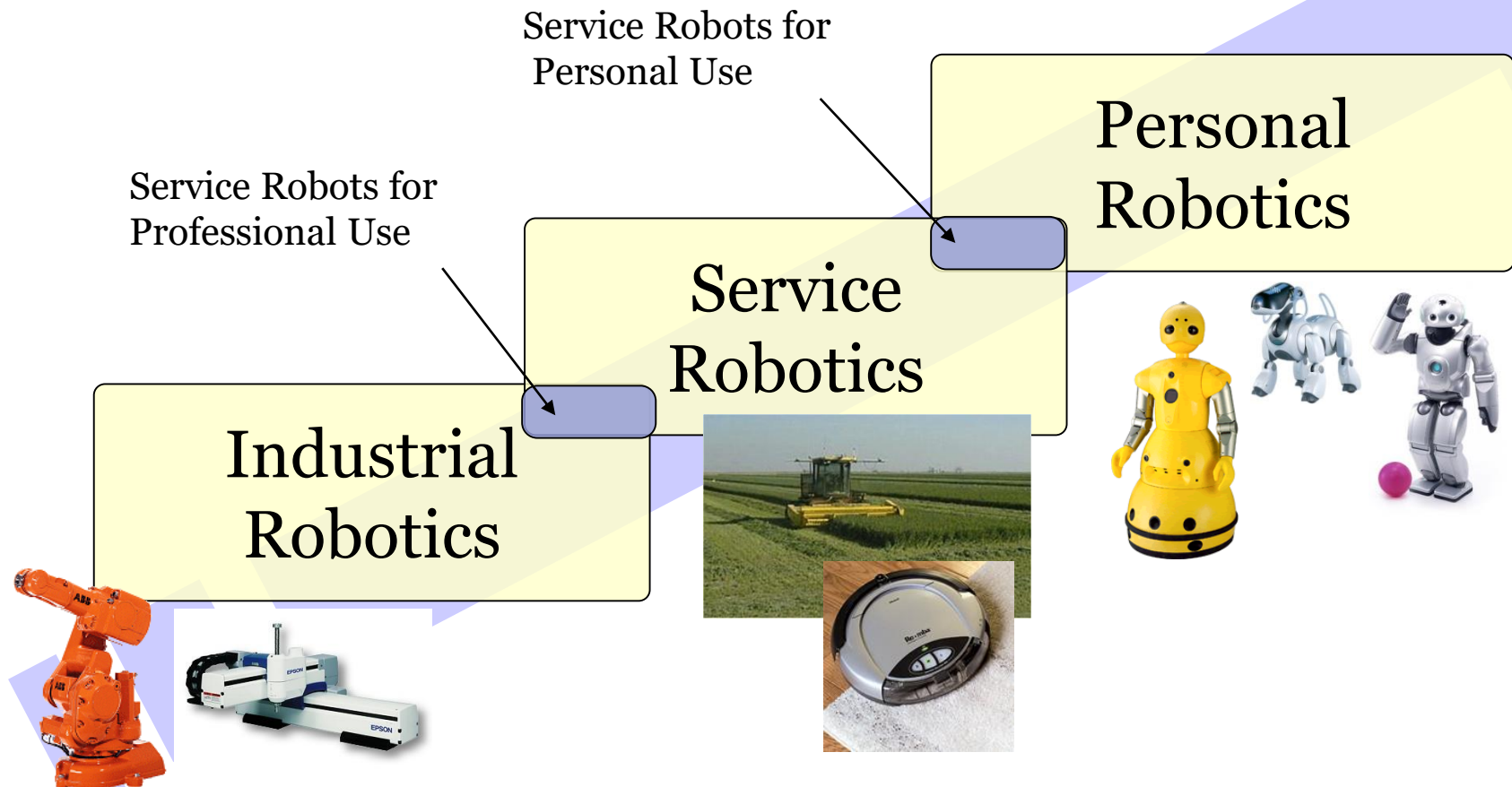
Introduction to Multirobot Systems

- **Body/Brain Evolution**



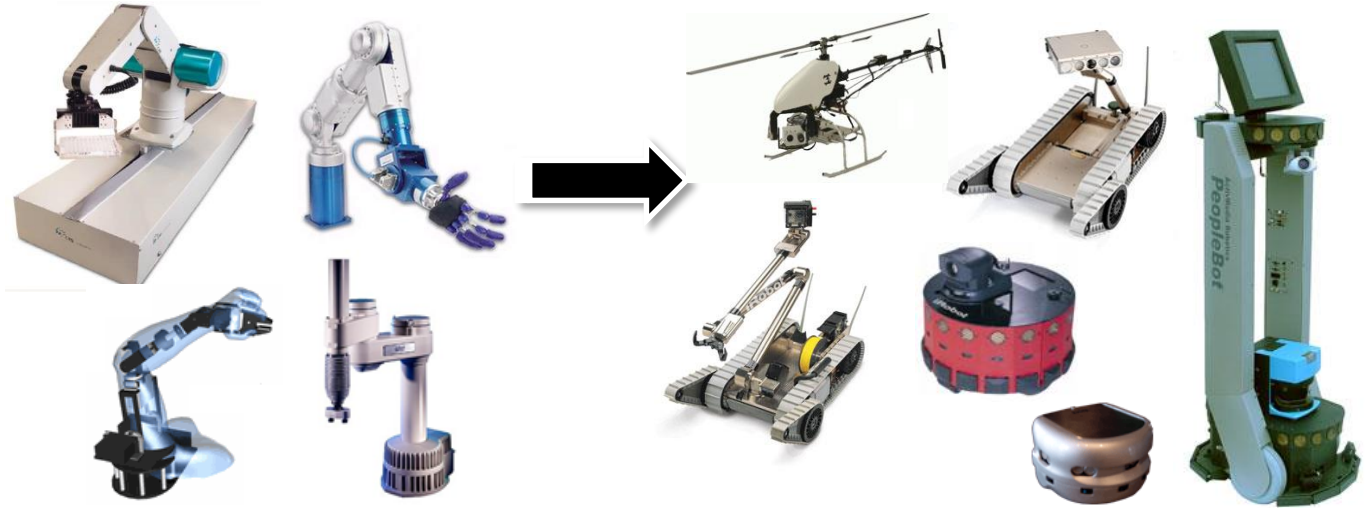
Introduction to Multirobot Systems

- **Traditional Robotics: Evolutionary Stages**



Introduction to Multirobot Systems

- **Traditional Robotics: From Arms to Mobile Robots**



Aspect	Industrial Manipulators	Mobile Robots
Environment	Used to be highly structured and static	Used to be highly unstructured and dynamic
Mobility	Lack or limited	High
Human interaction	Used to be uni-modal and rigid	Used to be multimodal and adaptive
Field maturity	High	Low
Market penetration	High	Low
Application field	Industry	Wide field

Introduction to Multirobot Systems

- **Traditional Robotics: Service Robots**

Service robots assist human beings, typically by performing a job that is dirty, dull, distant or dangerous.



Service Robots

Service Robots for Professional Use

(Construction robots, demolition, medical, space robots, underwater exploration, cleaning, demining, inspection, fire fighting,...)

Service Robots for Personal Use

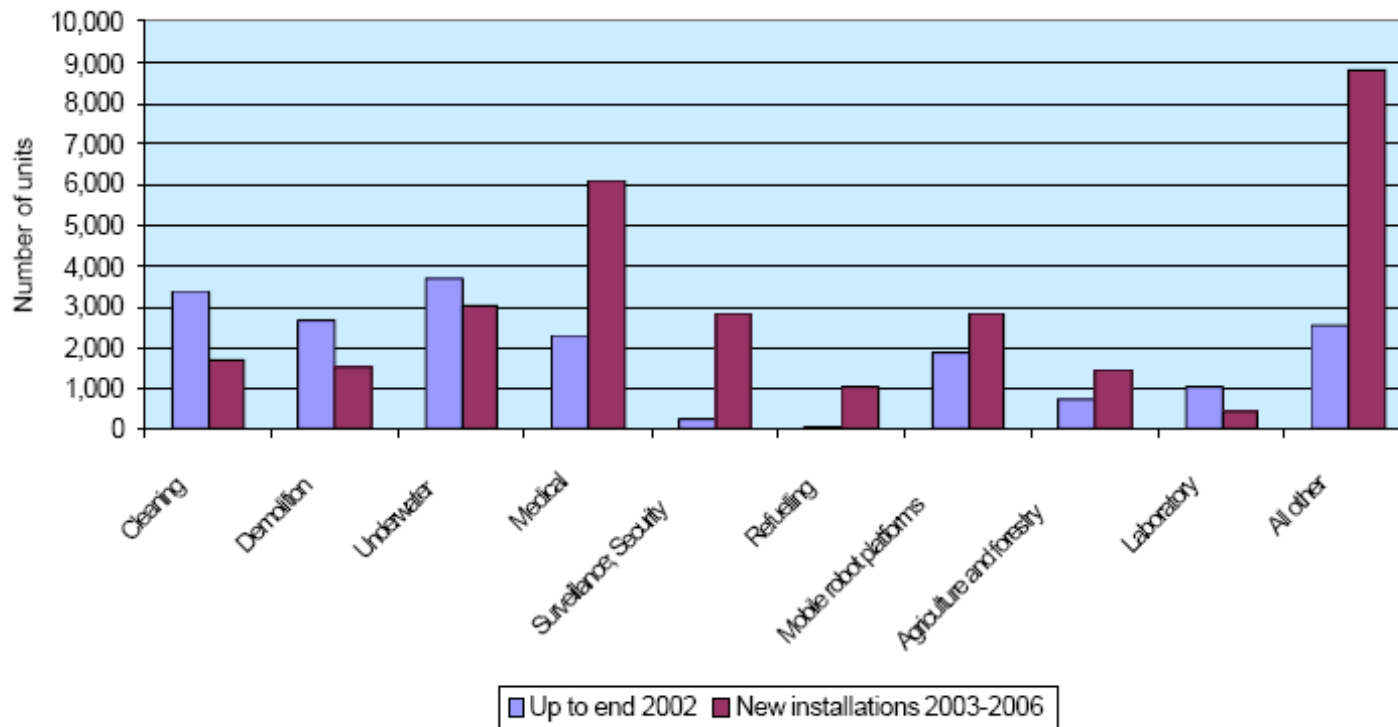
(domestic, personal assistant, entertainment robots, vacuum cleaners, etc.)



Introduction to Multirobot Systems

- **Traditional Robotics: Service Robots**

Service robots for professional use. Stock at the end of 2002 and projected installations in 2003-2006

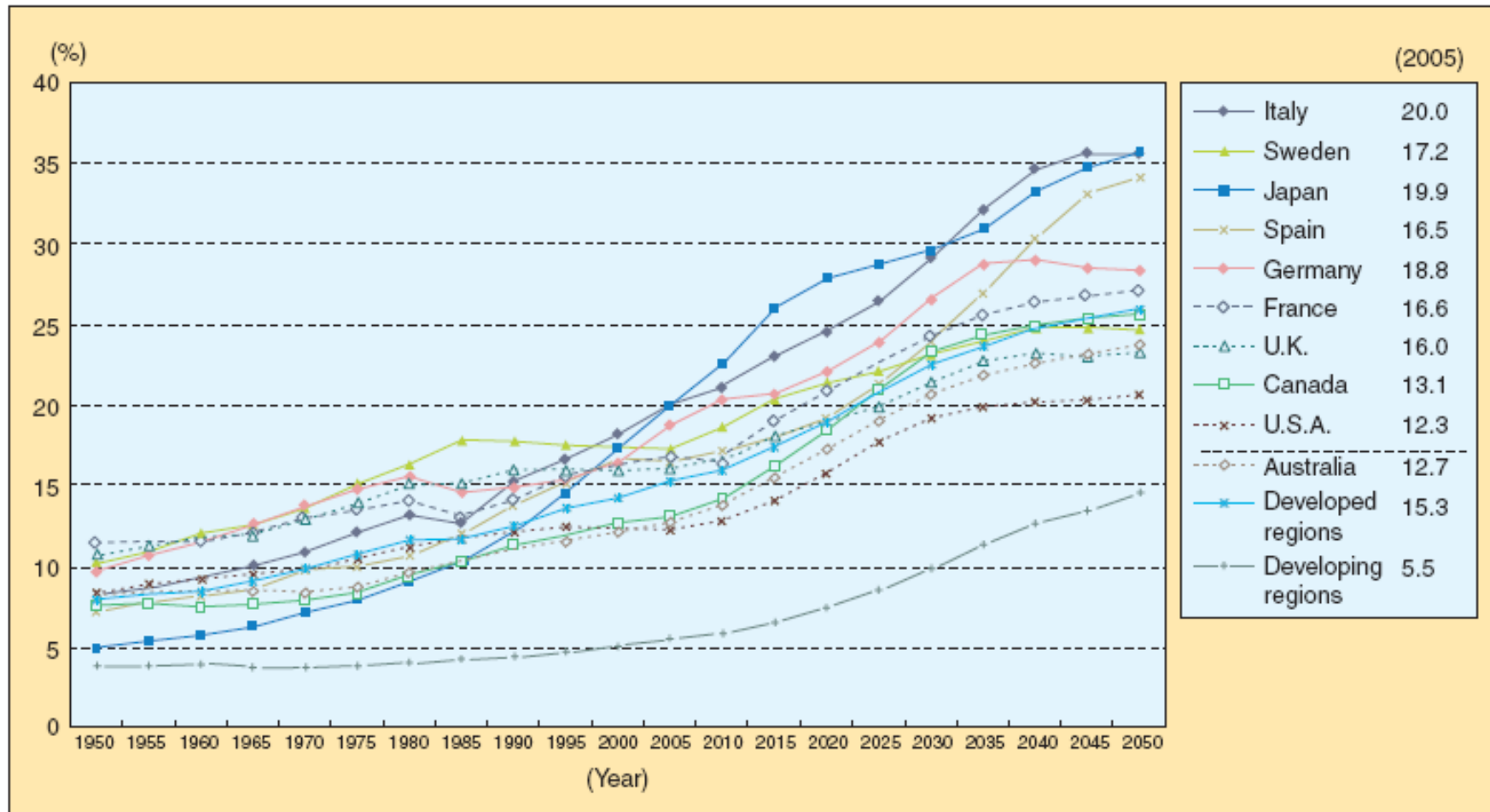


IFR, October 2002

Introduction to Multirobot Systems

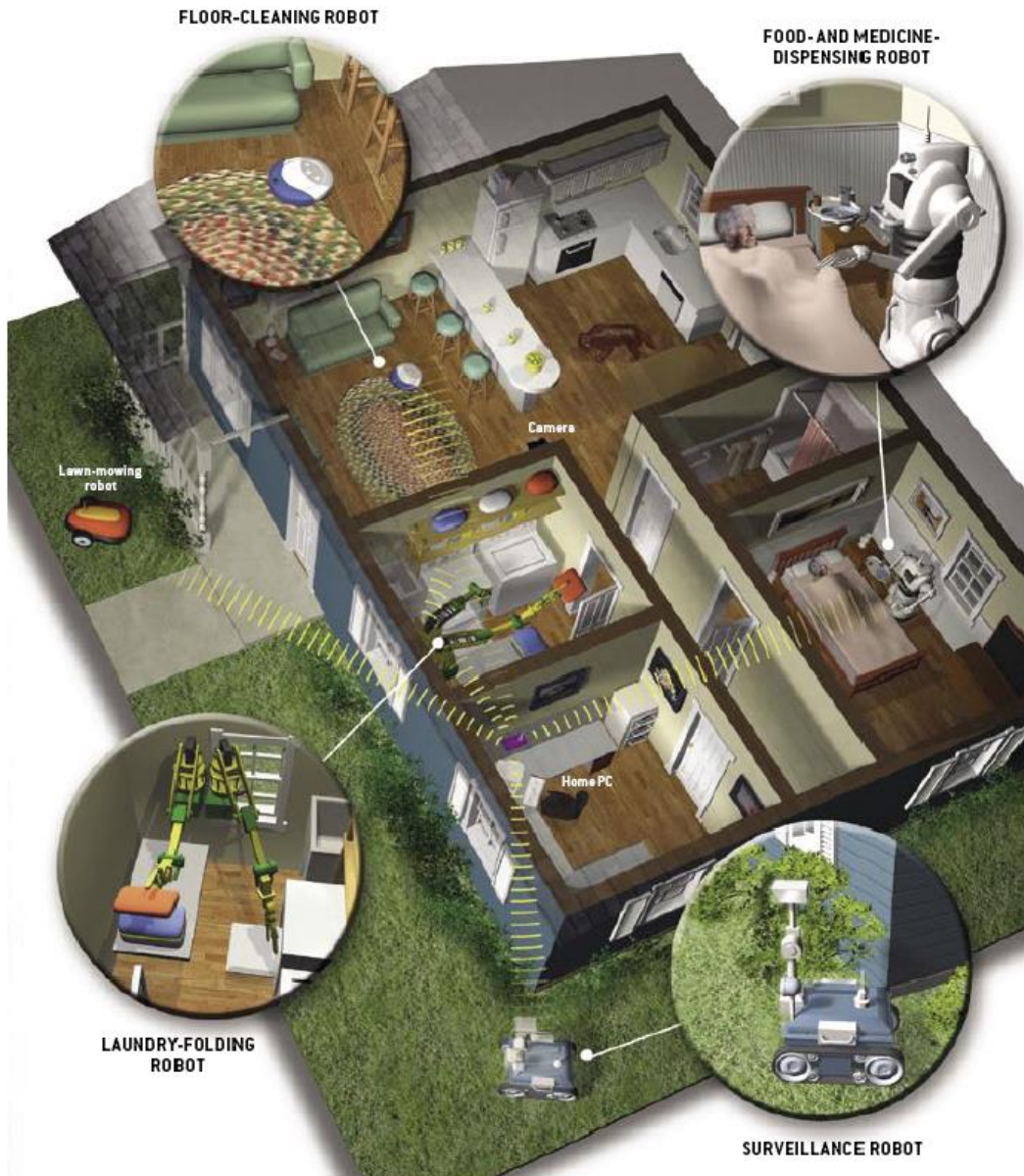
- **Traditional Robotics: Service Robots**

Trends in percentage of the elderly (over age 85) in the world



Source: ADRIANA TAPUS, MAJA J. MATARIC', AND BRIAN SCASSELLATI, "Socially Assistive Robotics: The Grand Challenges in Helping Humans Through Social Interaction," IEEE Robotics & Automation Magazine, MARCH 2007

Introduction to Multirobot Systems



- **A robot in every home**

“As I look at the trends that are now starting to converge, I can envision a future in which robotic devices will become a nearly ubiquitous part of our day-to-day lives.

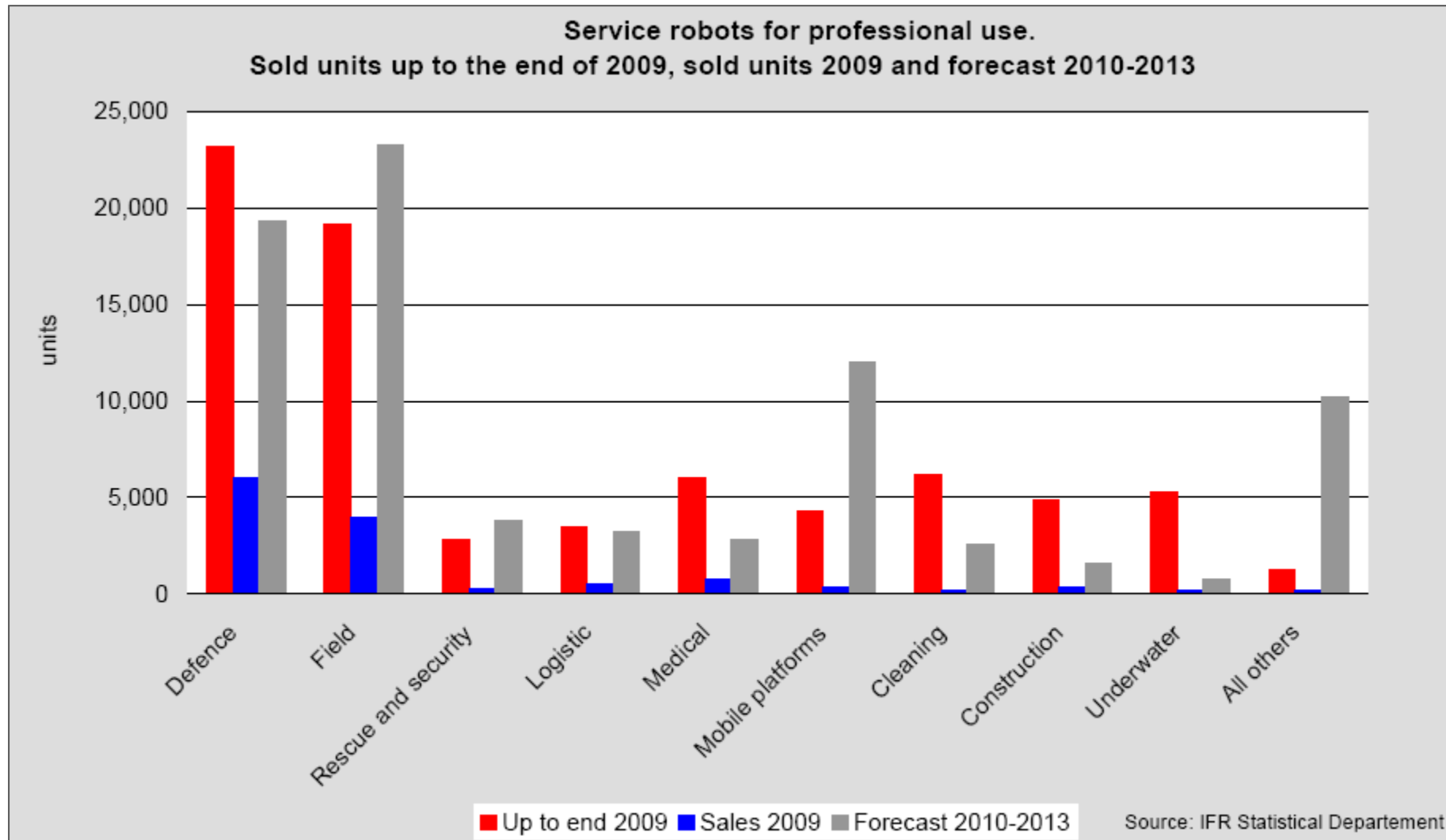
The challenges facing the robotics industry are similar to those we tackled in computing three decades ago.”

Bill Gates, 2007

Scientific American

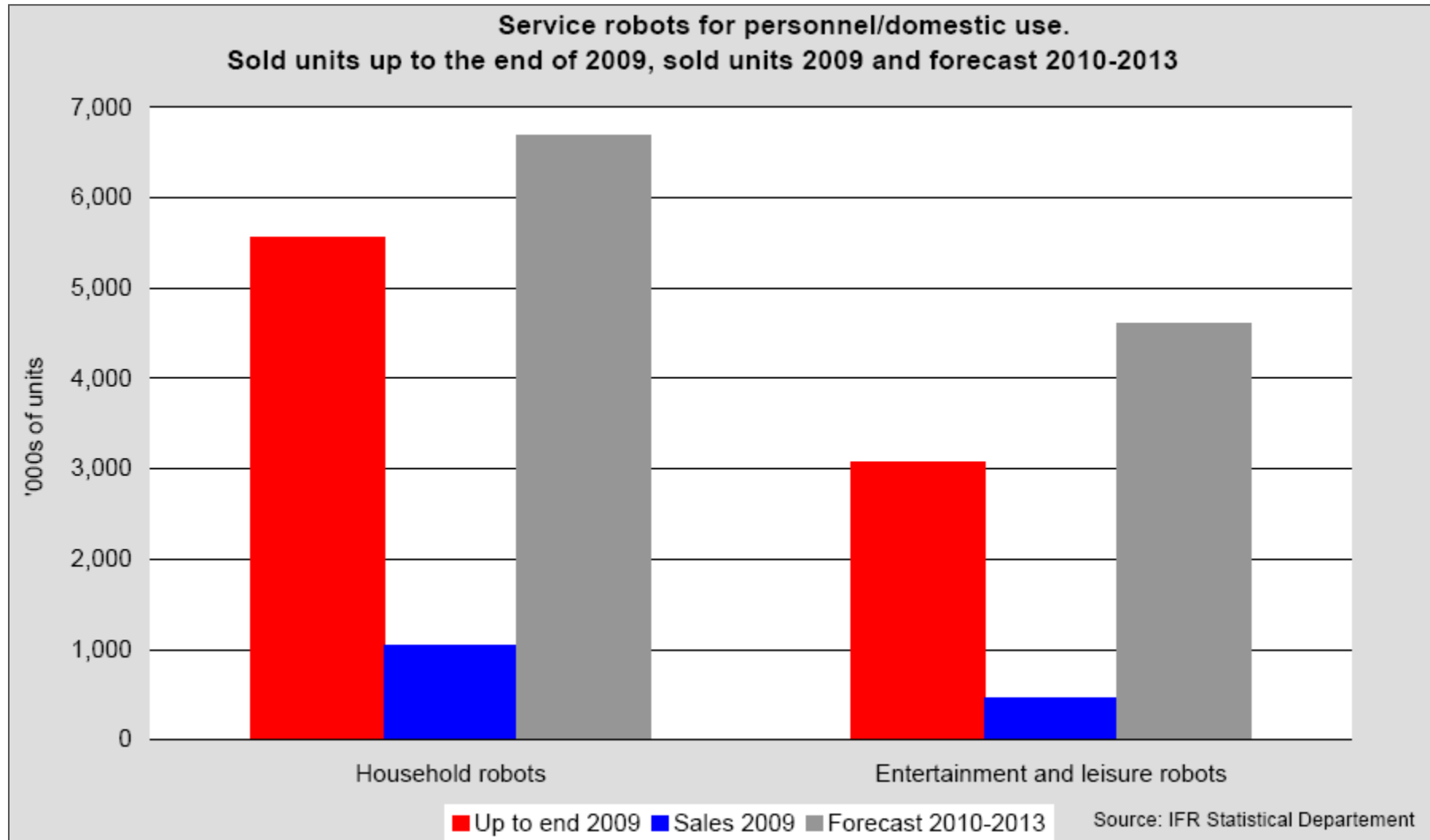
Introduction to Multirobot Systems

- **Traditional Robotics: Service Robots**



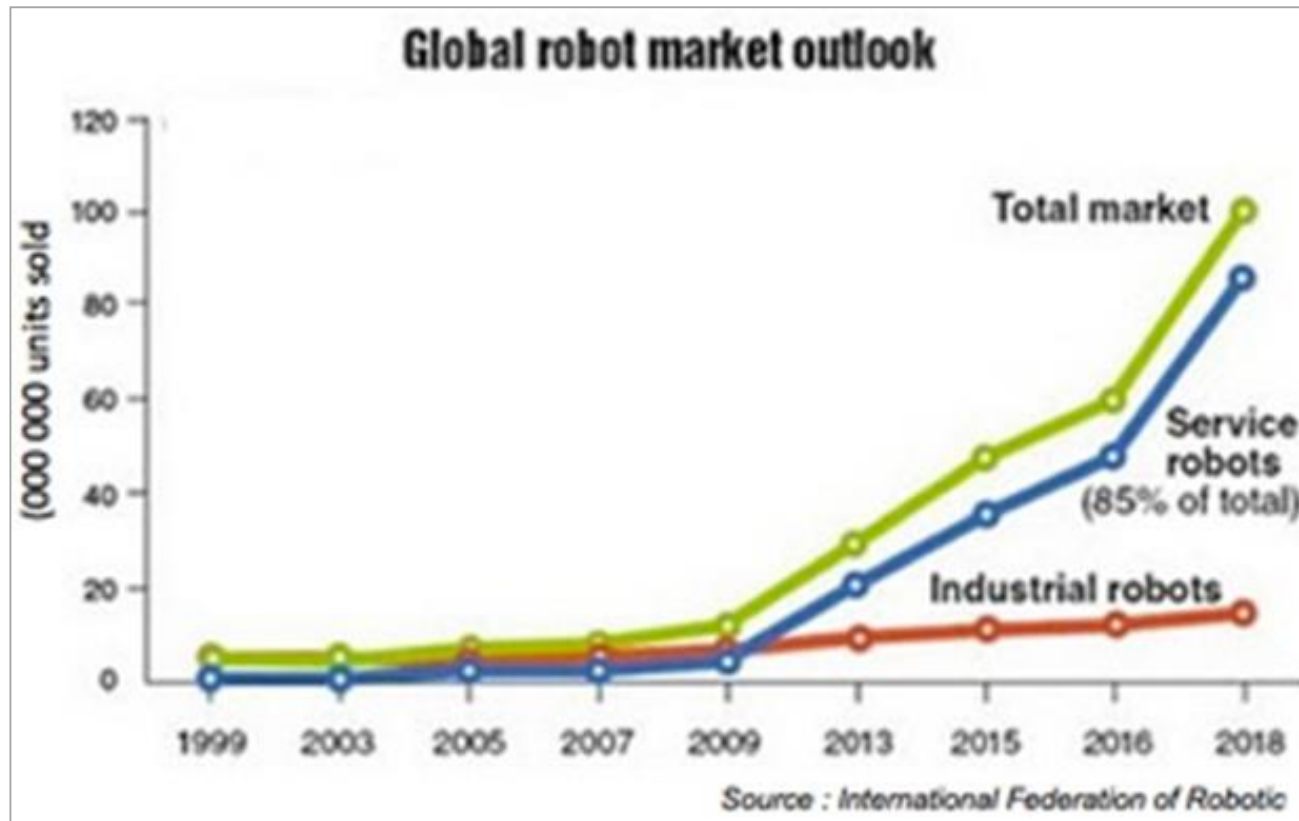
Introduction to Multirobot Systems

- **Traditional Robotics: Service Robots**



Introduction to Multirobot Systems

- **Traditional Robotics: Service Robots**



Service robotics is planned to be a major driver of economic growth in the 21st century. It offers high potential growth, revenue and employment perspectives.

Introduction to Multirobot Systems

- **Traditional Robotics: Traditional Problems**

- ◇ **Q1:** Where am I? Where am I going?
- ◇ **A1:** Localization
- ◇ **Q2:** What does the world look like?
- ◇ **A2:** Environment Mapping or Modeling
- ◇ **Q3:** How to explore an unknown environment?
- ◇ **A3:** Simultaneous Localization and Mapping “SLAM”
- ◇ **Q4:** How can I get there from here?
- ◇ **A4:** Planning
- ◇ **Q5:** How to achieve intelligent connection between perception and action?
- ◇ **A5:** Control Architecture

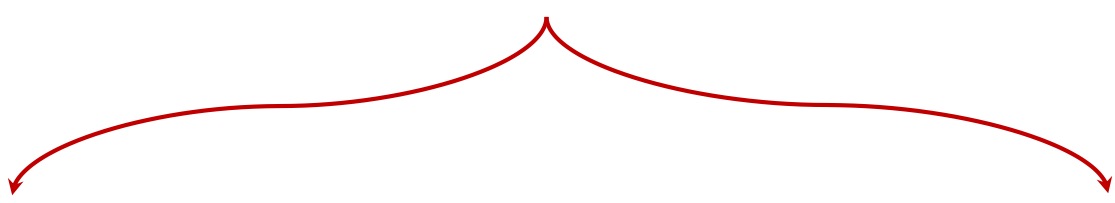


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- **Traditional Robotics: Localization**

The localization process allows a mobile robot to know where it is at any moment relative to its environment.

Localization



Local Localization

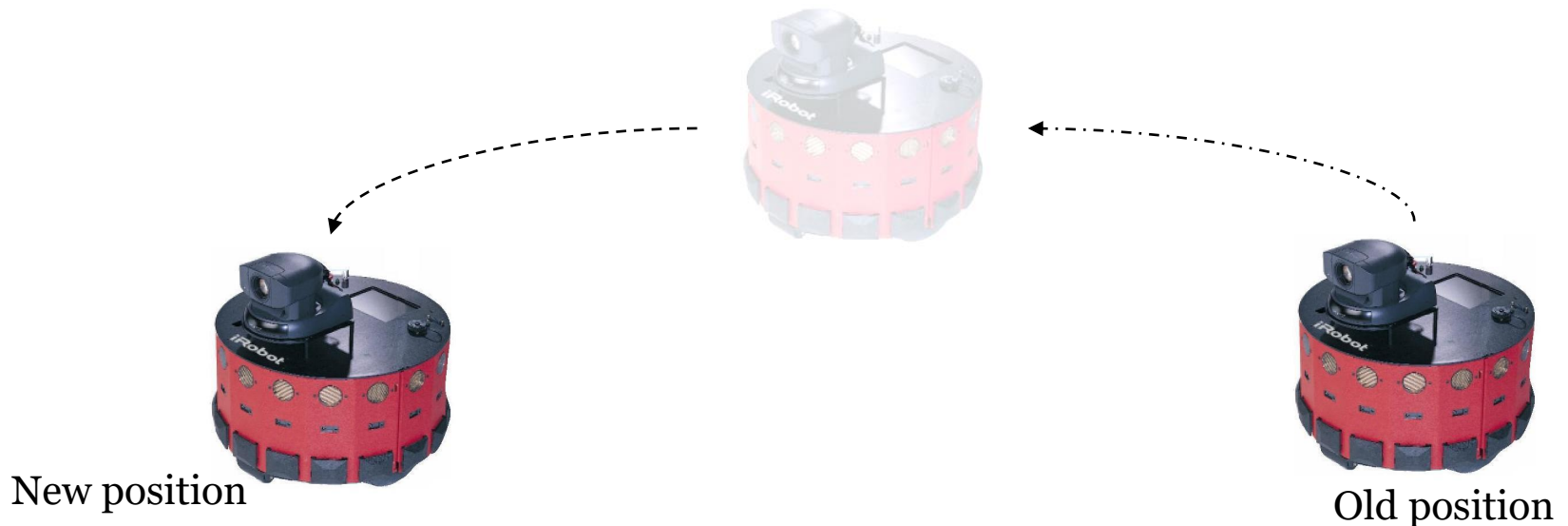
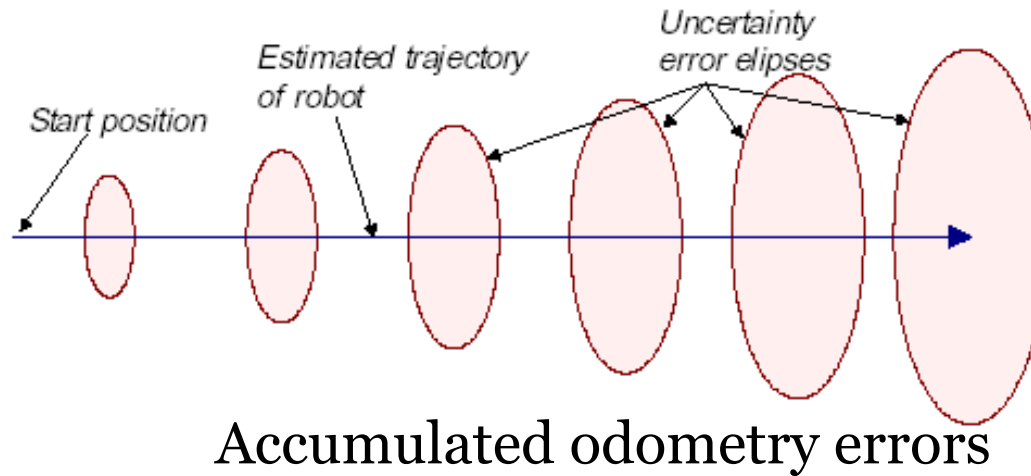
The robot incrementally corrects its position relative to an initial location

Global Localization

The robot's initial position is not needed. In addition, the location process can be based on the sensorial identification of **landmarks** in the environment whose location is well known, or it can be based on maps or models of the environment and identify characteristic elements of the mapped environment. Probabilistic approaches are used to solve the problem of uncertainty in the sensorial information.

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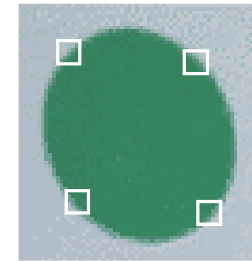
- **Traditional Robotics: Local Localization**



Introduction to Multirobot Systems

- **Traditional Robotics: Global Localization**

- Given a map, determine the robot's location
- Landmark locations are known, but the robot's position is not.



- From sensor readings, the robot must be able to infer its most likely position on the field.
- ***Example:*** where are the AIBOs on the soccer field?



Introduction to Multirobot Systems

- **Traditional Robotics: Mapping**

The information provided by the sensors has to be interpreted in a way that is sufficiently a structured knowledge of the space that surrounds the robot is obtained by the robot. Structuring this external information is accomplished through the construction of a **model or a map** of the environment.

This model is used to make decisions and to fuse the new information that the robot receives from its sensors, either from the **same sensor in several time instants** or from **several sensors** of the same or different type.



Introduction to Multirobot Systems

- **Traditional Robotics: Mapping**

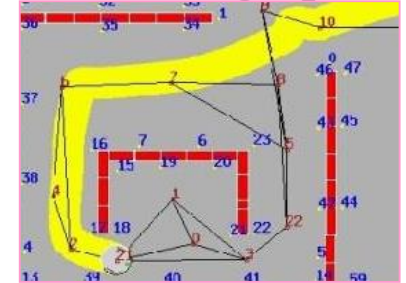
Metric Map

Metric maps capture the geometric properties of the environment.



Topological Map

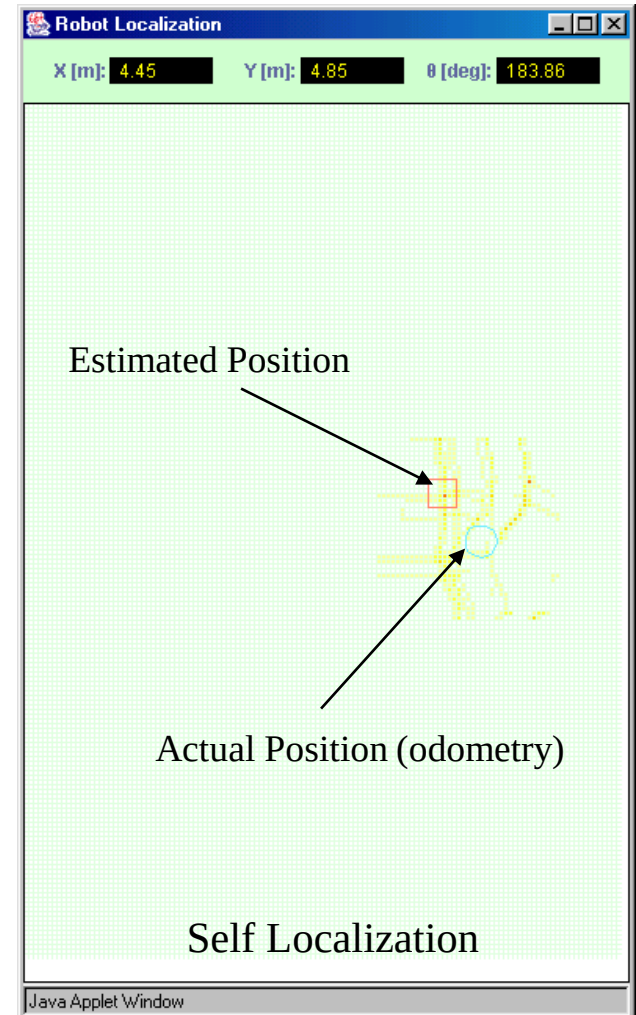
Topological maps describe the connectivity of different places by means of nodes-and-arcs graphs



Metric maps are finer grained than topological maps, but higher resolution comes at a computational burden. Metric maps can be discretized based on the probability of space occupation. The resulting mapping approaches are known as **occupancy-grid mapping**.

Introduction to Multirobot Systems

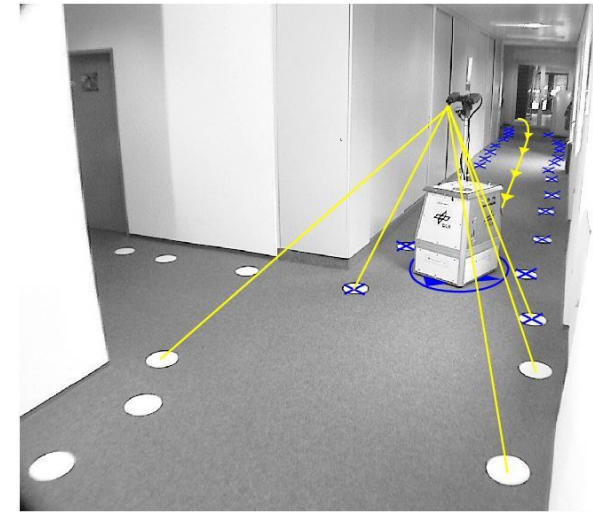
- **Traditional Robotics: Mapping and Localization**



Introduction to Multirobot Systems

- **Traditional Robotics: SLAM**

Simultaneous localization and mapping (SLAM) is a technique used by robots and autonomous vehicles to build up a map within an unknown environment while at the same time keeping track of its current position.



This is not as straightforward as it might sound due to inherent uncertainties in discerning the robot's relative movement from its various sensors.

Introduction to Multirobot Systems

- **Traditional Robotics: Motion Planning**



Path planning is a process to compute a **collision-free path** for a robot from a start position (s) to a given goal position (G), amidst a collection of obstacles.

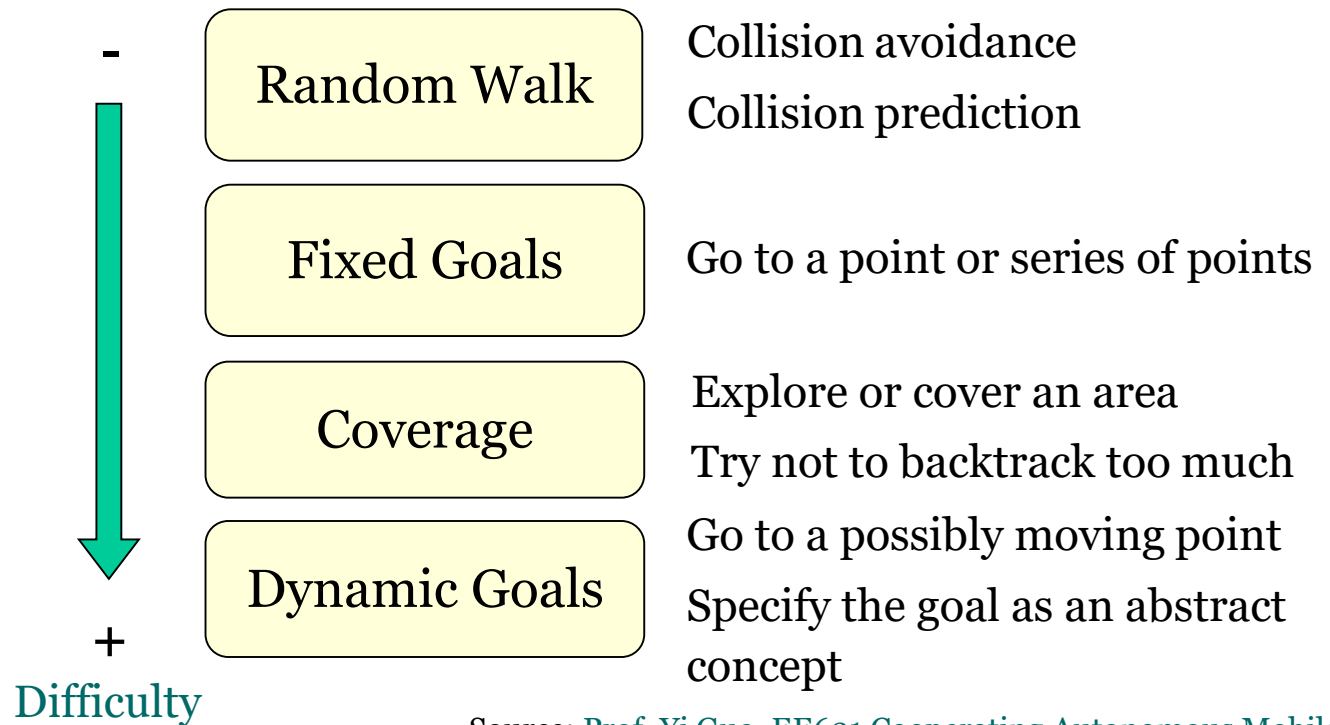
Introduction to Multirobot Systems

- **Traditional Robotics: Navigation**

Navigation is the action of **directing the course** of a mobile robot as it traverses the environment to make it reach a destination without getting lost or crashing with objects.

How difficult
is the problem
of navigation?

Depends on
what you want
to do...



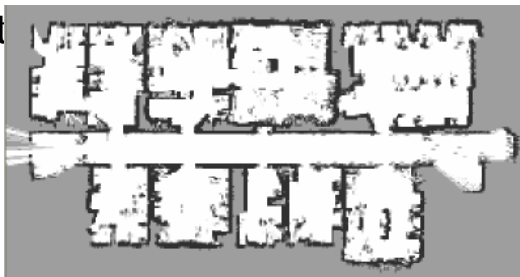
Source: Prof. Yi Guo, EE631 Cooperating Autonomous Mobile Robots, Stevens Institute of Technology, Hoboken, USA.

Introduction to Multirobot Systems

• Traditional Robotics: Navigation

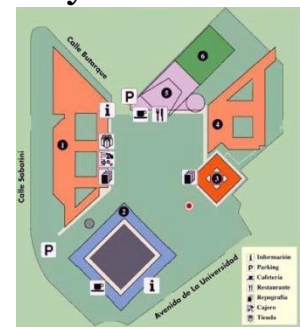
Geometric Navigation

The robot pose, the environment elements and the goal are defined by **its coordinates**. The localization is calculated by **odometry**. The inevitable accumulated errors are reduced by measuring the relative position of the robot in relation to known elements of the environment elements. The environment is usually modeled by **occupancy grids** and by geometric maps which incorporate some elements of the environment (e.g. doors, walls, corners, artificial beacons, etc.). Occupancy grids are mainly used for path planning and geometric localization.



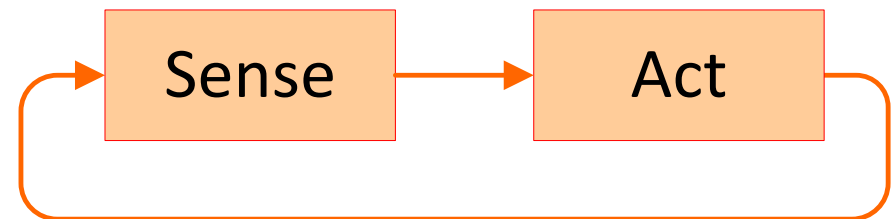
Topological Navigation

Topologic navigation, also called **qualitative navigation**, is an alternative to geometric navigation. The information is established as relations, and usually represented with graphs. A main drawback of pure topological navigation is that robot localization is hard to manage and the robot can be lost quite easily.



Introduction to Multirobot Systems

- **Traditional Robotics: Navigation Methods**
 1. **Reactive method: Sense-Decide-Act or Sense-Act**
 - Simplify the sensing: use a lower level representation.
 - Simplify the decision-making: use sub-symbolic or simple computation.
 - Let the decision-making and analysis work in parallel.
 - Combine the resulting “behaviors” in a useful way.
 - Actions are much faster.

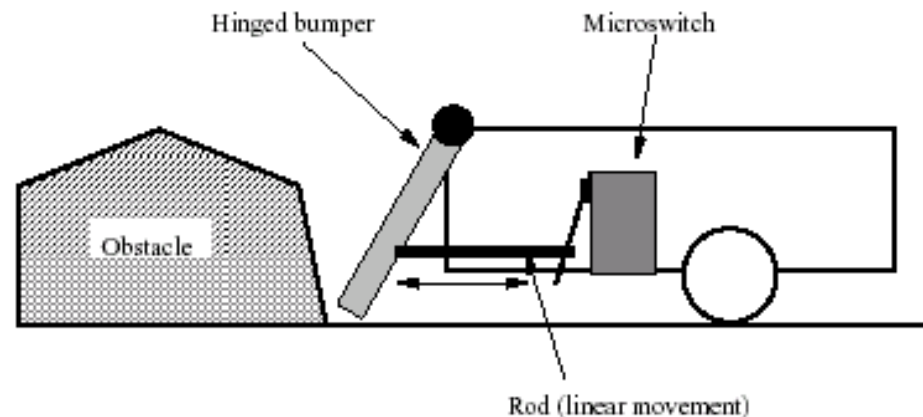


Introduction to Multirobot Systems

- **Traditional Robotics: Navigation Methods**

1. **Reactive method: Sense-Decide-Act or Sense-Act**

Robot Primitives	Input	Output
Sense	Sensor data	Sensed information
Act	Sensed information	Actuator commands

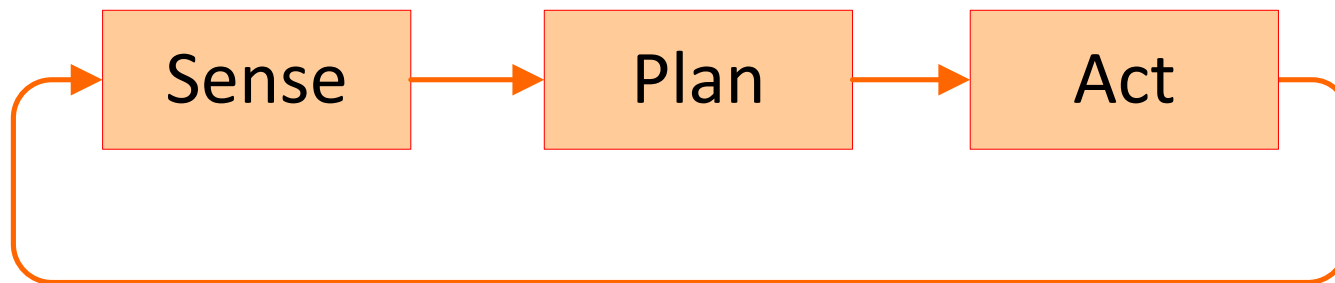


Introduction to Multirobot Systems

- **Traditional Robotics: Navigation Methods**

2. Deliberative method: Sense-Plan-Act

- Sensor analysis takes time: attempted to build high-level representations
- Planning is deliberative and takes time
- Actions are slow



Introduction to Multirobot Systems

- **Traditional Robotics: Navigation Methods**

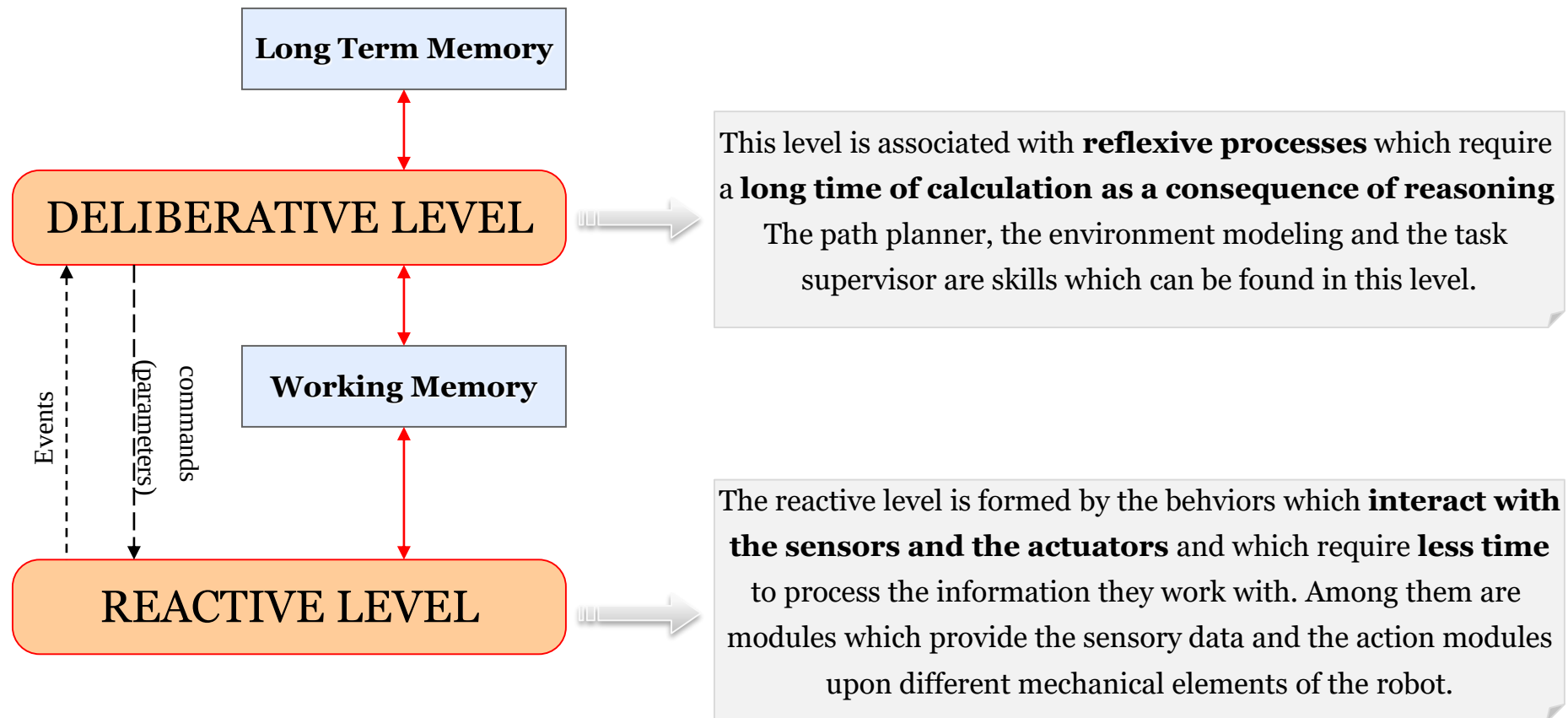
2. Deliberative method: Sense-Plan-Act

Robot Primitives	Input	Output
Sense	Sensor data	Sensed information
Plan	Information (sensed and/or cognitive)	Directives
Act	Sensed information or directives	Actuator commands

Introduction to Multirobot Systems

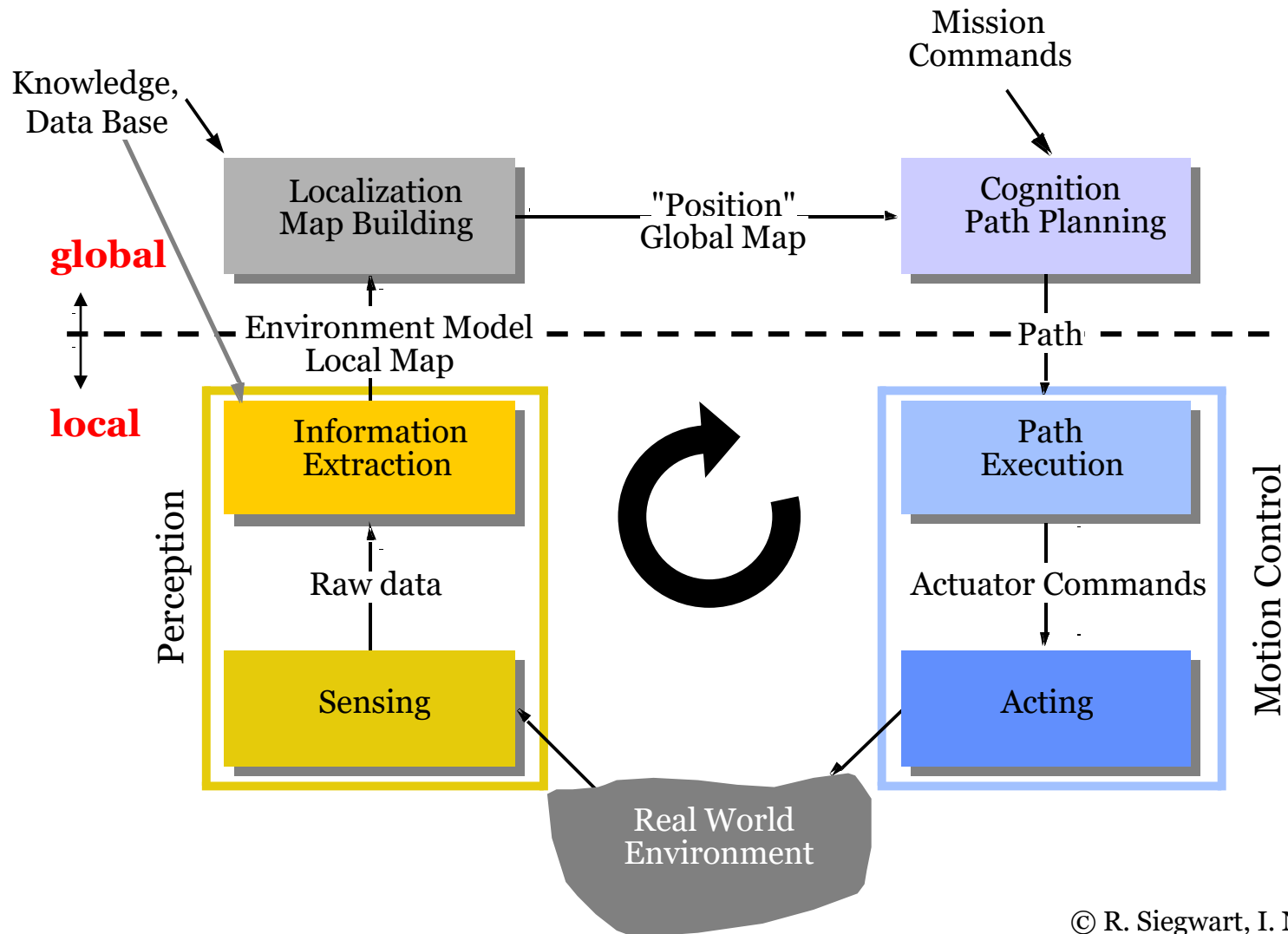
- **Traditional Robotics: Navigation Methods**

3. Combined Methods



Introduction to Multirobot Systems

- Traditional Robotics: Control Architecture



Introduction to Multirobot Systems

- **Traditional Robotics: Other problems**
 - ❖ Situation Awareness
 - ❖ Autonomy
 - ❖ Decision making under uncertainty
 - ❖ Interaction with human and/or non-human actors
 - ❖ Learning
 - ❖ ...

Introduction to Multirobot Systems

- **Multirobot Systems**

Multirobot systems (**MRS**) are **a group of robots** that are designed aiming to perform some **collective behavior**.

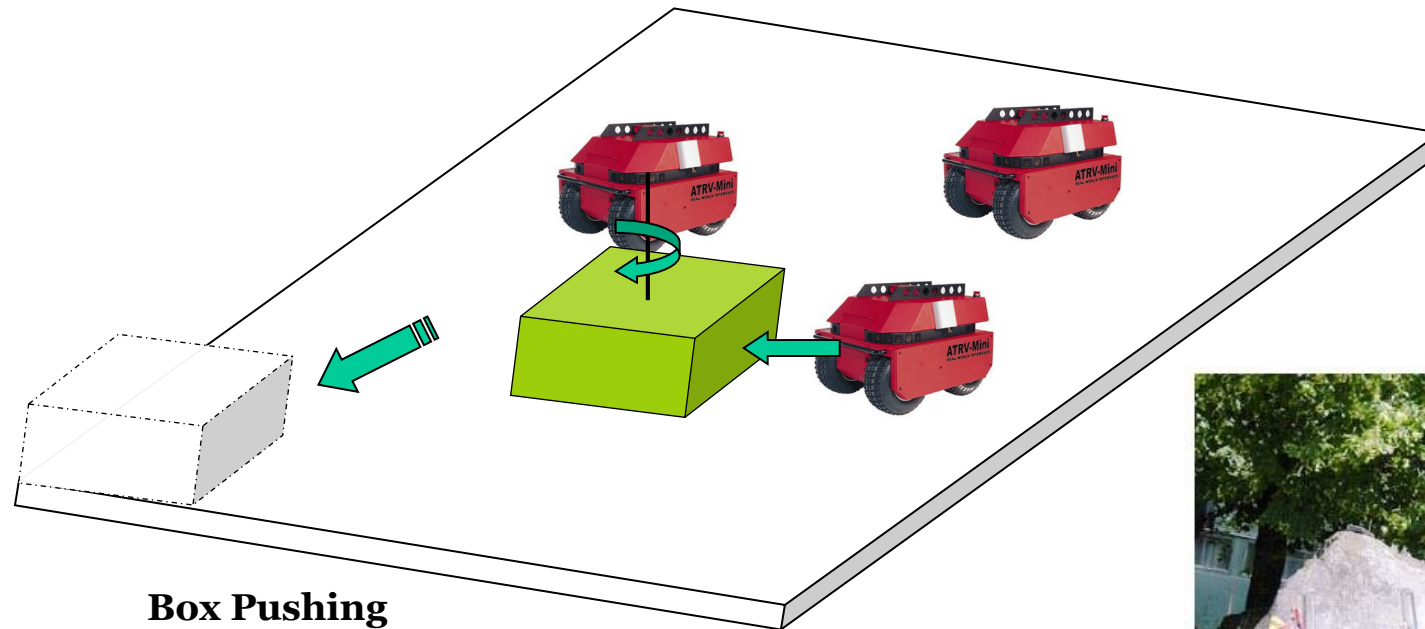
The MRS is gaining great interest because of the following reasons:

- ◇ Resolving task complexity
- ◇ Increasing performance
- ◇ Reliability
- ◇ Simplicity in design

Introduction to Multirobot Systems

- **Multirobot Systems: Resolving task complexity**

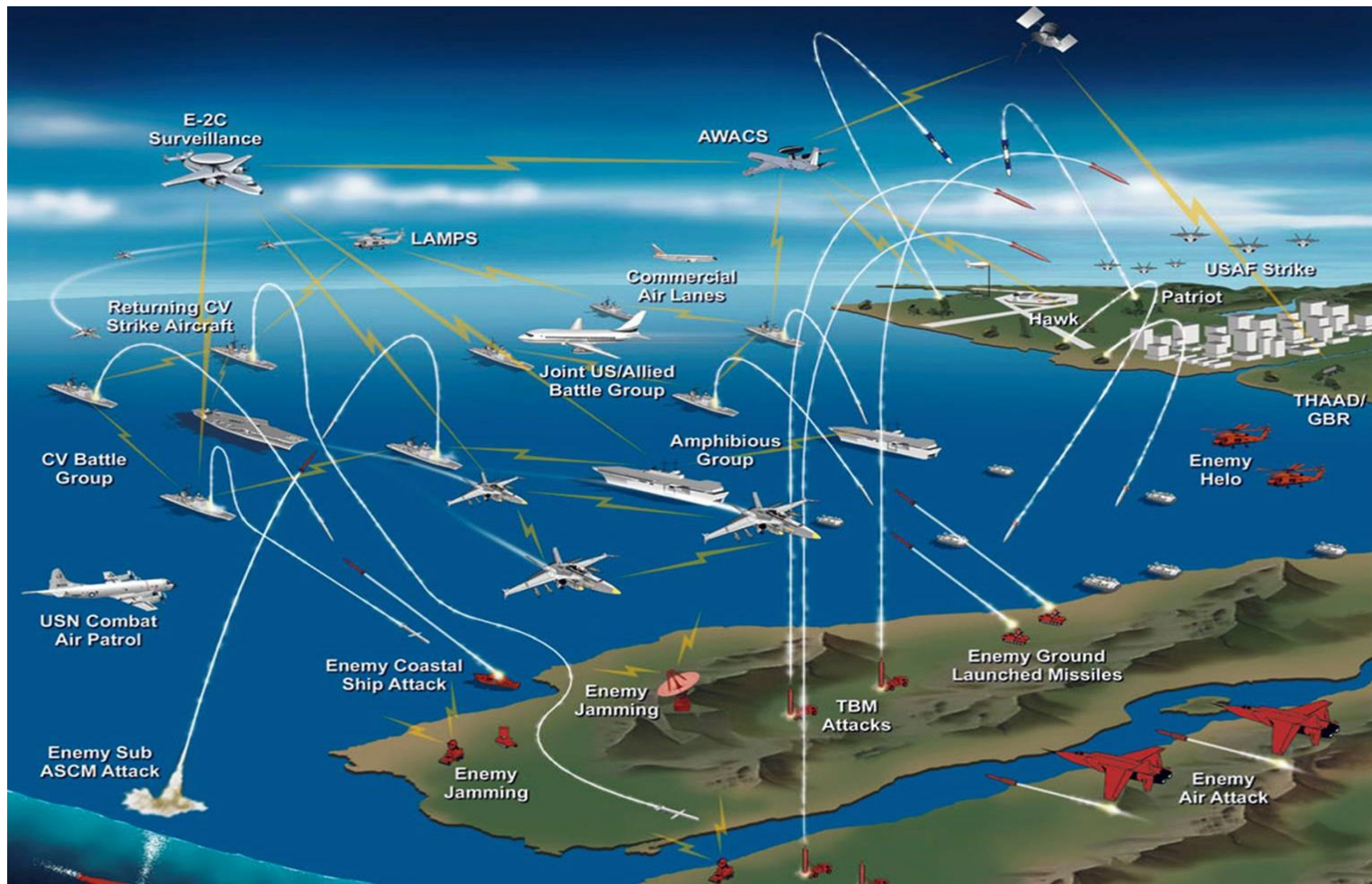
Some tasks may be quite complex for a single robot to do or even it might be impossible.



Crossing a gap

Introduction to Multirobot Systems

- **Multirobot Systems: Resolving task complexity**
Some tasks are inherently distributed.



Future battlespace with semi-autonomous and fully autonomous systems

Introduction to Multirobot Systems

- **Multirobot Systems: Resolving task complexity**

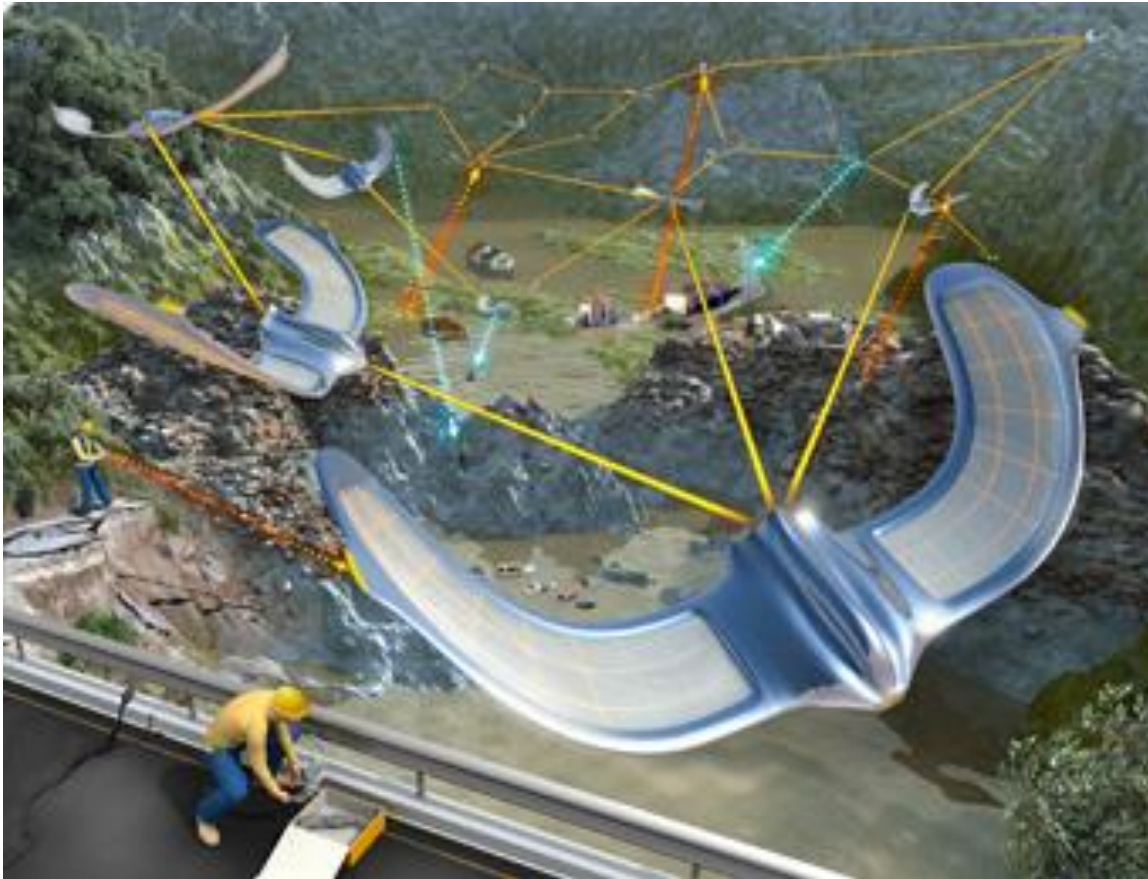
Some tasks are inherently distributed.



Heterogeneous team of an air and two ground vehicles that can perform cooperative reconnaissance and surveillance

Introduction to Multirobot Systems

- **Multirobot Systems: Increasing performance**



Maximize:

- Area Coverage
- Object Coverage
- Radio Coverage

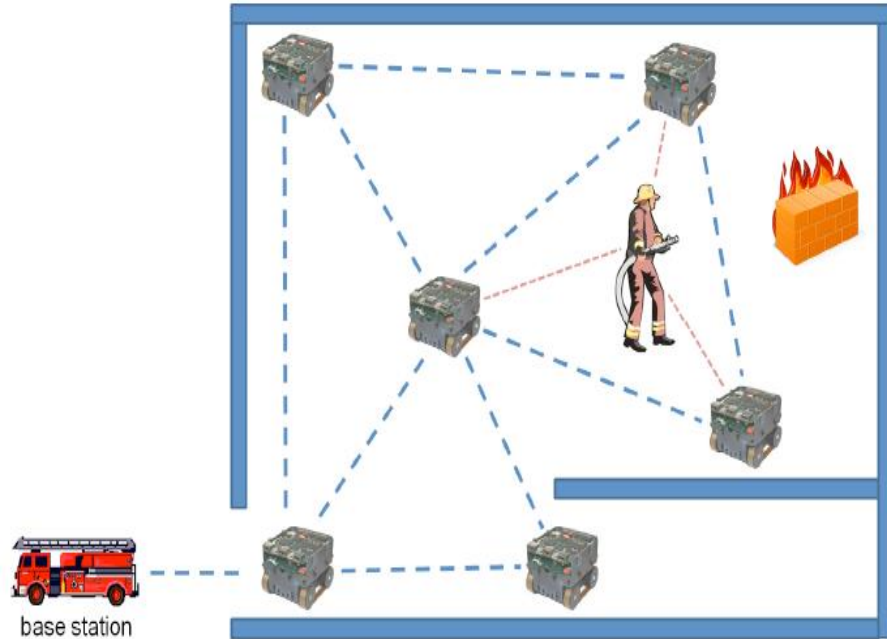
Minimize:

- Task completion time

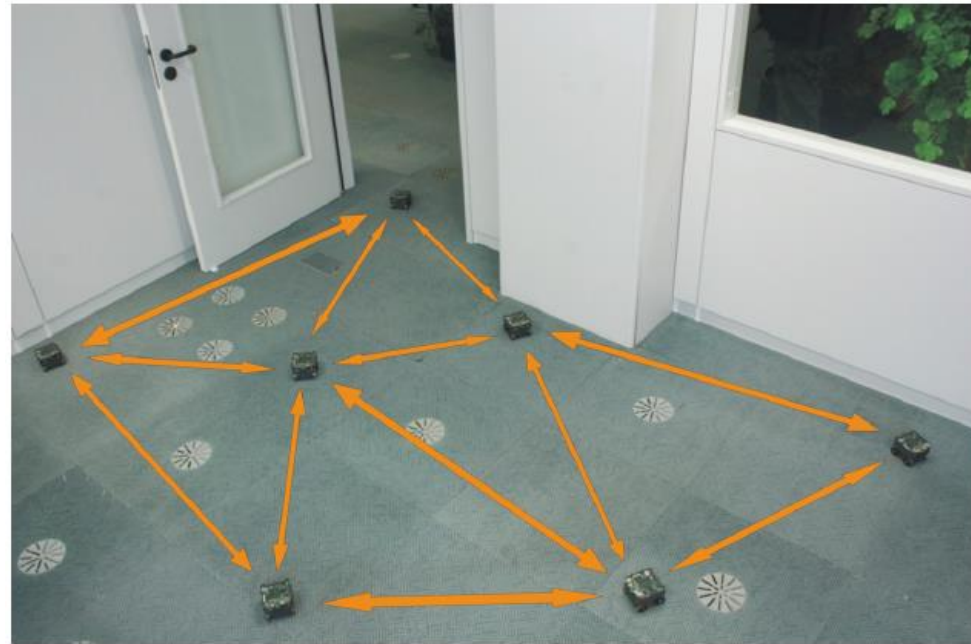
Multiple robots can solve problems faster using parallelism.

Introduction to Multirobot Systems

- **Multirobot Systems: Increasing performance**



Mobile ad-hoc communication network with integrated human team member and link to base station for monitoring the operation



Robots are spanning a communication network with equilateral triangles after they drove into a room

Ulf Witkowski et al, "Ad-hoc network communication infrastructure for multirobot systems in disaster scenarios",

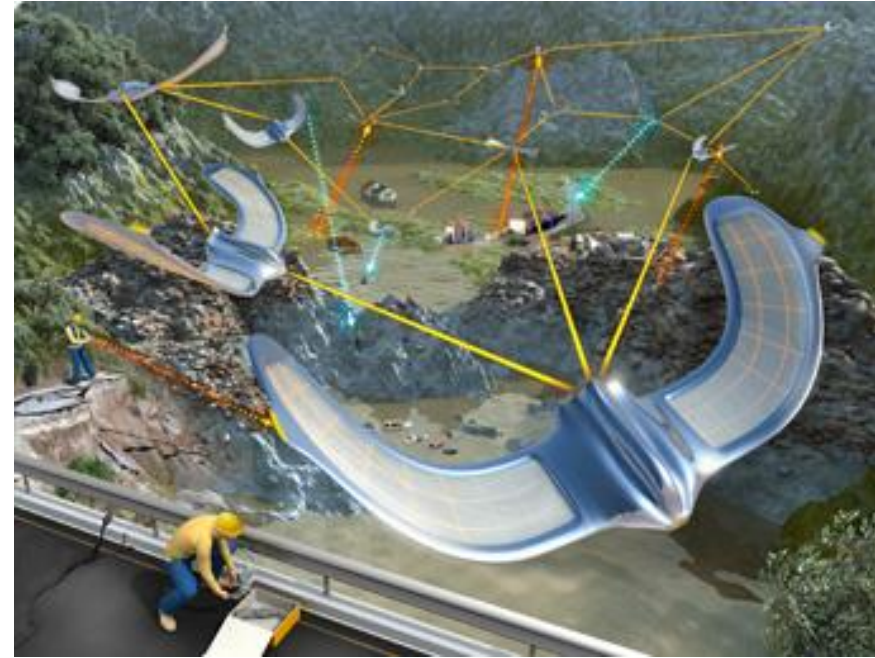
Introduction to Multirobot Systems

- **Multirobot Systems: Reliability**

Increasing the system reliability because having only one robot may work as a bottleneck for the whole system especially in critical times.

But when having multiple robots doing a task and one fails, others could still do the job.

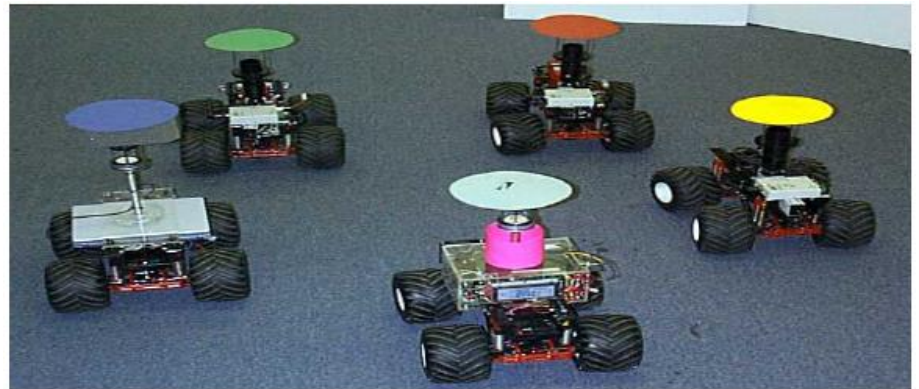
The introduction of multiple robots **increases robustness** through **redundancy**.



Introduction to Multirobot Systems

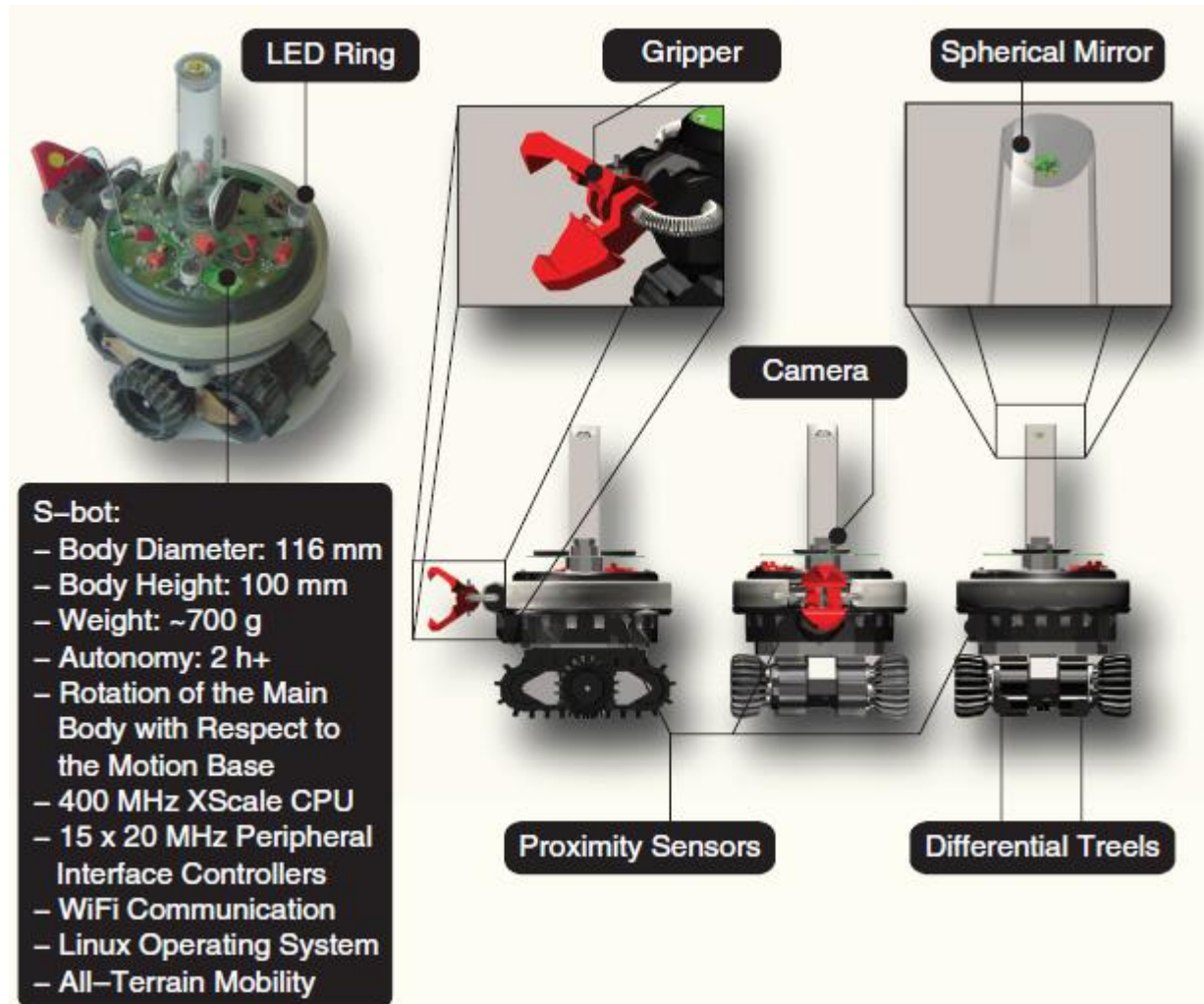
- **Multirobot Systems: Simplicity in design**

Building several resource-bounded robots is much easier than having a single powerful robot



Introduction to Multirobot Systems

- **Multirobot Systems: Simplicity in design**

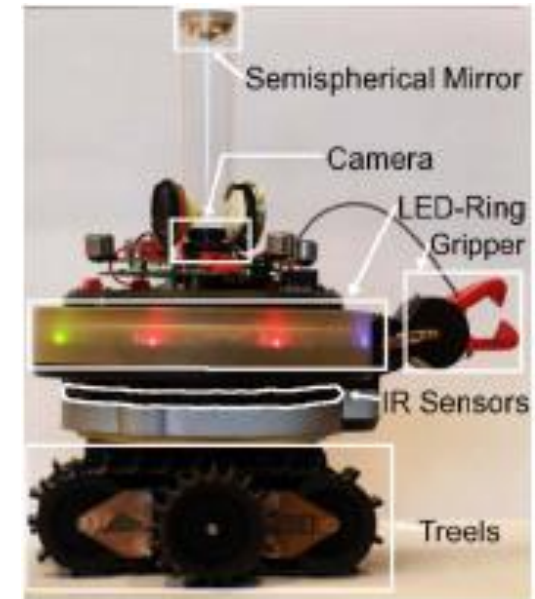


S-bot: an autonomous, mobile robot capable of self-assembly

ANDERS LYHNE CHRISTENSEN, REHAN O'GRADY, AND MARCO DORIGO, "Morphology Control in a Multirobot System: Distributed Growth of Specific Structures Using Directional Self-Assembly", IEEE Robotics & Automation Magazine, December 2007.

Introduction to Multirobot Systems

- **Multirobot Systems: Simplicity in design**



s-bot

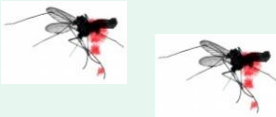
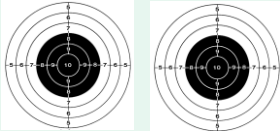
Small Mobile Robot

Dario Floreano and Claudio Mattiussi. *Bio-Inspired Artificial Intelligence: Theories, Methods, and Technologies*. MIT Press

Introduction to Multirobot Systems

- **Swarm Robotics**

How large number of relatively **simple physically embodied agents** can be designed such that a desired **collective behavior** emerges from the local interactions among agents and between the agents and the environment.

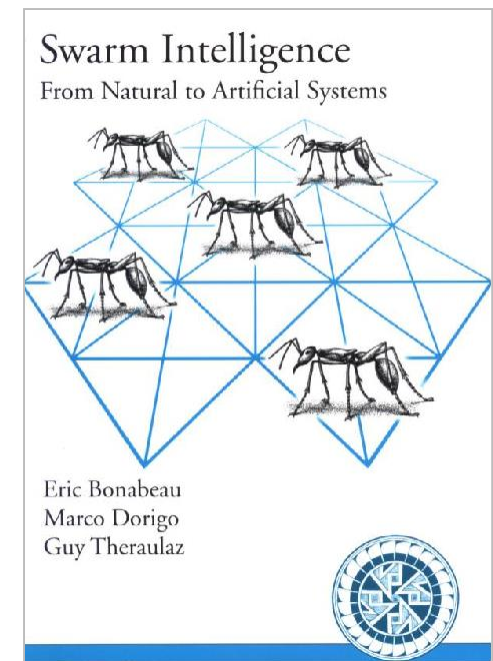
Analogies	Nature	Unmanned Aerial/Ground Vehicles
Ants/UAVs		
Prey/Target		
Predators/Threats		
Environment/ Battlefield		

Introduction to Multirobot Systems

- **Swarm Robotics and Swarm Intelligence**

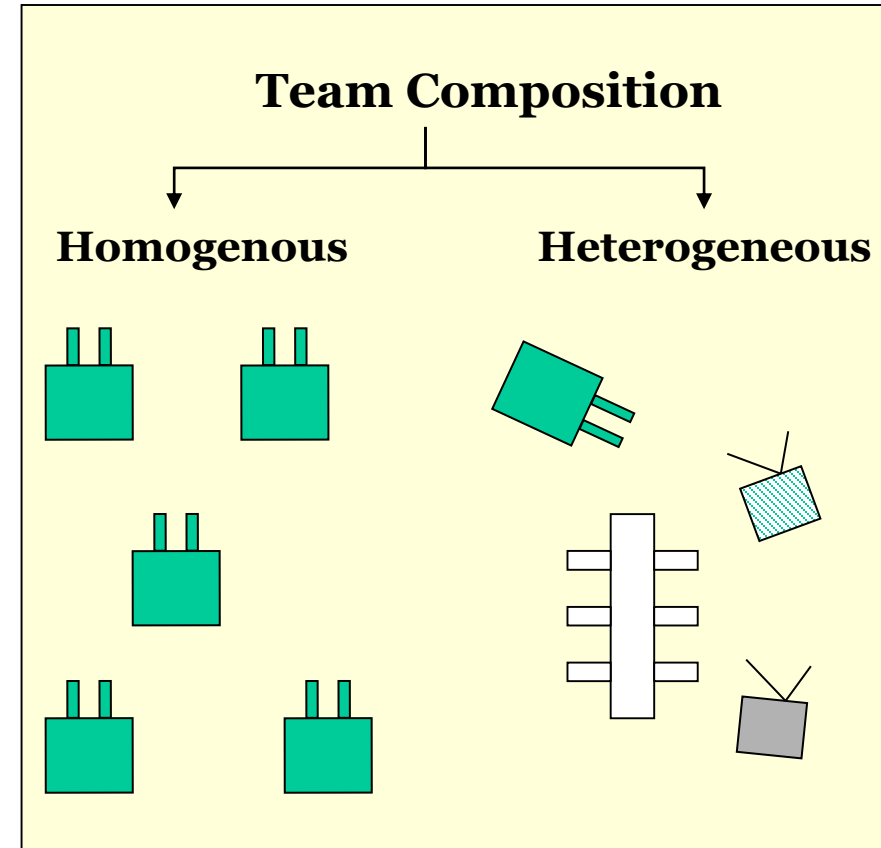
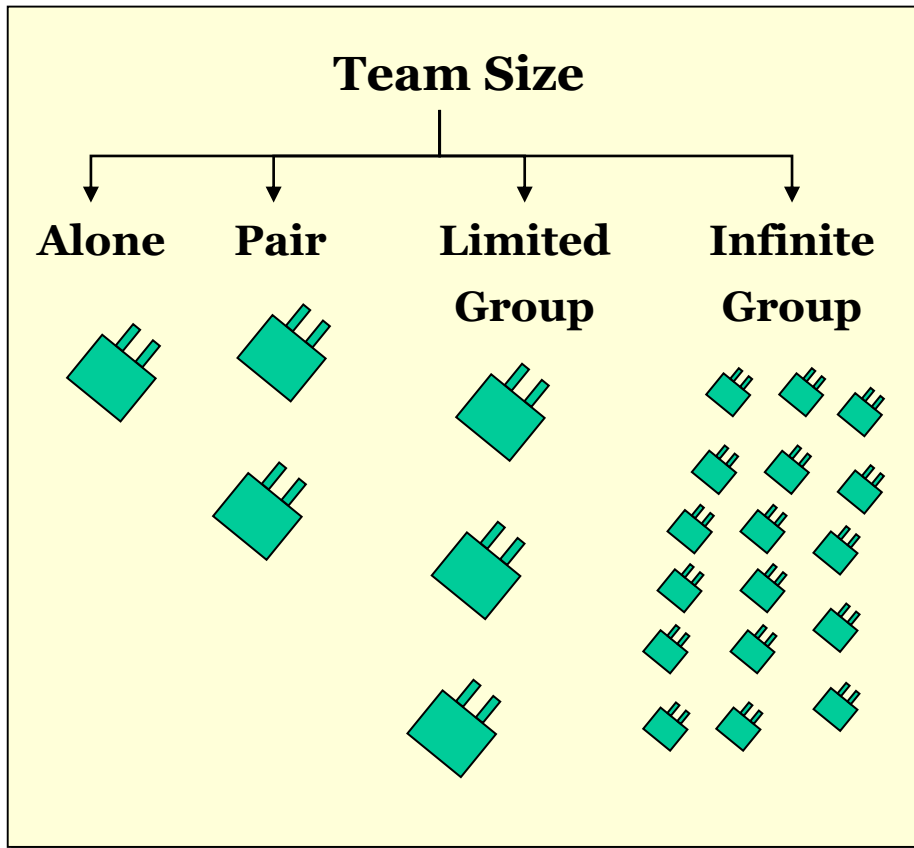
Swarm intelligence (SI) refers to the phenomenon of a system of **spatially distributed individuals** coordinating their actions in a **decentralized and self-organized** manner so as to exhibit **complex collective behavior**.

“SI is the property of a system whereby the **collective behaviors** of (unsophisticated) agents interacting locally with their environment cause coherent functional global patterns to emerge.”



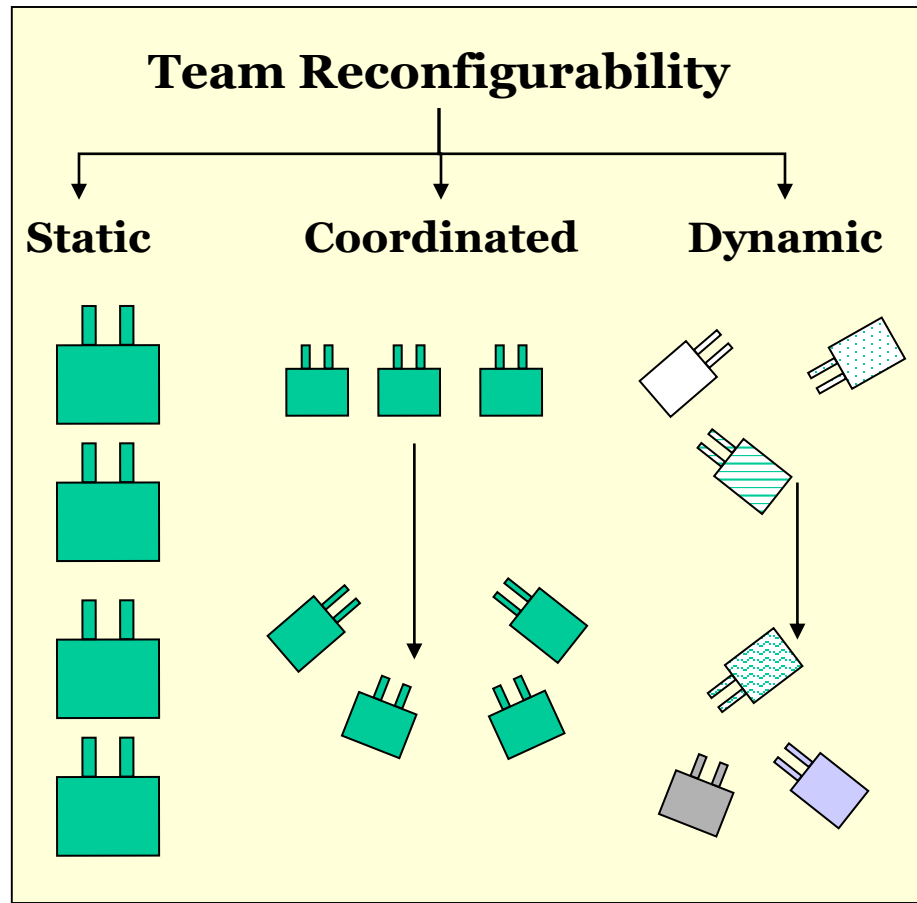
Introduction to Multirobot Systems

- Aspects of MRS: Team Size and Composition



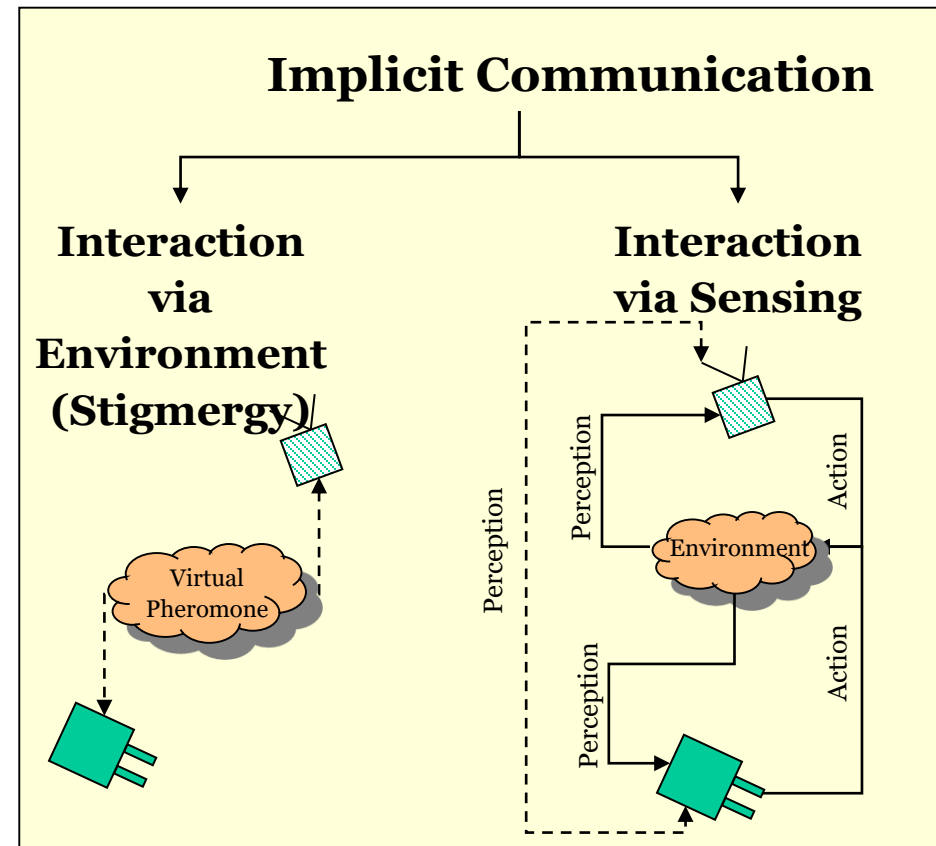
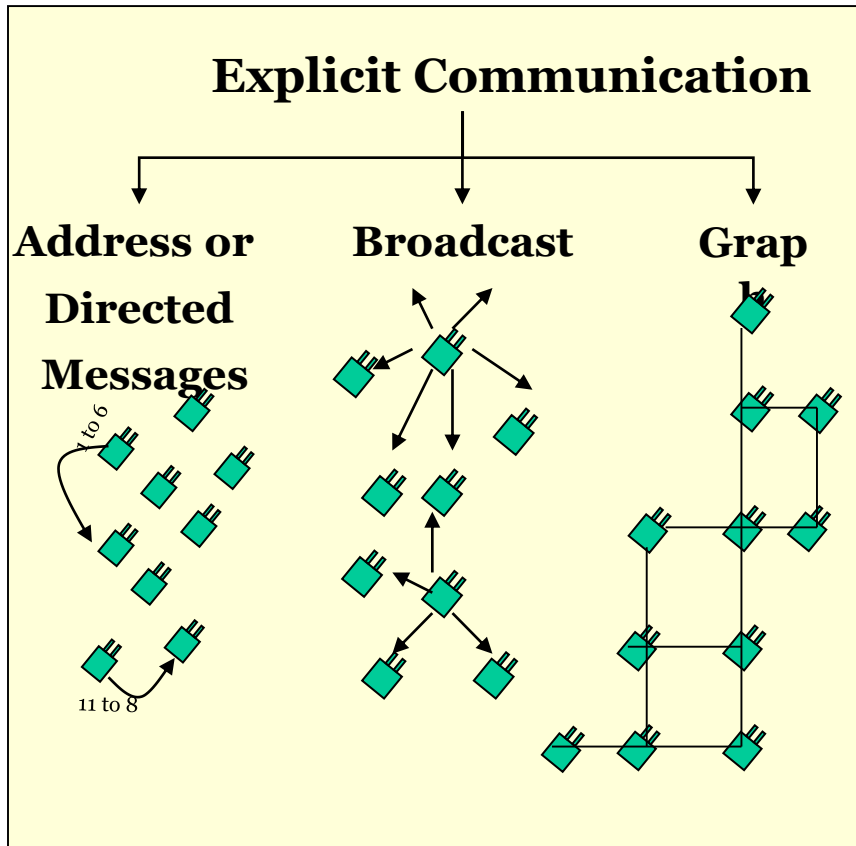
Introduction to Multirobot Systems

- **Aspects of MRS: Team Reconfigurability**



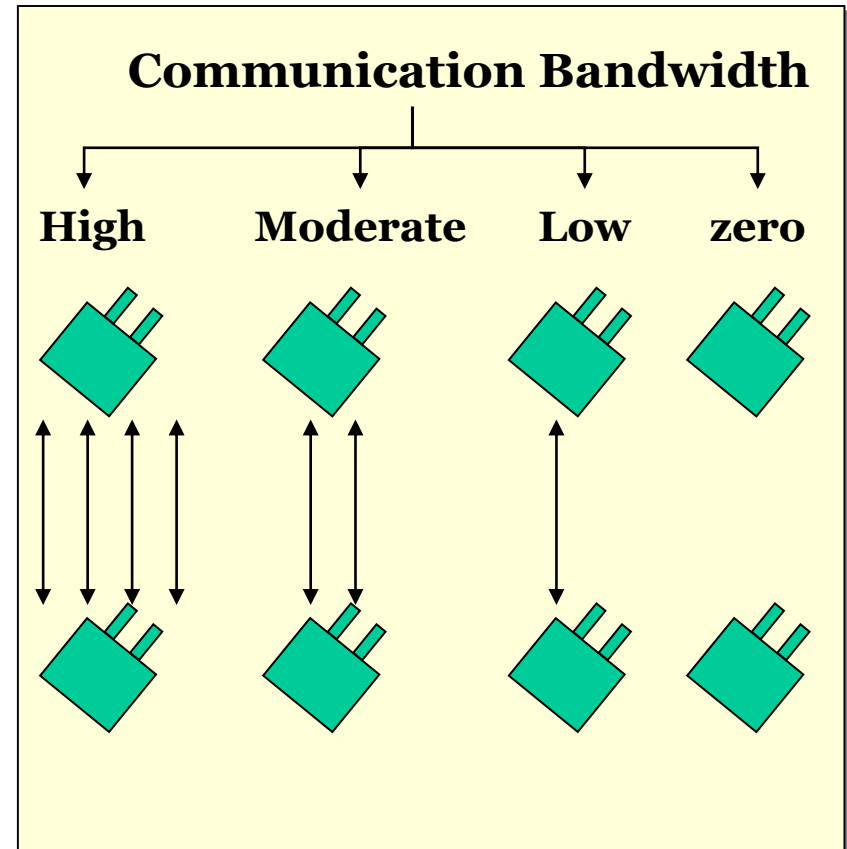
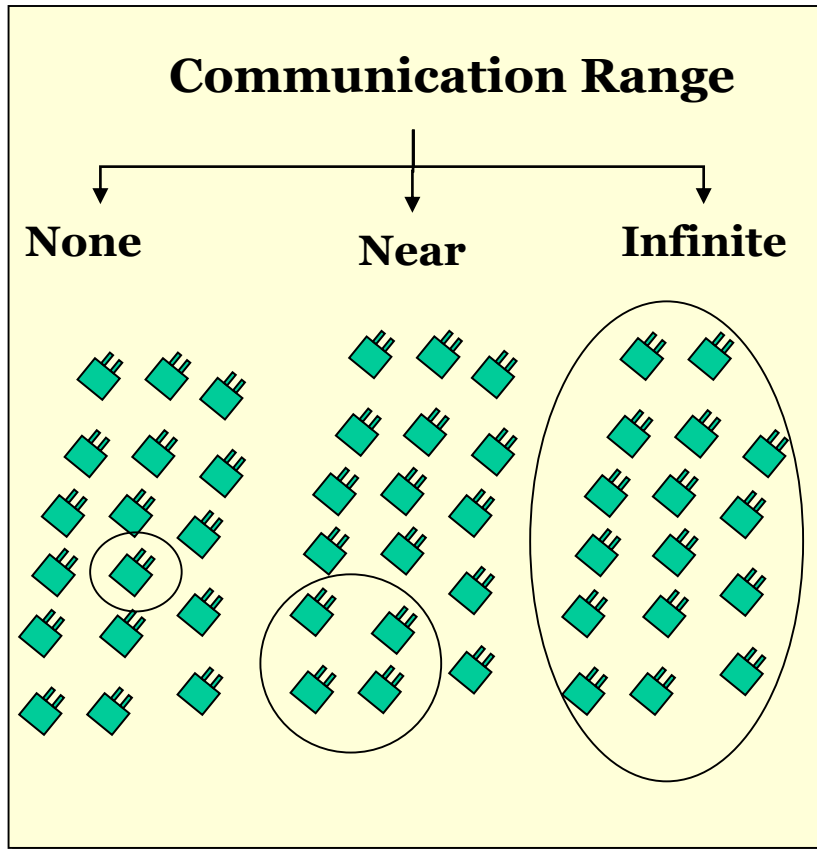
Introduction to Multirobot Systems

- Aspects of MRS: Communication Pattern



Introduction to Multirobot Systems

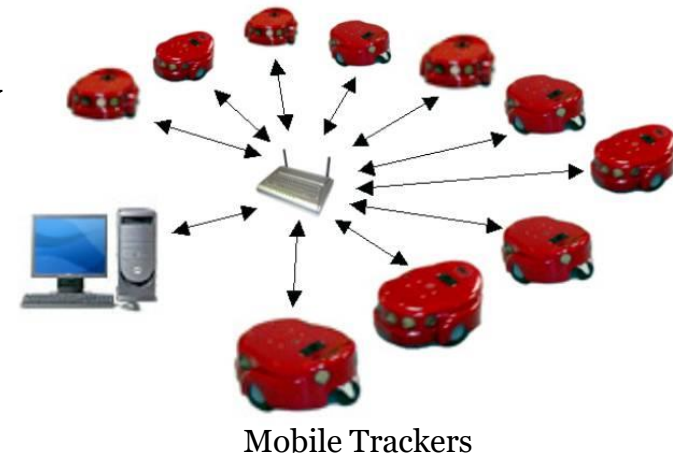
- Aspects of MRS: Comm. Range and Bandwidth



Introduction to Multirobot Systems

• Applications

- ◇ Autonomous inspection of complex engineered structures.
- ◇ Distributed sensing tasks in micromachinery or the human body.
- ◇ Killing Cancer Tumors in Human Body
- ◇ Mining
- ◇ Agricultural Foraging
- ◇ Cooperative Tracking
- ◇ Surveillance, Reconnaissance and Intelligence.



Spy robots

Introduction to Multirobot Systems

- **Applications: Intelligent Carts**

Unintelligent carts are commonly found in large airports. Travelers pick up carts at designated points and leave them in arbitrary places. It is a considerable task to re-collect them.



It is, therefore, desirable that intelligent carts (intelligent robots) **draw themselves** together autonomously.

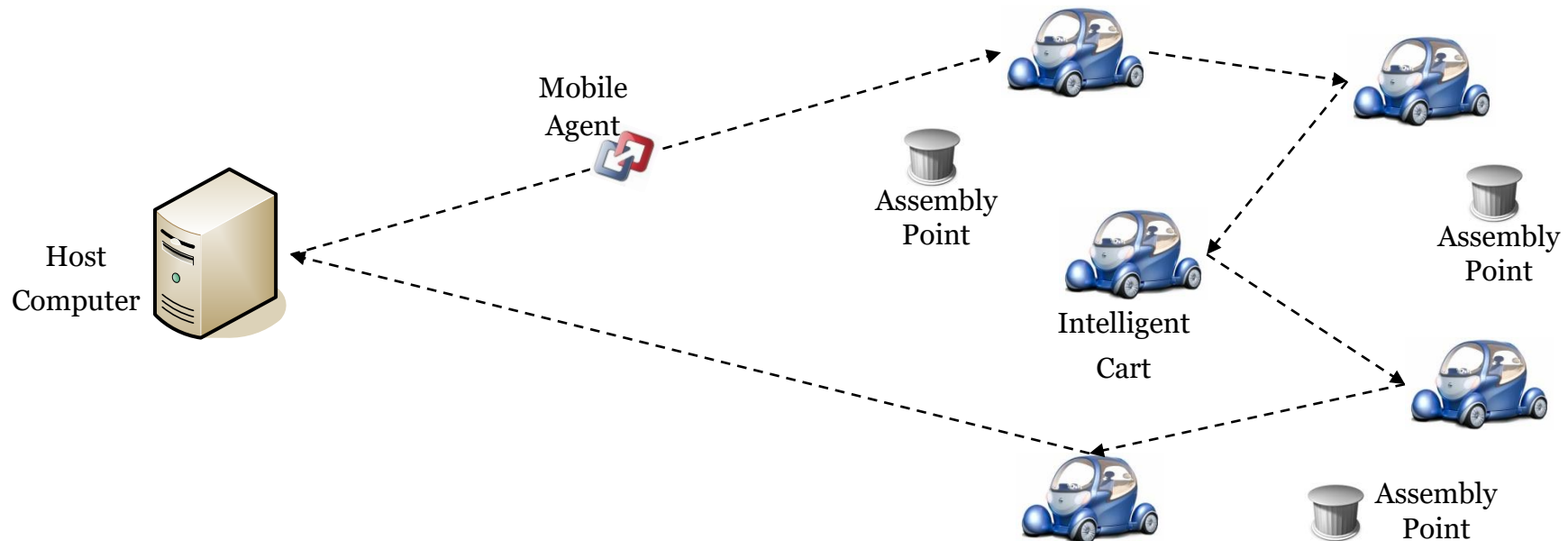


Introduction to Multirobot Systems

- Applications: Intelligent Carts

- Quasi Optimal Robots Collection

Step 1: One **mobile agent** issued from the host computer **visits scattered robots** one by one and **collects the positions** of them.

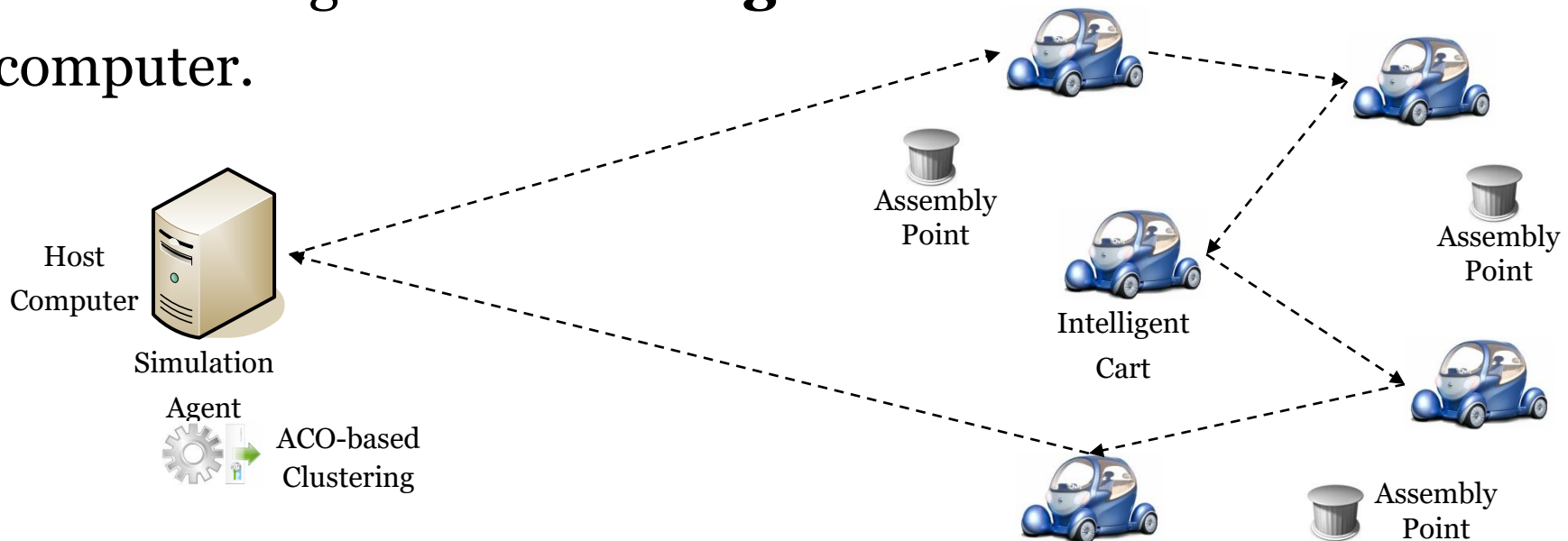


Introduction to Multirobot Systems

- Applications: Intelligent Carts

- Quasi Optimal Robots Collection

Step 2: Another agent called the **simulation agent**, performs **clustering algorithm** and produces the quasi optimal gathering positions for the mobile robots. The simulation agent is a **static agent** that resides in the host computer.

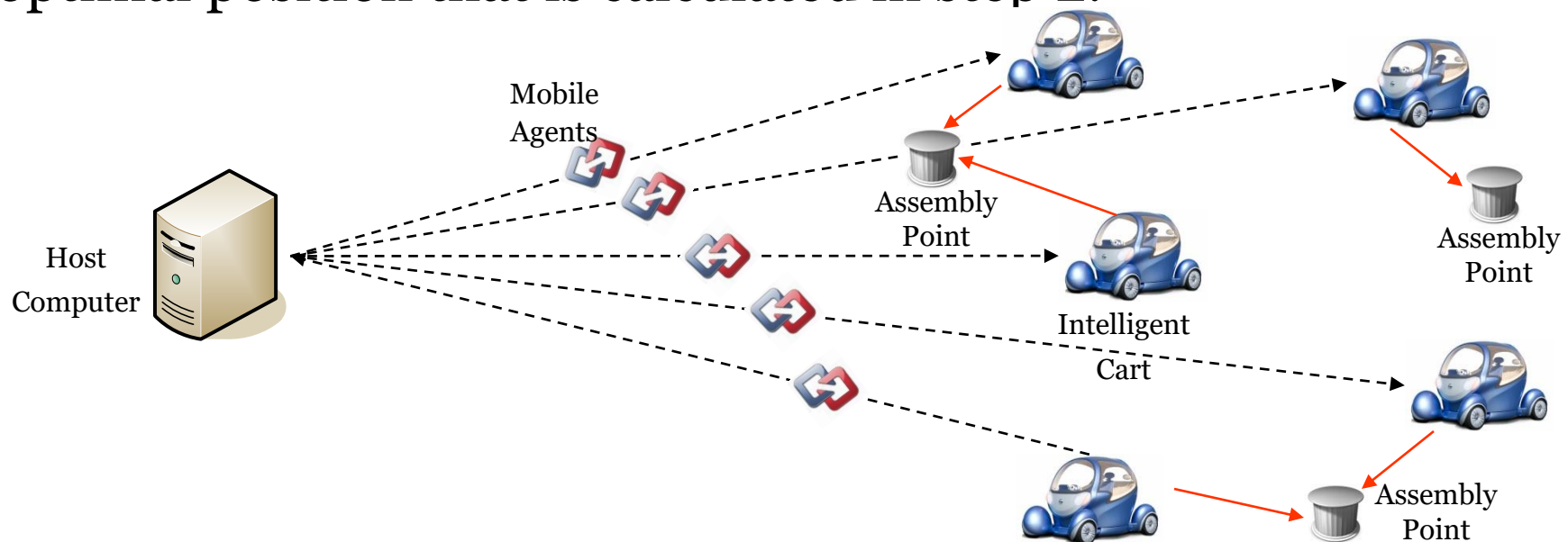


Introduction to Multirobot Systems

- Applications: Intelligent Carts

- Quasi Optimal Robots Collection

Step 3: A number of **mobile agents** are issued from the host computer. One mobile agent migrates to a designated mobile robot, and **drives the robot** to the assigned quasi optimal position that is calculated in step 2.

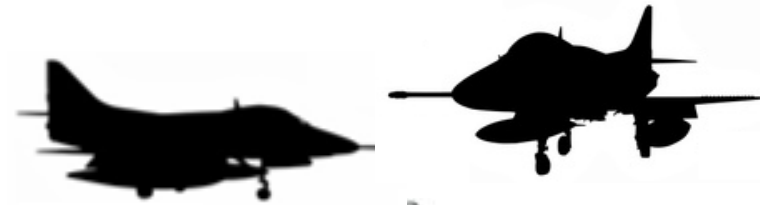


Introduction to Multirobot Systems

- **Applications**

- ◇ Military Applications: Drones

- Congress has mandated that by 2015, 1/3rd of all US military missions should be unmanned.
 - There are 17,300 drones in the US army inventory.
 - These drones can carry up to 3000 pounds of weapons.
 - Fabricated by Boeing



UAVs



UGV

UAV carrying
Viper Strike
Weapon
System



A forward looking
infrared (FLIR) camera
mounted on the side of an
UAV

Introduction to Multirobot Systems

- **Applications**

- ◇ Military Applications: UGV



Boston Dynamics BigDog
Robot The Army mule



Boston Dynamics
RHex

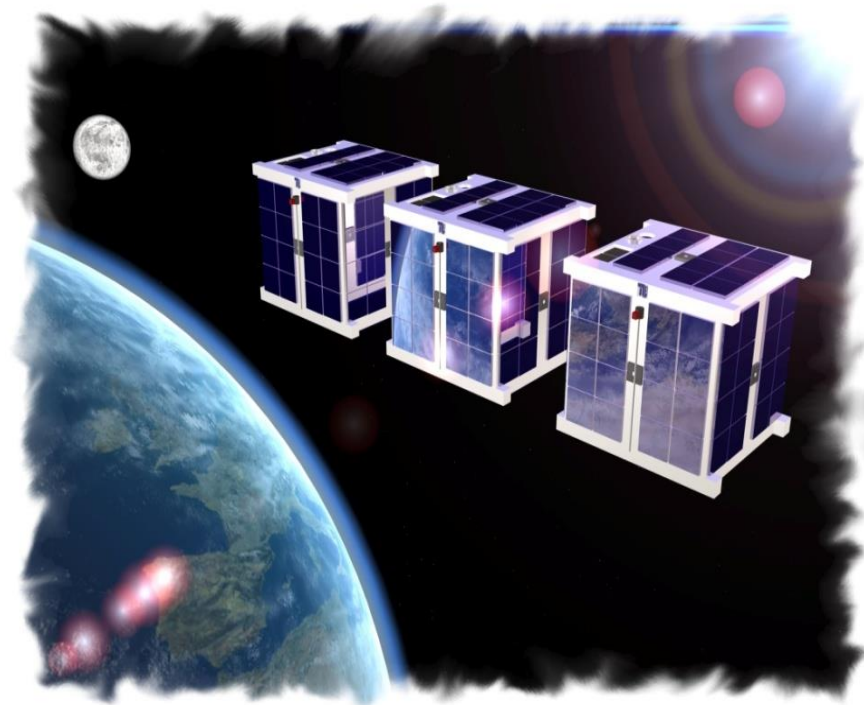


Introduction to Multirobot Systems

- **Applications**

- ◇ Small and Pico Satellites

- Small satellites are artificial satellites of lower weights and smaller sizes (under 500 kg).
 - Small satellites can be Minisatellite, Microsatellite, Nanosatellite, Picosatellite or Molecularsatellite.
 - Picosatellite or “picosat” is an artificial satellite with a wet mass between 0.1 and 1 kg.



CubeSat

Introduction to Multirobot Systems

- **Applications: Homeland Security**

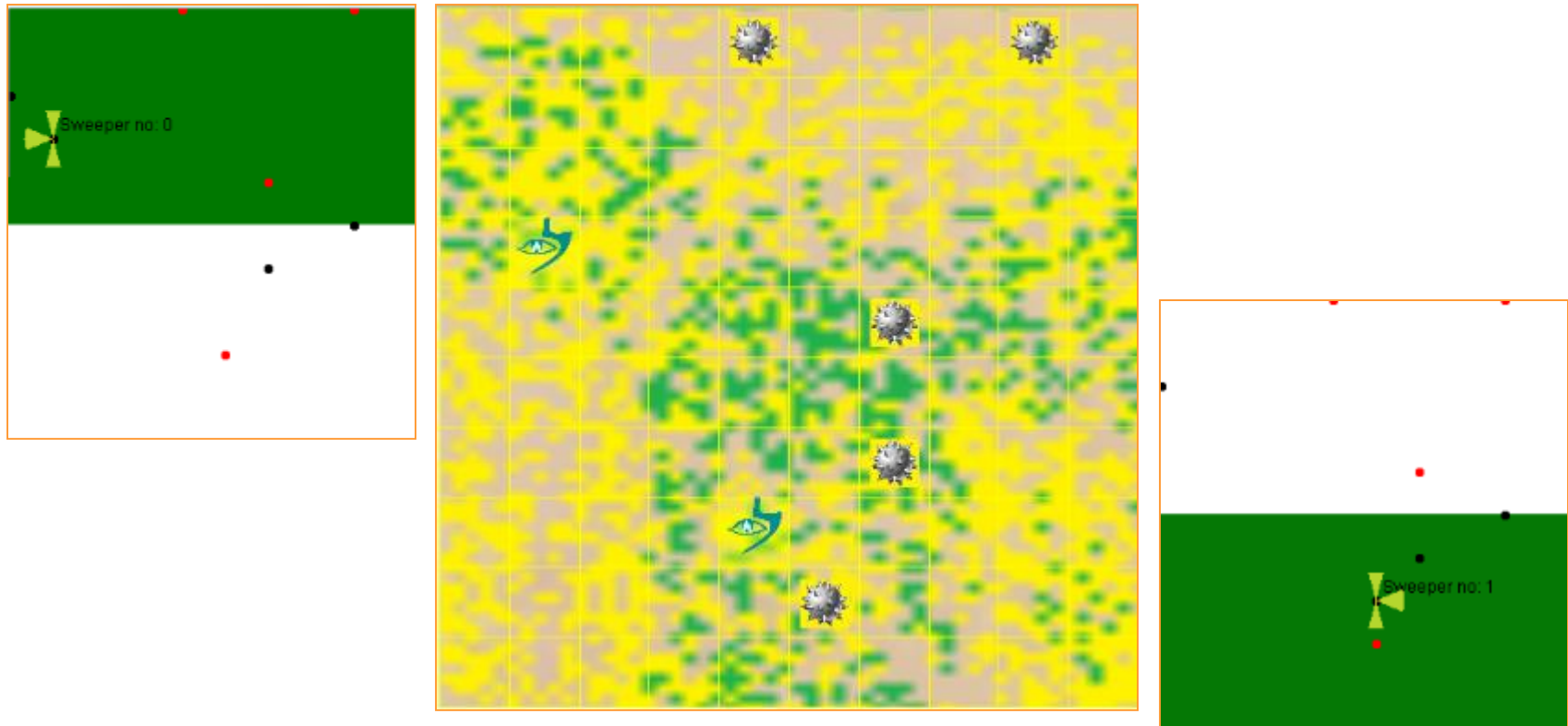


- Border protection
- Territorial protection
- Critical infrastructures protection
- Transport infrastructure protection
- Crisis management and civil protection
- Large event management

Source: <http://www.selex-si.com/SelexSI/EN//index.sdo>

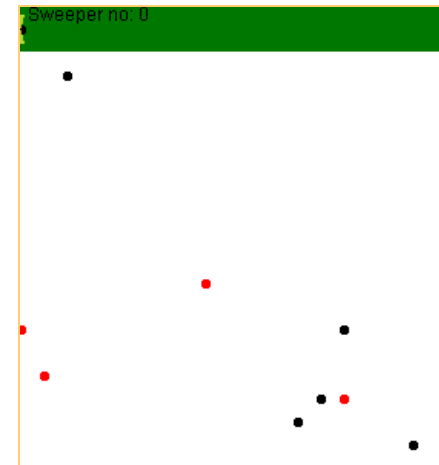
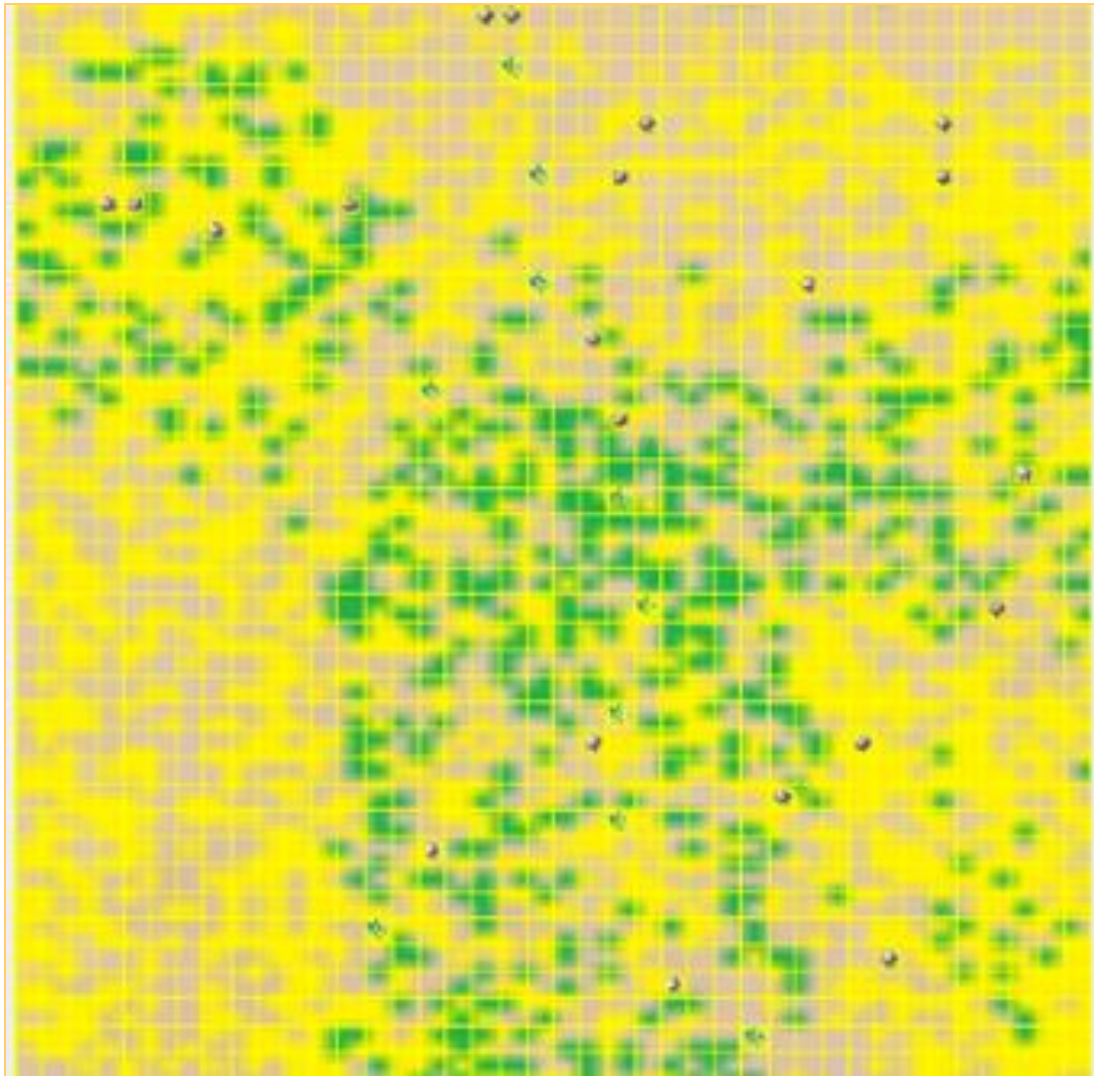
Introduction to Multirobot Systems

- **Applications: Humanitarian Demining**

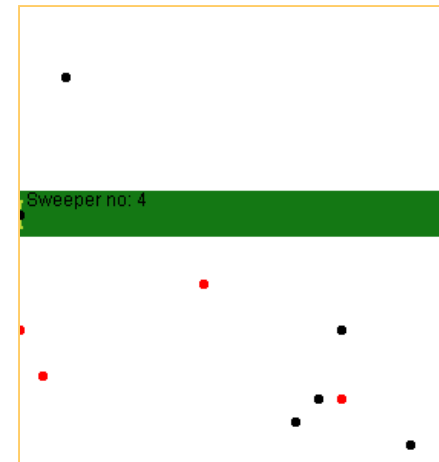


Introduction to Multirobot Systems

- **Applications: Humanitarian Demining**



...

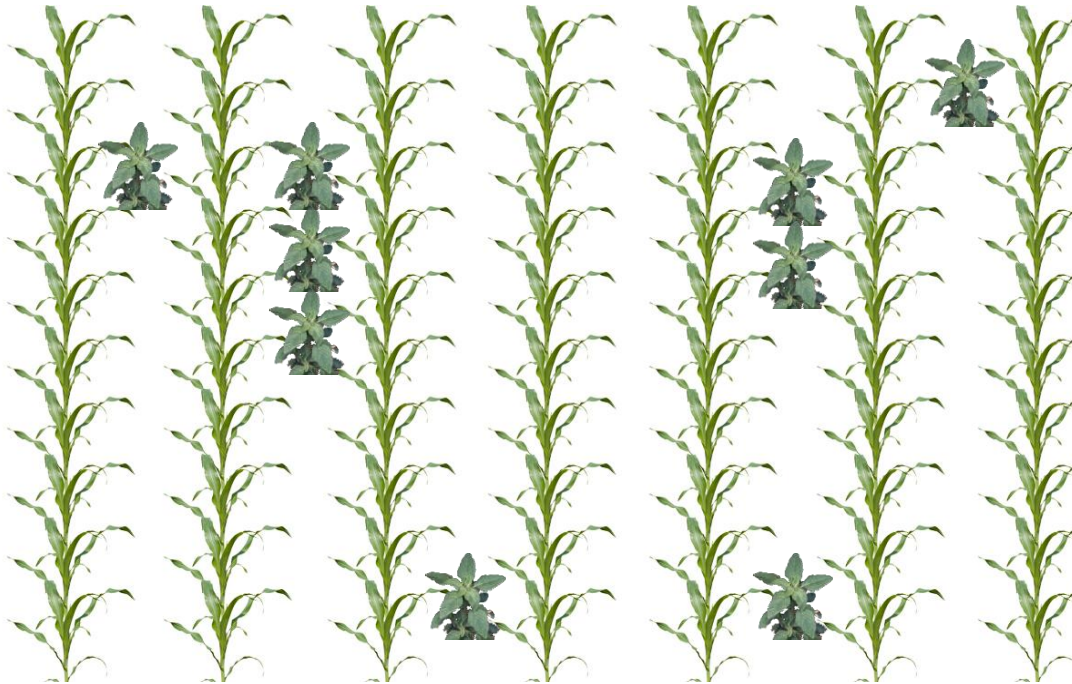


<http://www.landminefree.org/> © Asser El-Gindy
and Alaa Khamis, German University in Cairo

Introduction to Multirobot Systems

- **Applications: Agriculture**

Arable Field

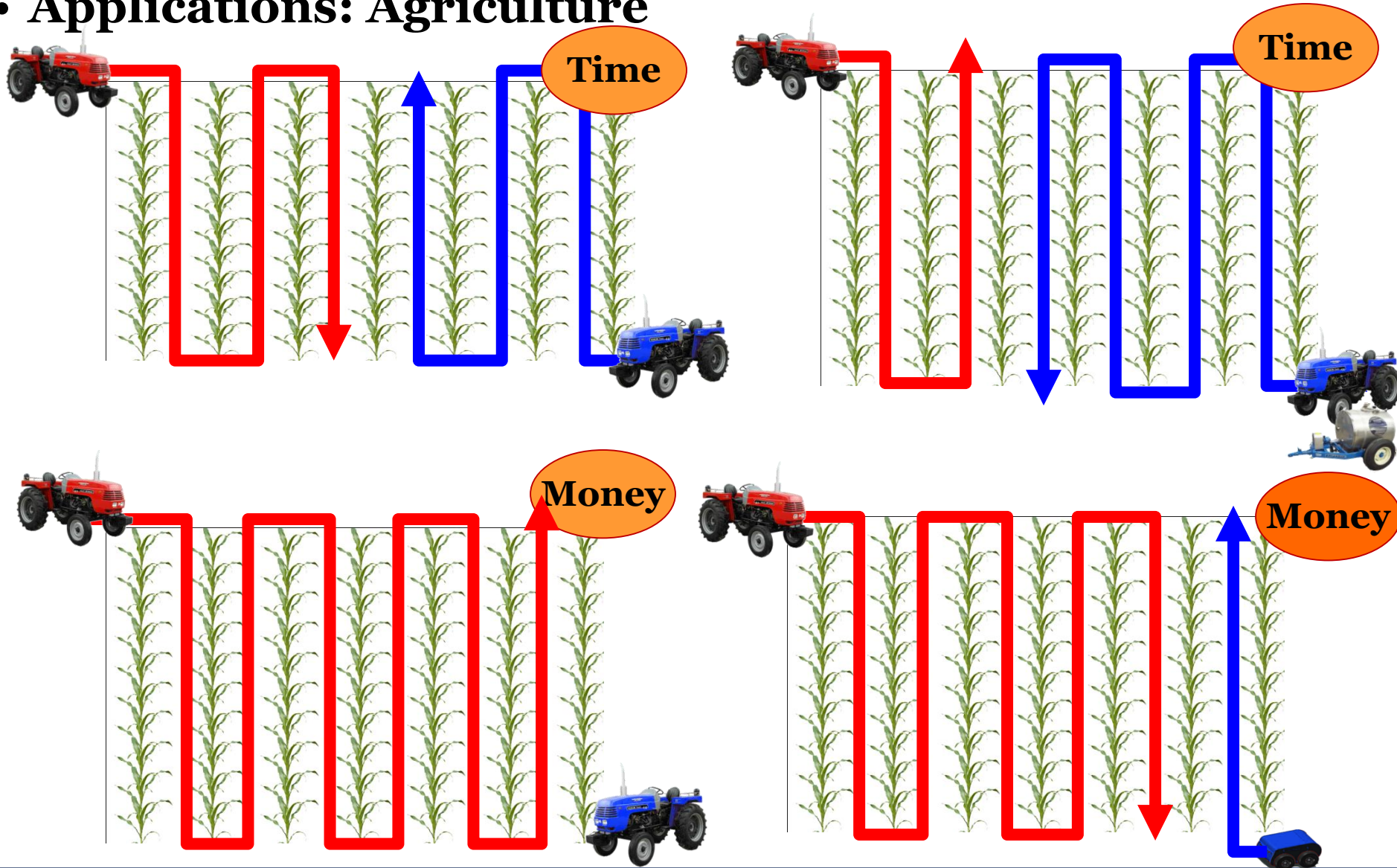


Fleet of heterogeneous robots

Jesús Conesa-Muñoz, Ángela Ribeiro, and Gonzalo Pajares, "A Multi-path Planning Approach Based on a Genetic Algorithm for a Robot Fleet Working in Arable Crops", EPIA'2011

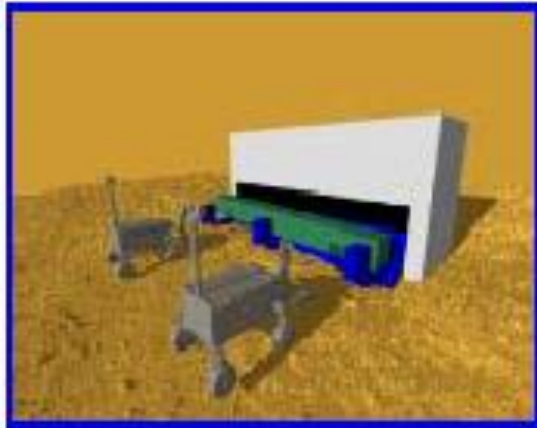
Introduction to Multirobot Systems

- Applications: Agriculture

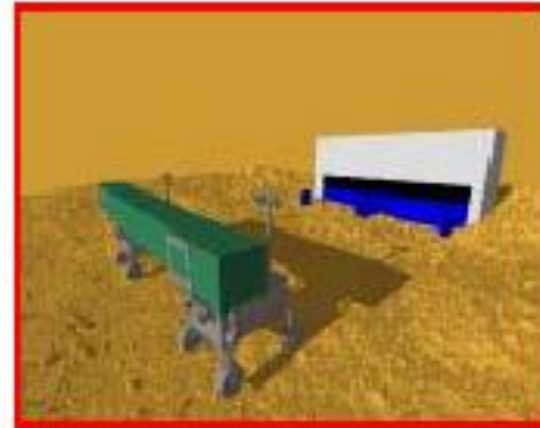


Introduction to Multirobot Systems

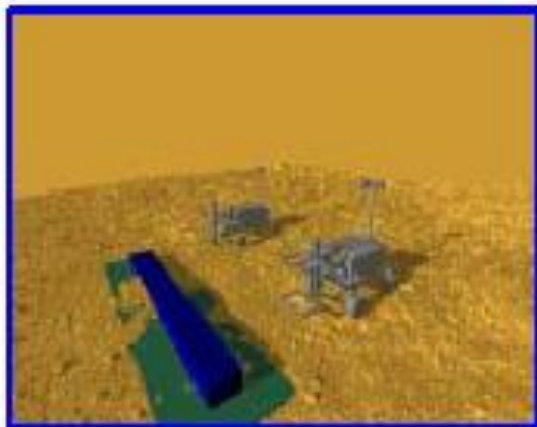
- **Applications: Space-based Construction**



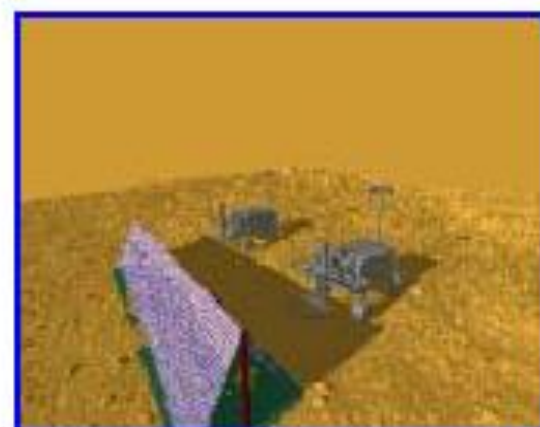
1. Unload container from Container Storage Unit (CSU)



2. Traverse to deployment site



3. Position and open container



4. Deploy PV tent

Introduction to Multirobot Systems

- **Applications: Search & Rescue Operations**



Companion slides for the book *Bio-Inspired Artificial Intelligence: Theories, Methods, and Technologies* by Dario Floreano and Claudio Mattiussi, MIT Press

Introduction to Multirobot Systems

- **Projects**

<http://www.swarm-bots.org/>

The SWARM-BOT project aims to study a novel swarm robotics system.

- ◇ It is directly inspired by the **collective behavior** of social insects and other animal societies.
- ◇ It focuses on **self-organization** and **self-assembling** of autonomous agents.
- ◇ Its main scientific challenge lays in the development of a **novel hardware and of innovative control** solutions.



Swarm-bots, Marco Dorigo, 2005

Introduction to Multirobot Systems

- **Projects**

The Centibots project:

The Centibots are a team of **100 autonomous robots** (97

ActivMedia Amigobot and 6

ActivMedia Pioneer 2 AT). The

goal of the project is to

demonstrate 100 robots

mapping, tracking, guarding

in a coherent fashion during a

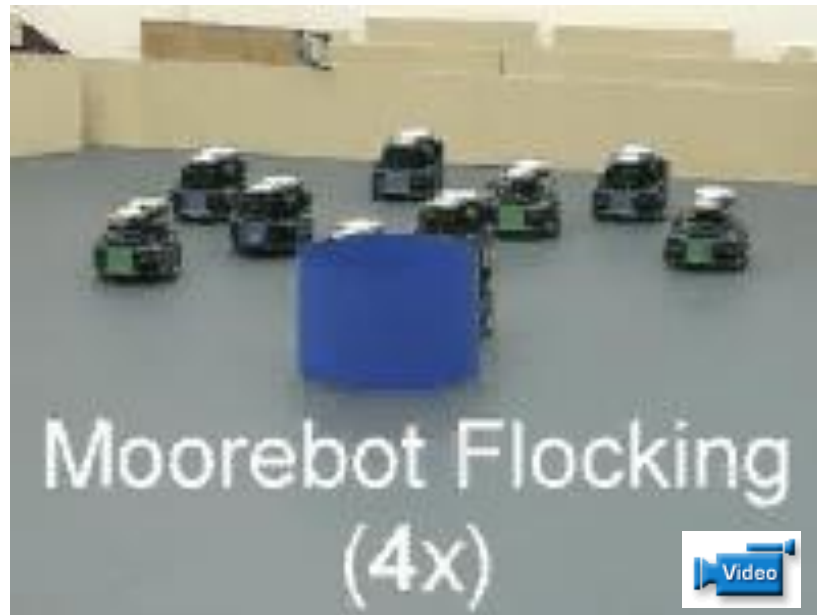
period of 24 hours.

<http://www.ai.sri.com/centibots/index.html>



Introduction to Multirobot Systems

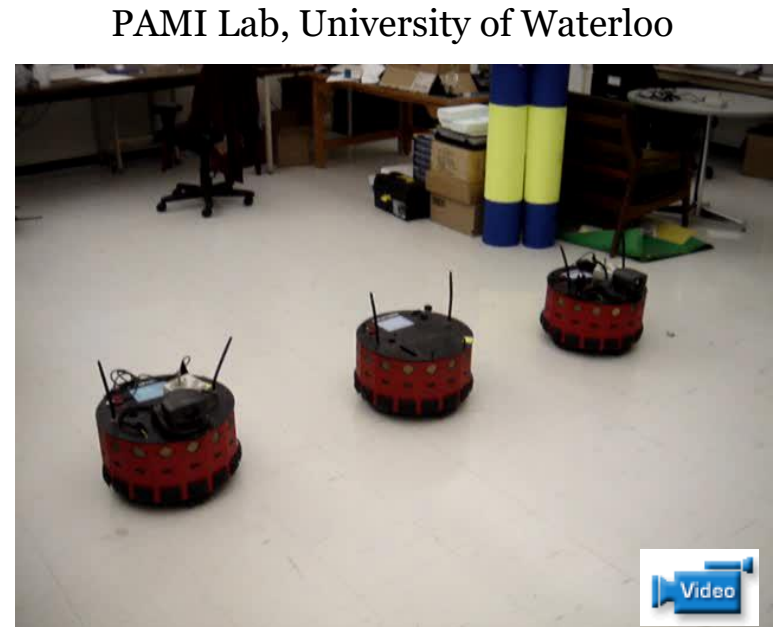
- **Projects**



Leaderless Distributed (LD) Flocking Algorithm

Flocking in Embedded Robotic Systems

CORO - Caltech

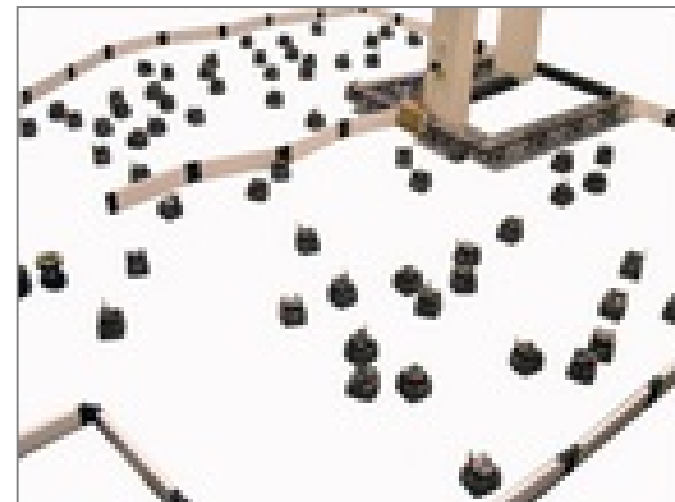


Introduction to Multirobot Systems

- **Projects**

Multi-Robot Systems Lab at Rice University develops distributed algorithms for multi-robot systems, with a focus on both algorithm and systems design.

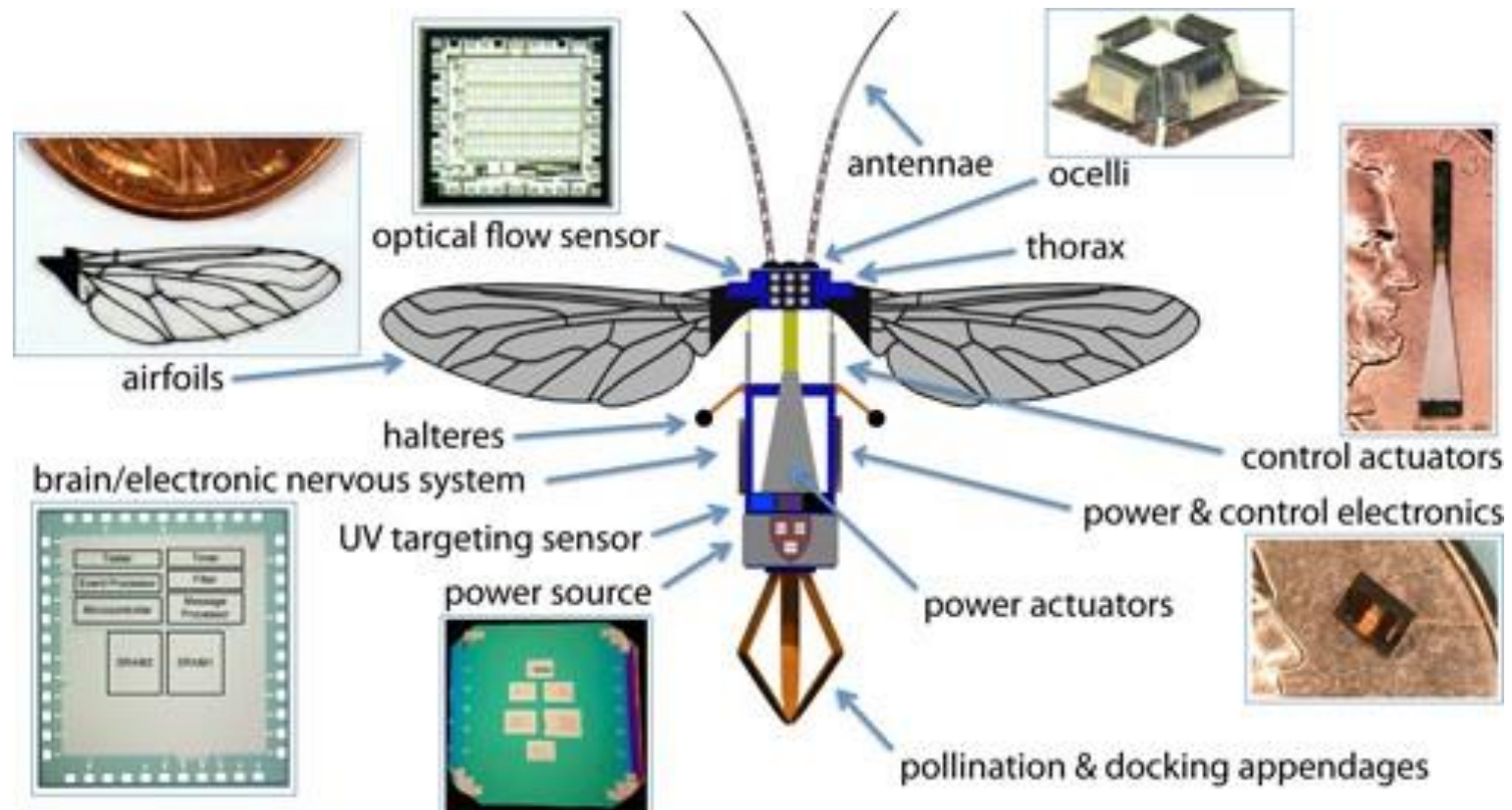
- ◇ Hexagonal Lattice Formation in MRS
- ◇ Agreement on Stochastic MRS with Communication Failures
- ◇ MRS Engineering
- ◇ Dynamic Task Assignment
- ◇ Boundary Detection



Introduction to Multirobot Systems

- **Projects: ROBOBEEES**

The objective of this project is to design “**smart**” **sensors**; and refine **coordination algorithms** to manage multiple, independent machines.



Introduction to Multirobot Systems

- **Projects: Cloud Robotics**

Cloud computing is the **delivery of computing as a service** rather than a product, whereby shared resources, software, and information are provided to computers and other devices as a **utility (like the electricity grid)** over a network (typically the Internet).

Using the cloud, a robot could **improve capabilities** such as speech recognition, language translation, path planning, and 3D mapping.



<http://spectrum.ieee.org/automaton/robotics/robotics-software/cloud-robotics>

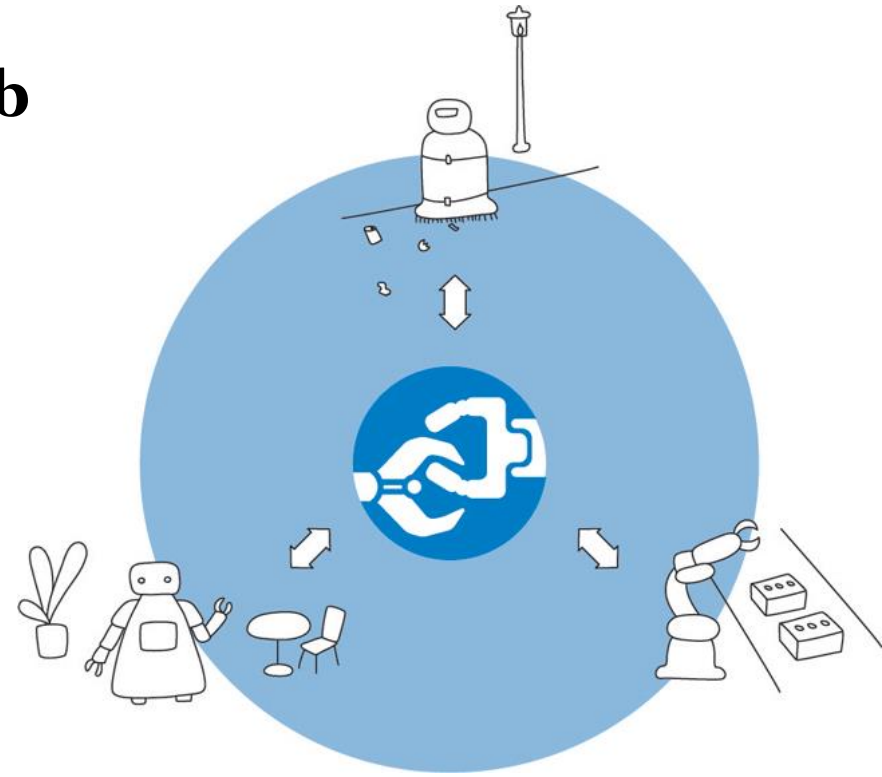
<http://www.google.com/events/io/2011/sessions/cloud-robotics.html>

Introduction to Multirobot Systems

- **Projects: RoboEarth**

RoboEarth is a **World Wide Web for robots**: a giant network and database repository where robots can share information and learn from each other about their behavior and their environment.

RoboEarth offers a complete **Cloud Robotics infrastructure**, which includes everything needed to close the loop from robot to RoboEarth to robot.



<http://www.roboearth.org/>

Introduction to Multirobot Systems

• Commercial Robots: Wifibots

- ◇ Flexible and open modular architecture
- ◇ Easy to operate
- ◇ Fully programmable
- ◇ Low cost
- ◇ Small size
- ◇ Low weight
- ◇ Man portable
- ◇ Great indoor and outdoor mobility
- ◇ Transparent integration within existing Wi-Fi networks
- ◇ Mesh networking support for Windows and Linux (BATMAN / OPENWRT)
- ◇ Great for multi-robot applications



<http://www.wifibot.com/>

Introduction to Multirobot Systems

- **Commercial Robots: K-Team Mobile Robots**



Kilobot



Koala II



K-Junior



Khepera II



Khepera III



Hemisson



KoreBot II

<http://www.k-team.com/>

Outline

- Seminar Description
- Introduction to Multirobot Systems
- **Benchmark Problems of Multirobot Systems**
- Organizational paradigms of Multirobot Systems
- Challenging Problems of Multirobot Systems
- Summary

Benchmark Problems of MRS

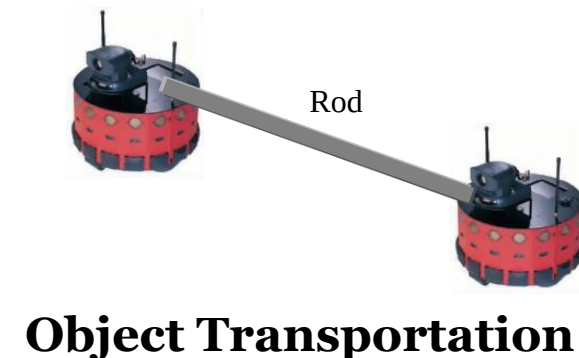
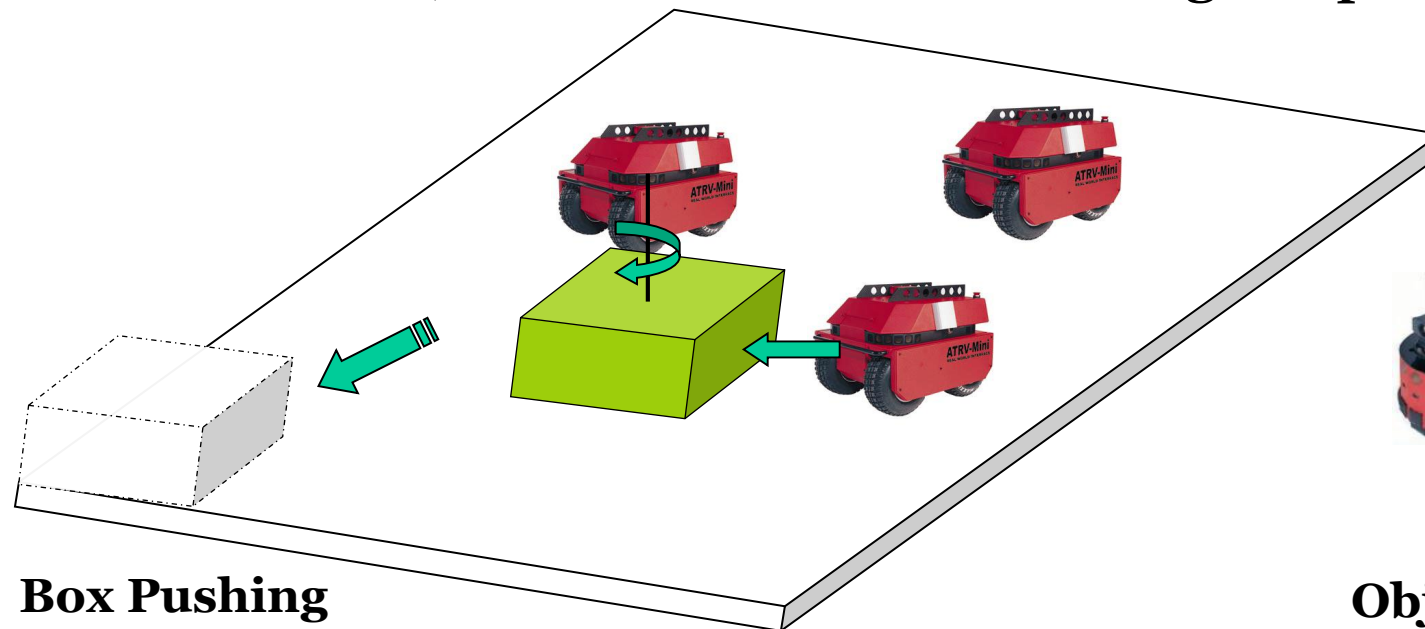
- Box Pushing and Object Transportation
- Exploration and Formation Control
- Division of Labor
- Foraging
- Object/Area/Radio Coverage
- Soccer Tournaments
- Cooperative perception
- Cooperative Target Cueing and Handoff
- Cooperative Mapping
- ...

Benchmark Problems of MRS

- **Box Pushing and Object Transportation**

This problem's concern is about a group of robots try to **push a box** to a certain point.

Applications include **transportation of heavy objects** in industrial environments or **assembly of large-scale structures**, such as terrestrial buildings or planetary habitat.



Object Transportation

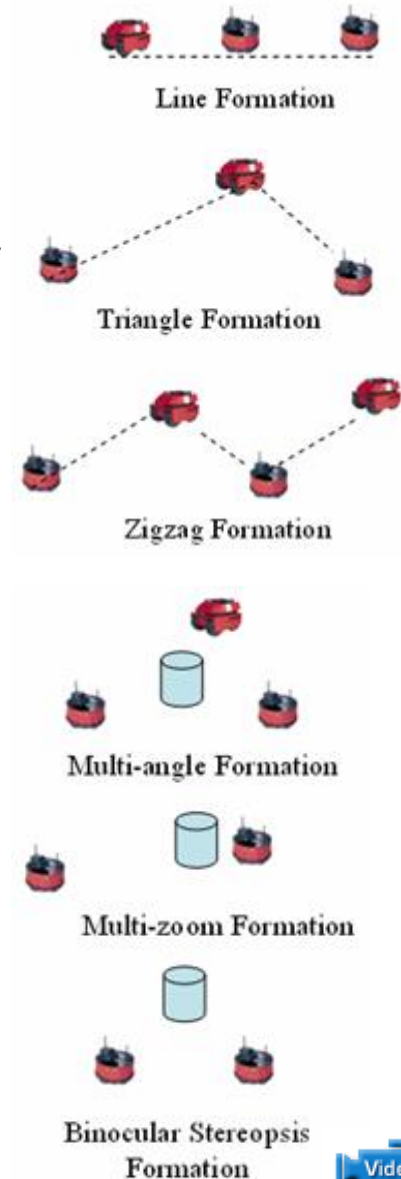
Box Pushing

Benchmark Problems of MRS

- **Exploration and Formation Control**

In the **exploration** task the robots must be **spread in the environment** in order to collect as much information as possible about the surrounding area.

The **formation** task is focused on having the robots move in the environment **forming particular shapes**.



Benchmark Problems of MRS

- **Exploration and Formation Control**

The problem of exploration and flocking is related with several applications such as:

- ◇ Transshipment operations in harbors,
- ◇ Airports and marshalling yards,
- ◇ Motion coordination in industrial application,
- ◇ Exploration of dangerous environments and searching victims in an unknown environment representing a disaster scenario.

Benchmark Problems of MRS

- **Foraging**

The foraging task requires robots to **pick up objects scattered** in the environment.

Although this task could be done by having only a single robot, using cooperative foraging enhance the overall performance of the system.

Applications of this cooperative behavior include:

- ◇ Toxic waste cleaning,
- ◇ Mine cleaning, and
- ◇ Service robots

Benchmark Problems of MRS

- **Coverage**

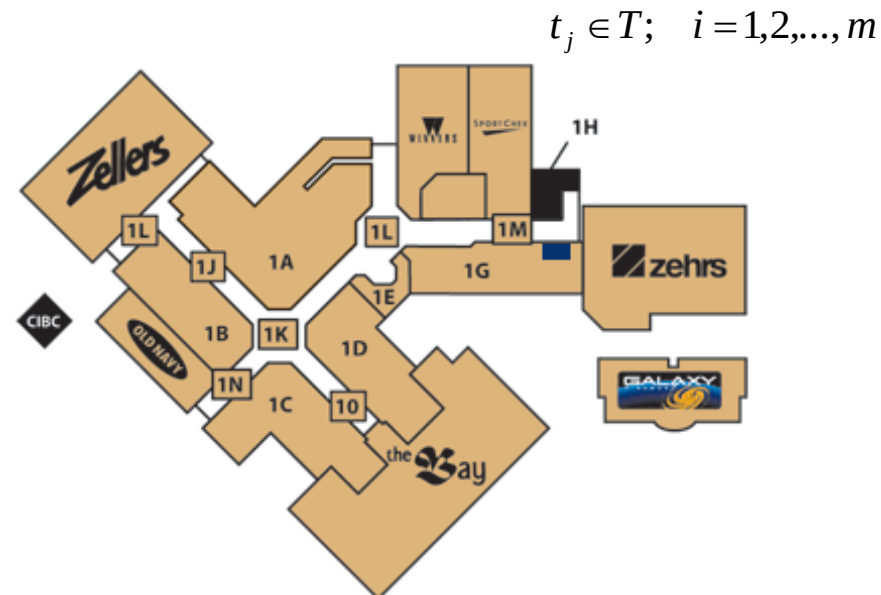
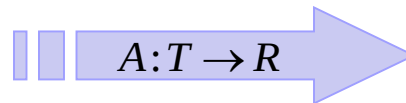
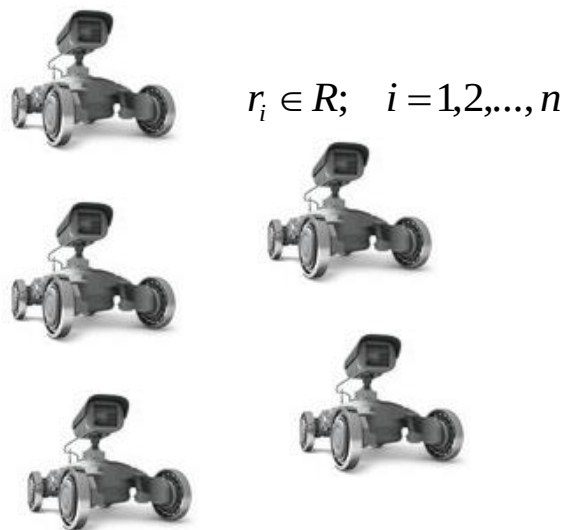
The coverage task is very similar to **foraging**, since it requires the robots to process all the points of the free space in the environment. The main issue for coverage is therefore, to find effective techniques for **cooperatively scanning all** the environment. Applications include:

- ◇ Distributed search,
- ◇ Demining,
- ◇ Snow removal,
- ◇ Lawn mowing,
- ◇ Car-body painting, etc.

Benchmark Problems of MRS

- **Division of Labor**

This cooperative behavior addresses how to dynamically assign a set of tasks to a set of robots to maximize overall expected performance.



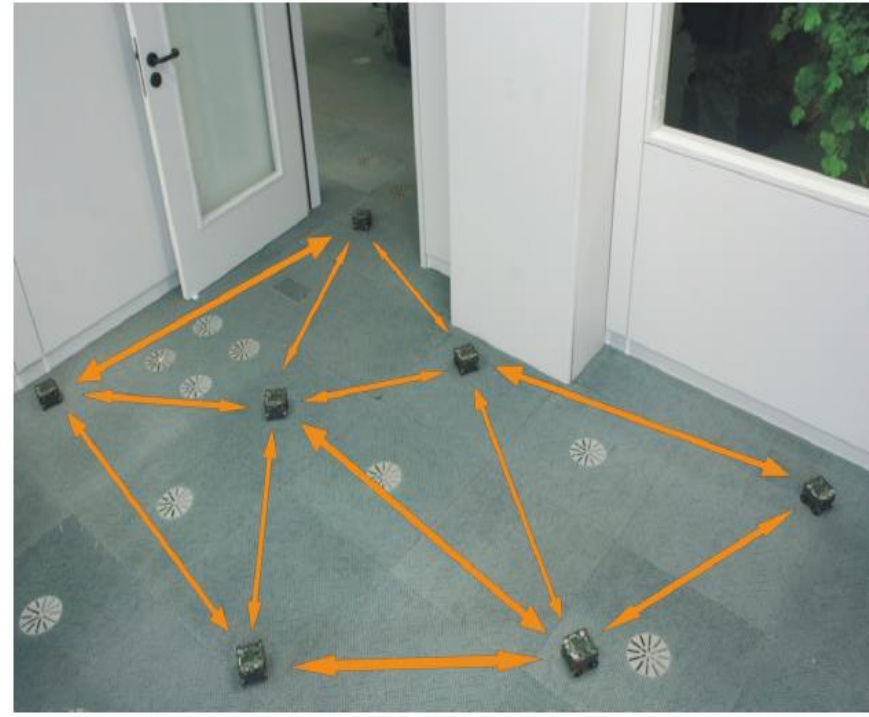
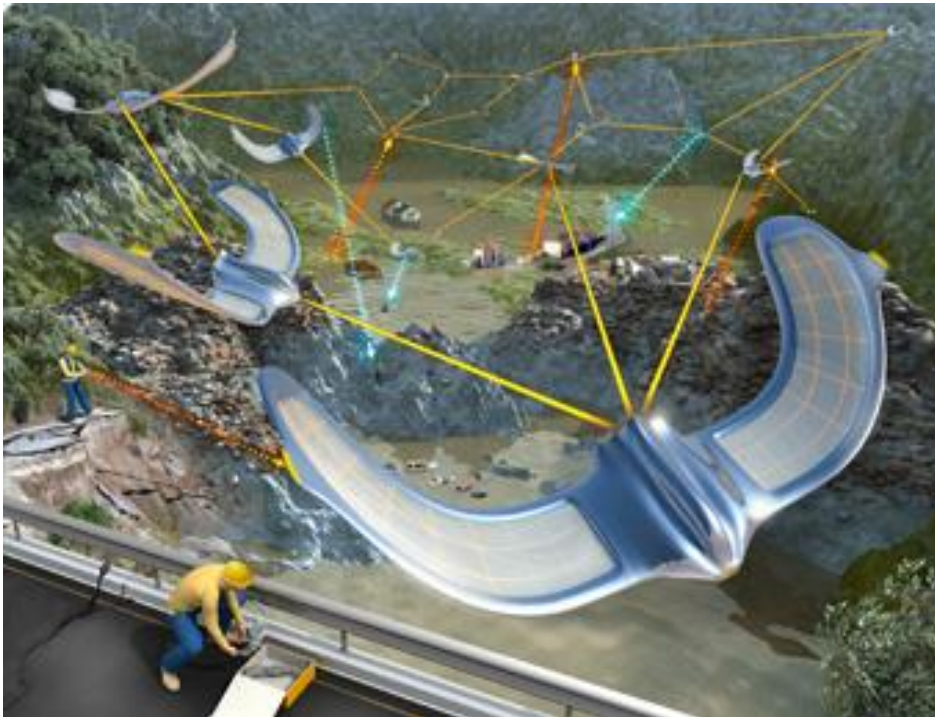
A set of n mobile Sensors: R

A set of m surveillance tasks: T

Benchmark Problems of MRS

- **Communication Relaying**

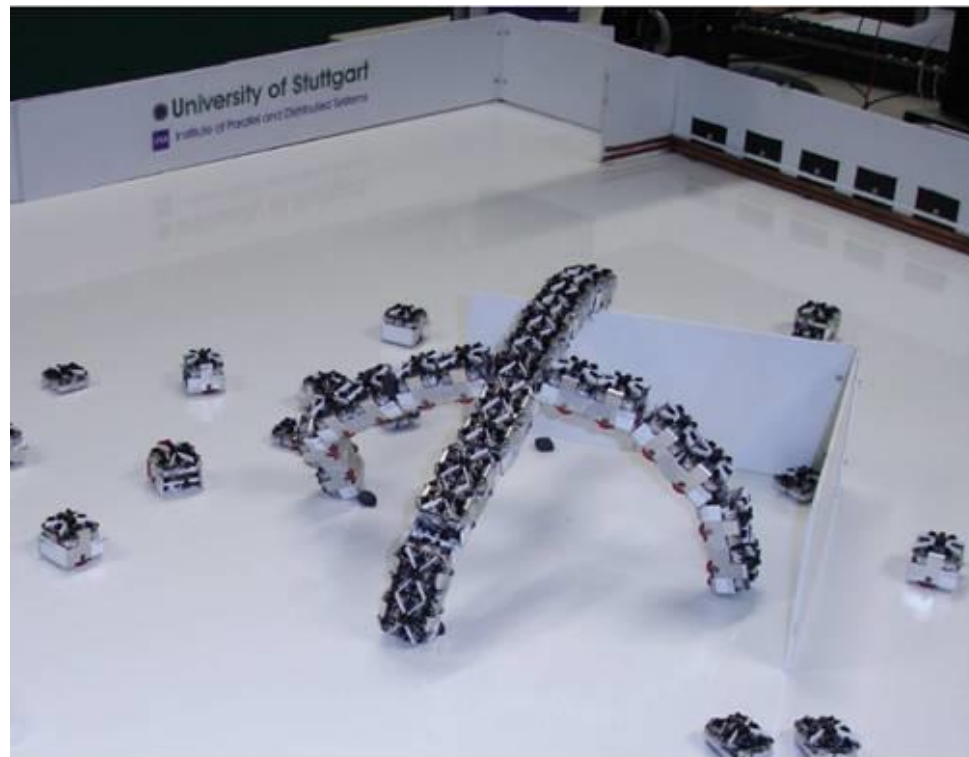
This cooperative behavior consists in establishing communication through relaying in order to dramatically increase **radio coverage** or expand communications links, primarily over rugged, mountainous or urban terrains.



Benchmark Problems of MRS

- **Reconfigurable Robotics**

Most of the work in this area involves identical modules with interconnection mechanisms that allow either **manual or automatic reconfiguration**.



Benchmark Problems of MRS

- **Reconfigurable Robotics**

Research in this area is **still very young**, and most of the systems developed are not yet able to perform beyond laboratory experiments.

While the potential of large numbers of robot modules has been **demonstrated in simulation**, it is still **uncommon to have implementations** involving more than a dozen or so physical modules.

The practical application of these systems is **yet to be demonstrated**, although progress is being made in that direction.

Benchmark Problems of MRS

- **Soccer**

Soccer playing is challenge problem for studying coordination and control in multirobot systems. This domain incorporates many challenging aspects of multirobot control, including:

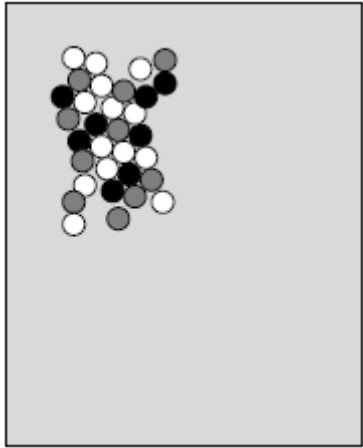
- ◇ Collaboration,
- ◇ Robot control architectures,
- ◇ Strategy acquisition,
- ◇ Real-time reasoning and action,
- ◇ Sensor fusion,
- ◇ Dealing with adversarial environments,
- ◇ Cognitive modeling, and
- ◇ Learning.



<http://www.robocup.org/> & <http://www.fira.net/>

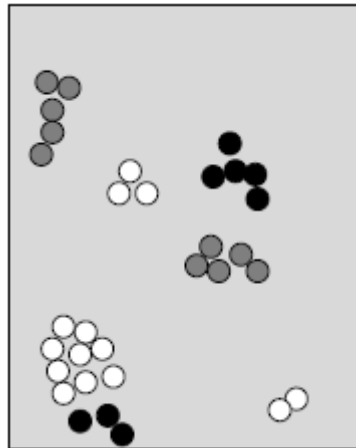
Benchmark Problems of MRS

- **Sorting**



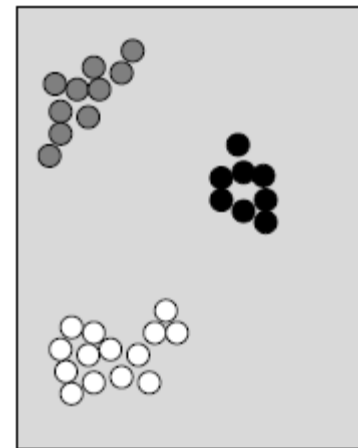
Clustering:

grouping a class of objects within a contiguous area that is a small fraction of the area of the available environment.



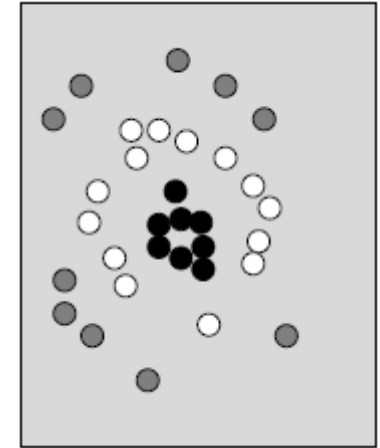
Segregation:

grouping two or more classes of objects so that each occupies a continuous area of the environment that is not occupied by members of any other class.



Patch sorting:

grouping two or more classes of objects so that each is both clustered and segregated, and each lies outside the boundary of each other.



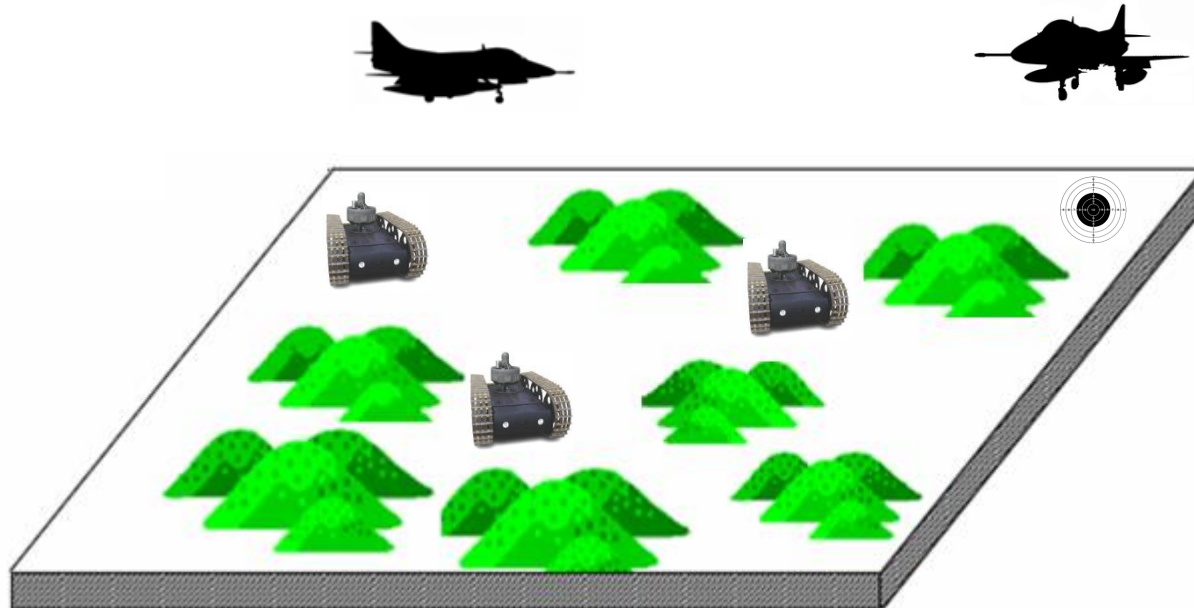
Annular sorting:

forming a cluster of one class of objects, and surrounding it with annular bands of the other classes, each band containing objects of only one type.

Benchmark Problems of MRS

- **Other problems**

- ◇ Cooperative perception in robotics
- ◇ Cooperative Mapping
- ◇ Collective Robotic Search
- ◇ ...



Outline

- Seminar Description
- Introduction to Multirobot Systems
- Benchmark Problems of Multirobot Systems
- **Organizational paradigms of Multirobot Systems**
- Challenging Problems of Multirobot Systems
- Summary

Organizational Paradigms

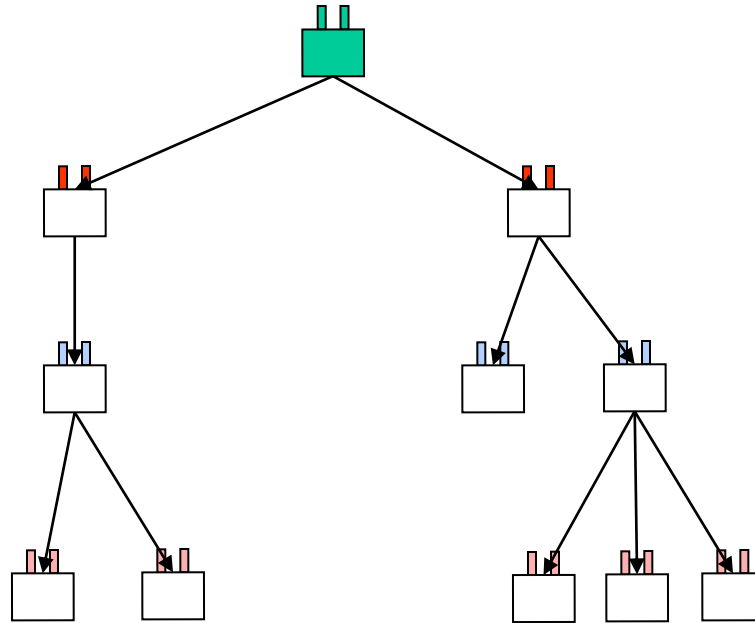
- The organizational paradigm or architecture of a system defines the **organizational behavior** of the nodes that comprise it and the **inter-node communications** pathways that enable **control** and **flow of data** [1].
- The organization of a multi-agent system is the collection of **roles, relationships, and authority structures** which **govern its behavior** [2].

[1] A. R. Benaskeur, H. Irandoust, and P. McGuire, "Holon control-based sensor management," in Proceedings of the International Conference on Information Fusion, 2007, pp. 1-8.

[2] B. Horling and V. Lesser, "A Survey of Multi-Agent Organizational Paradigms," The Knowledge Engineering Review, Vol: 19, Num: 4, pp. 281 - 316, Cambridge University Press, 2005.

Organizational Paradigms

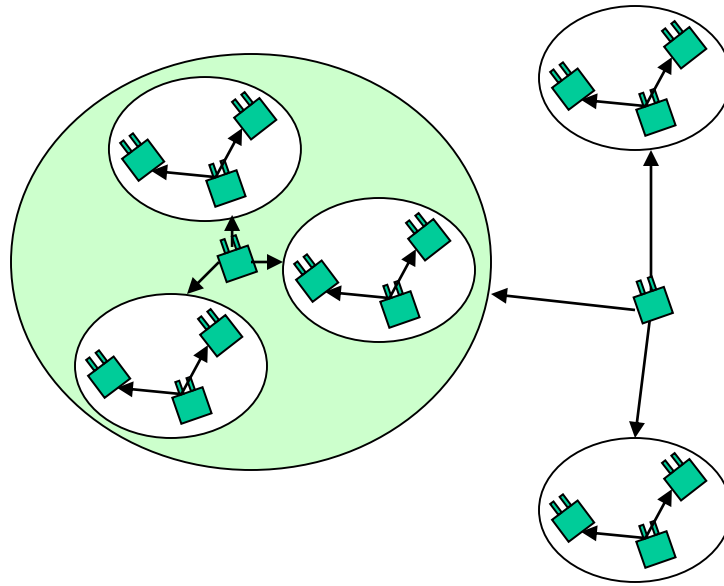
- Hierarchical Organization



Paradigm	Key Characteristic	Benefits	Drawbacks
Hierarchy	Decomposition	Maps to many common domains; handles scale well	Potentially brittle; can lead to bottlenecks or delays

Organizational Paradigms

- Holarchical Organization

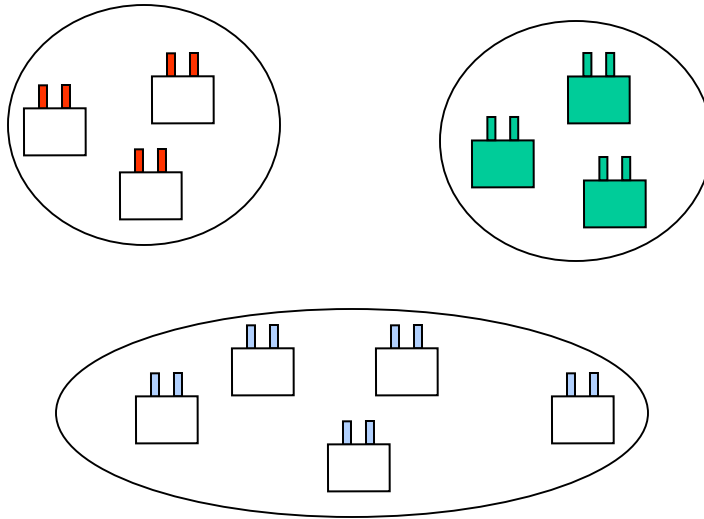


◇ Holarachy can be seen as hierarchical with cross-tree interaction, recursiveness and local autonomy.

Paradigm	Key Characteristic	Benefits	Drawbacks
Holarchy	Decomposition with autonomy	Exploit autonomy of functional units	Must organize holons; lack of predictable performance

Organizational Paradigms

- **Coalition-based Organization**

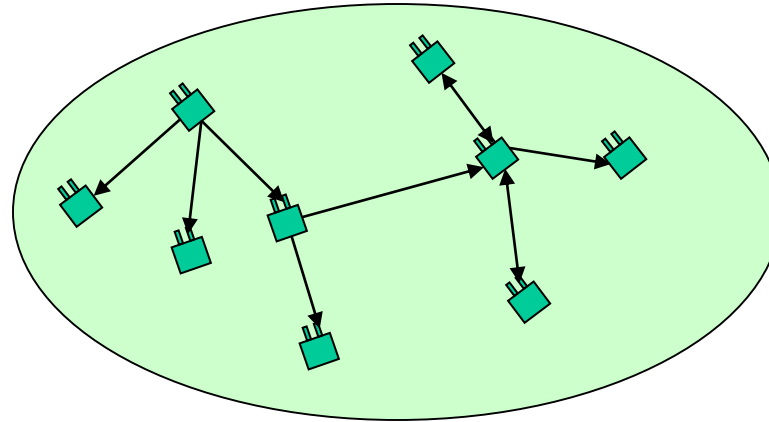


- ◇ Self-interested robots
- ◇ Cooperation to achieve individual goal
- ◇ Short-lived

Paradigm	Key Characteristic	Benefits	Drawbacks
Coalition	Dynamic, goal-directed	Exploit strength in numbers	Short term benefits may not outweigh organization construction costs

Organizational Paradigms

- **Team-based Organization**

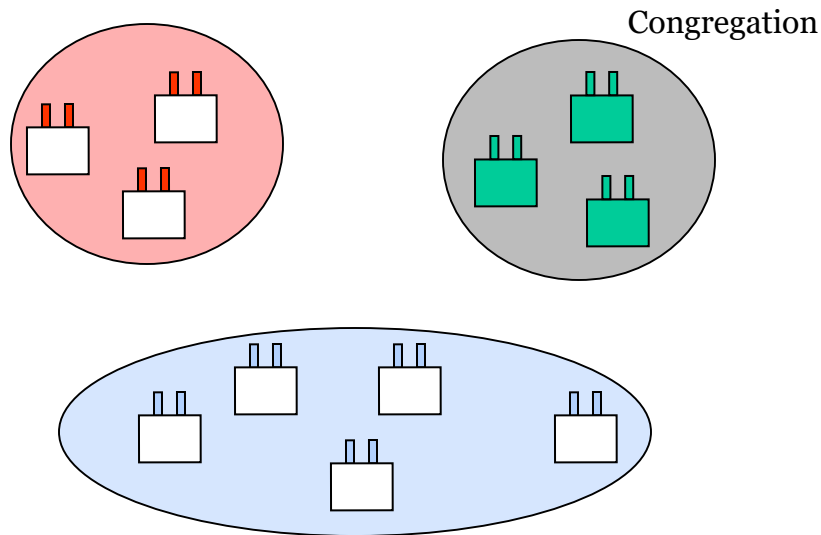


- ◇ Cooperation to achieve common goal
- ◇ Short-lived

Paradigm	Key Characteristic	Benefits	Drawbacks
Team	Group level cohesion	Address larger grained problems; task-centric	Increased communication

Organizational Paradigms

- **Congregations Organization**

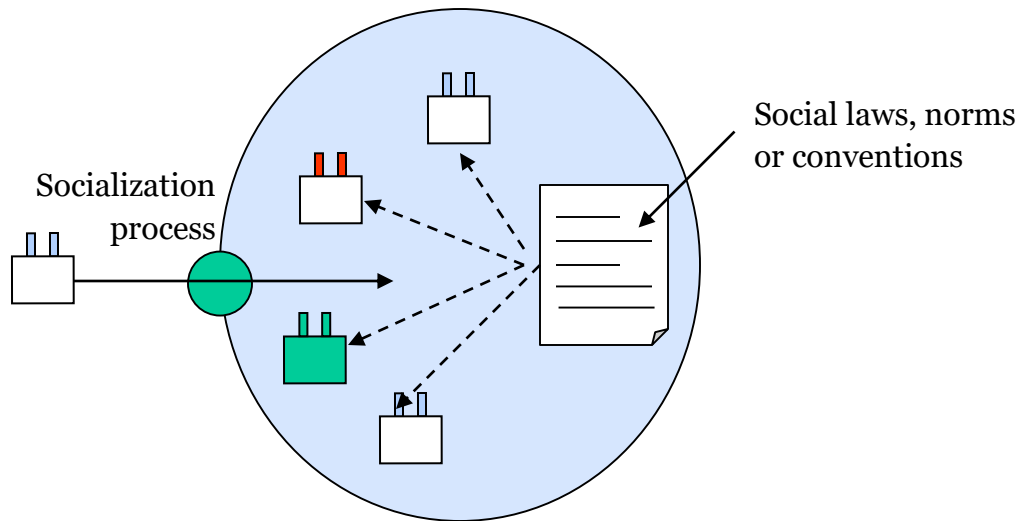


- ◇ Long-lived organization and
- ◇ Not formed with a single specific goal in mind.

Paradigm	Key Characteristic	Benefits	Drawbacks
Congregation	Long-lived, utility-directed	Facilitates agent discovery	Sets may be overly restrictive

Organizational Paradigms

- **Societies Organization**

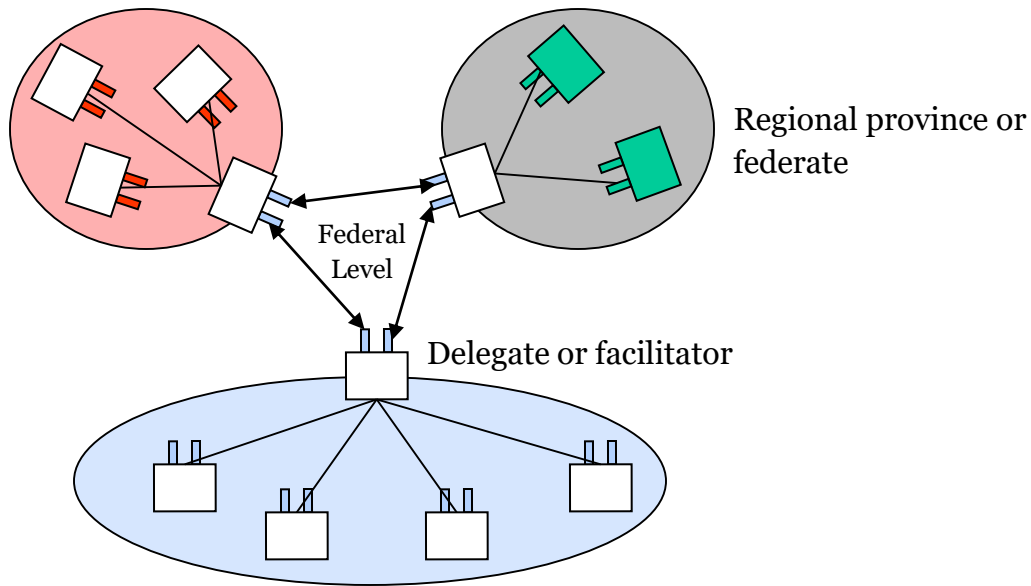


- ◇ A robot enters a society through an explicit negotiation process between the agent and a representative of the society.

Paradigm	Key Characteristic	Benefits	Drawbacks
Society	Open system	Public services; well defined conventions	Potentially complex, agents may require additional society-related capabilities

Organizational Paradigms

- **Federations Organization**

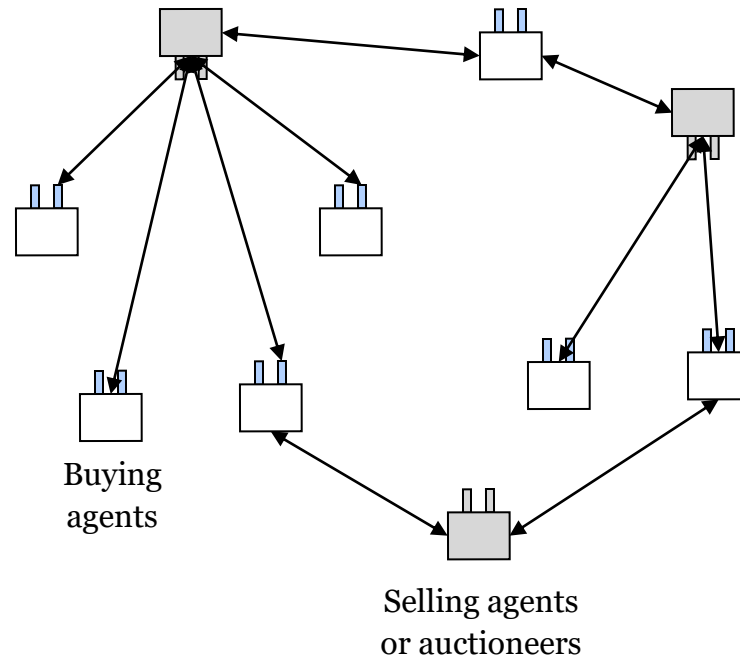


- ◇ Similar to holons
- ◇ A group of robots which have ceded some amount of autonomy to a single delegate which represents the group.

Paradigm	Key Characteristic	Benefits	Drawbacks
Federation	Middle-agents	Matchmaking, brokering, translation services; facilitates dynamic agent pool	Intermediaries become bottlenecks

Organizational Paradigms

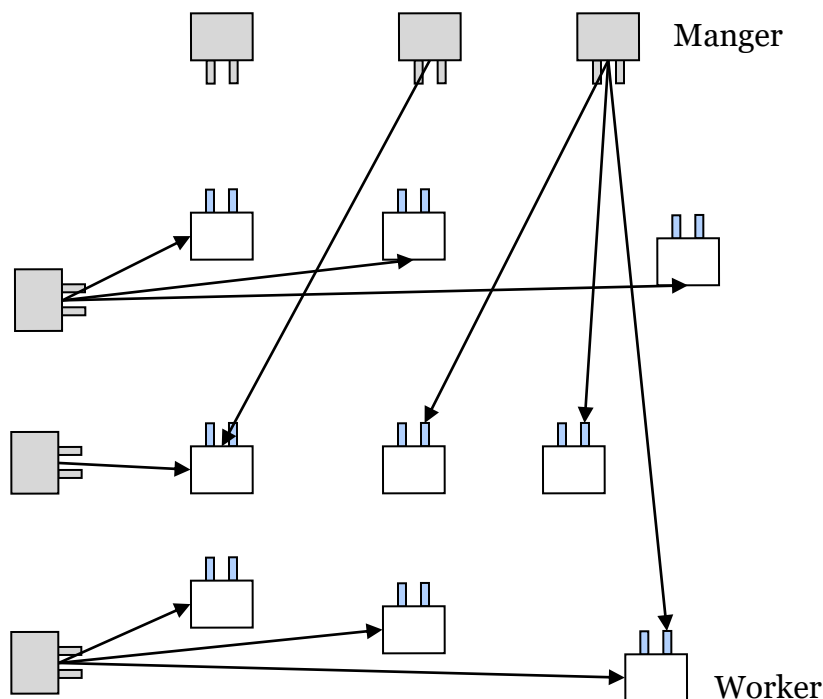
- Market-based Organization



Paradigm	Key Characteristic	Benefits	Drawbacks
Market	Competition through pricing	Good at allocation; increased utility through centralization; increased fairness through bidding	Potential for collusion, malicious behavior; allocation decision complexity can be high

Organizational Paradigms

- **Matrix Organization**

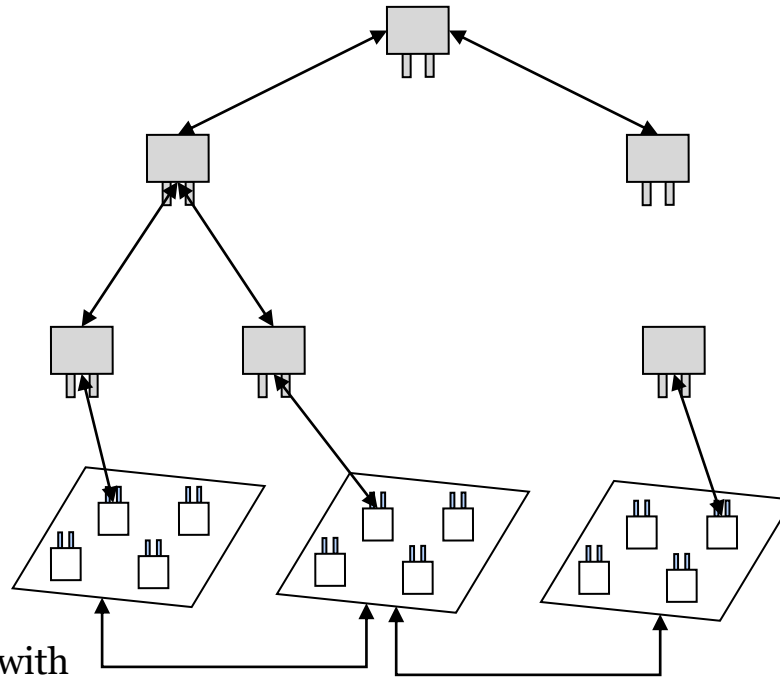


- ◇ Permitting many managers or peers to influence the activities of a robot.

Paradigm	Key Characteristic	Benefits	Drawbacks
Matrix	Multiple managers	Resource sharing; multiply-influenced agents	Potential for conflicts; need for increased agent sophistication

Organizational Paradigms

- **Compound Organization**



- ◇ A system may have one organization for **control**, another for **data flow**, a third for **discovery**, and so on.

This combines hierarchy with a set of coalitions

Paradigm	Key Characteristic	Benefits	Drawbacks
Compound	Concurrent organizations	Exploit benefits of several organizational styles	Increased sophistication; drawbacks of several organizational styles

Organizational Paradigms

Paradigm	Key Characteristic	Benefits	Drawbacks
Hierarchy	Decomposition	Maps to many common domains; handles scale well	Potentially brittle; can lead to bottlenecks or delays
Holarchy	Decomposition with autonomy	Exploit autonomy of functional units	Must organize holons; lack of predictable performance
Coalition	Dynamic, goal-directed	Exploit strength in numbers	Short term benefits may not outweigh organization construction costs
Team	Group level cohesion	Address larger grained problems; task-centric	Increased communication
Congregation	Long-lived, utility-directed	Facilitates agent discovery	Sets may be overly restrictive
Society	Open system	Public services; well defined conventions	Potentially complex, agents may require additional society-related capabilities
Federation	Middle-agents	Matchmaking, brokering, translation services; facilitates dynamic agent pool	Intermediaries become bottlenecks
Market	Competition through pricing	Good at allocation; increased utility through centralization; increased fairness through bidding	Potential for collusion, malicious behavior; allocation decision complexity can be high
Matrix	Multiple managers	Resource sharing; multiply-influenced agents	Potential for conflicts; need for increased agent sophistication
Compound	Concurrent organizations	Exploit benefits of several organizational styles	Increased sophistication; drawbacks of several organizational styles

Source: B. Horling and V. Lesser, "A Survey of Multi-Agent Organizational Paradigms," The Knowledge Engineering Review, Vol: 19, Num: 4, pp. 281 - 316, Cambridge University Press, 2005.

Organizational Paradigms

- **MRS Architecture Examples: Nerd Herd**

The Nerd Herd is representative of a pure swarm robotics approach using **large numbers of homogeneous robots**.

The decentralized control approach was based on the **subsumption architecture**, and assumed that all robots were homogeneous, but with relatively simple individual capabilities, such as detecting obstacles and kin (i. e., other robot team members).

Organizational Paradigms

- **MRS Architecture Examples: Nerd Herd**

- ◇ **Subsumption Architecture**

A subsumption architecture is a way of **decomposing one complex behavior into many “simple” layers of increasingly more abstract behaviors.**

Each layer can use (**subsume**) the underlying layer,

Example:

The decision to move forward by the **eat-food layer** takes in account the decision of the **lowest obstacle-avoidance layer.**



Rodney Brooks (1954-)

Organizational Paradigms

- **MRS Architecture Examples: Nerd Herd**

- ◇ **Subsumption Architecture**

Reactive:

- quick, instinctive
- stop when you get to the door

Executive:

- automatic, sequential
- stick out your hand, turn knob, push door

Deliberative:

- requires thought
- if knob doesn't turn, look for key
- if door doesn't push, try pulling

Organizational Paradigms

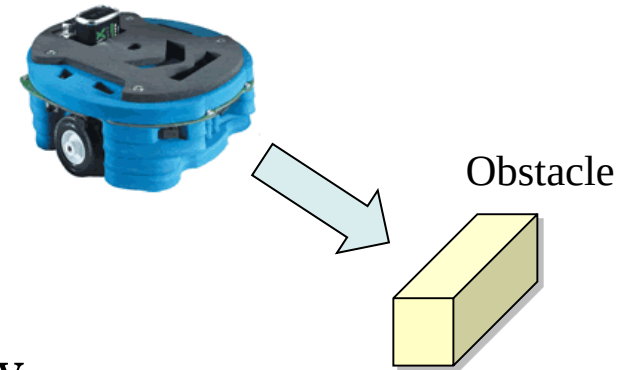
- **MRS Architecture Examples: Nerd Herd** Mobile Robot

- ◇ **Subsumption Architecture**

Example: A mobile robot that:

(1) will run forward until it hits an obstacle

(2) then avoids this obstacle and continues its way.



One Task

```
repeat forever
{
  go until you hit an obstacle,
  then do some avoiding movements,
}
```

Two Tasks

```
go_task
repeat forever
{
  go
}
```

```
avoid_task
repeat forever
{
  if hit then
  {
    stop go_task
    avoid
    start go_task
  }
}
```

This system works without ambiguity with two tasks, but what, if we add another mission.

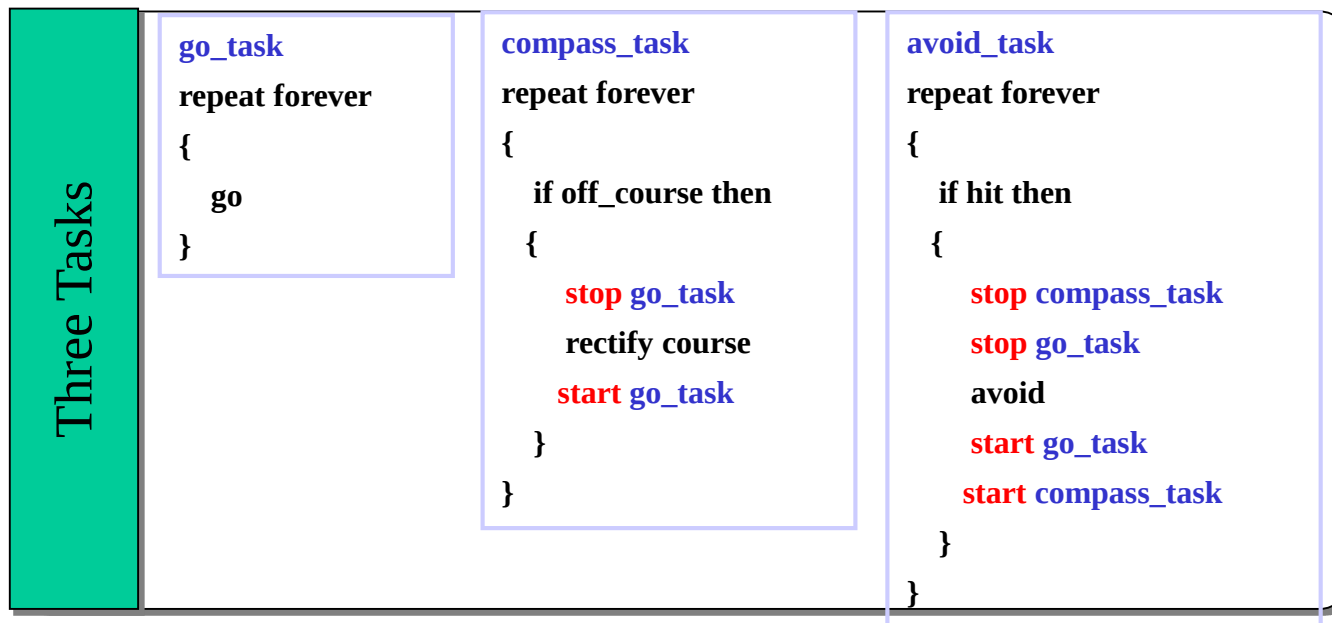
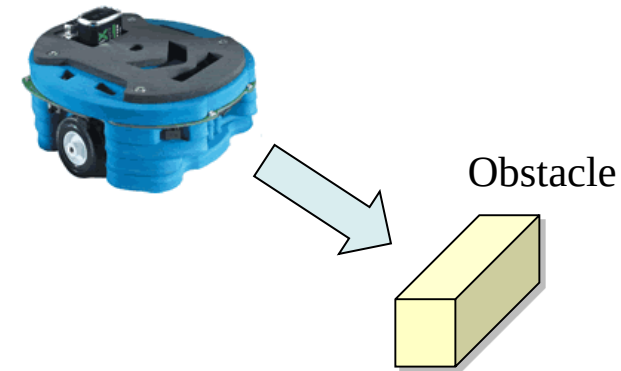
Organizational Paradigms

• MRS Architecture Examples: Nerd Herd Mobile Robot

◇ Subsumption Architecture

Example: A mobile robot that:

- (1) will run forward until it hits an obstacle
- (2) follows a given compass-direction
- (3) then avoids this obstacle and continues its way.

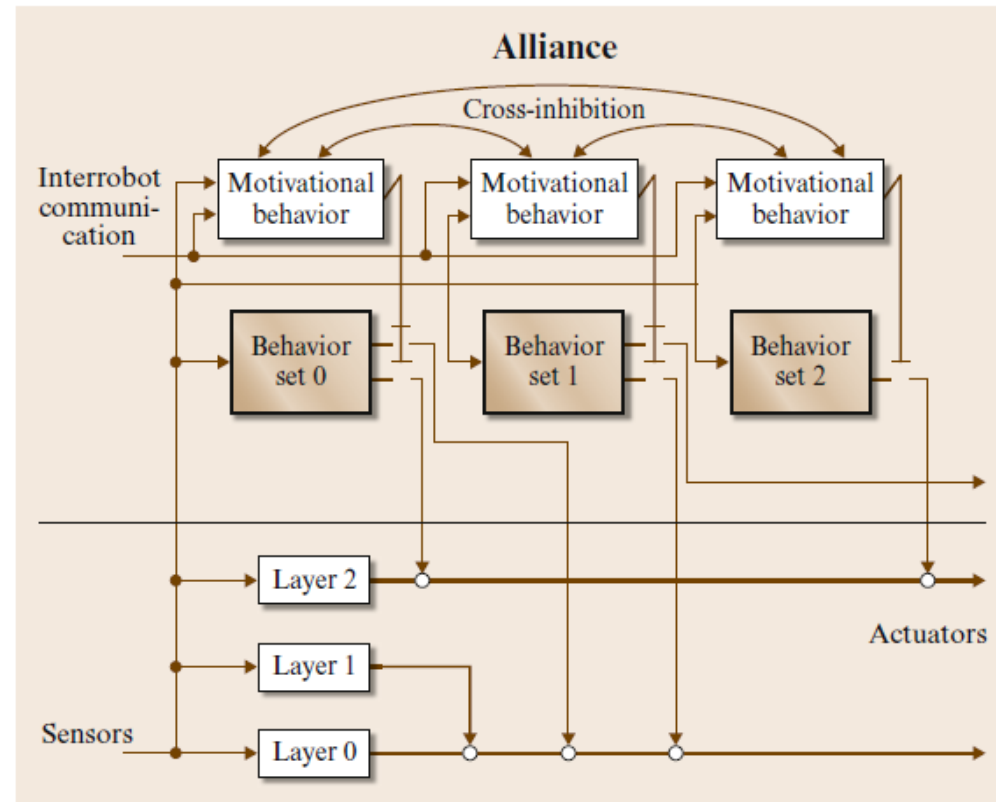


This scenario works, but shows some trouble during the transition time, as the motor-control passes from one task to another. This radical solution causes a lot of **jiggling**.

Organizational Paradigms

- **MRS Architecture Examples: ALLIANCE**

This approach builds on the **subsumption architecture** by adding **behavior sets** and **motivations** for achieving action selection without explicit negotiations between robots.

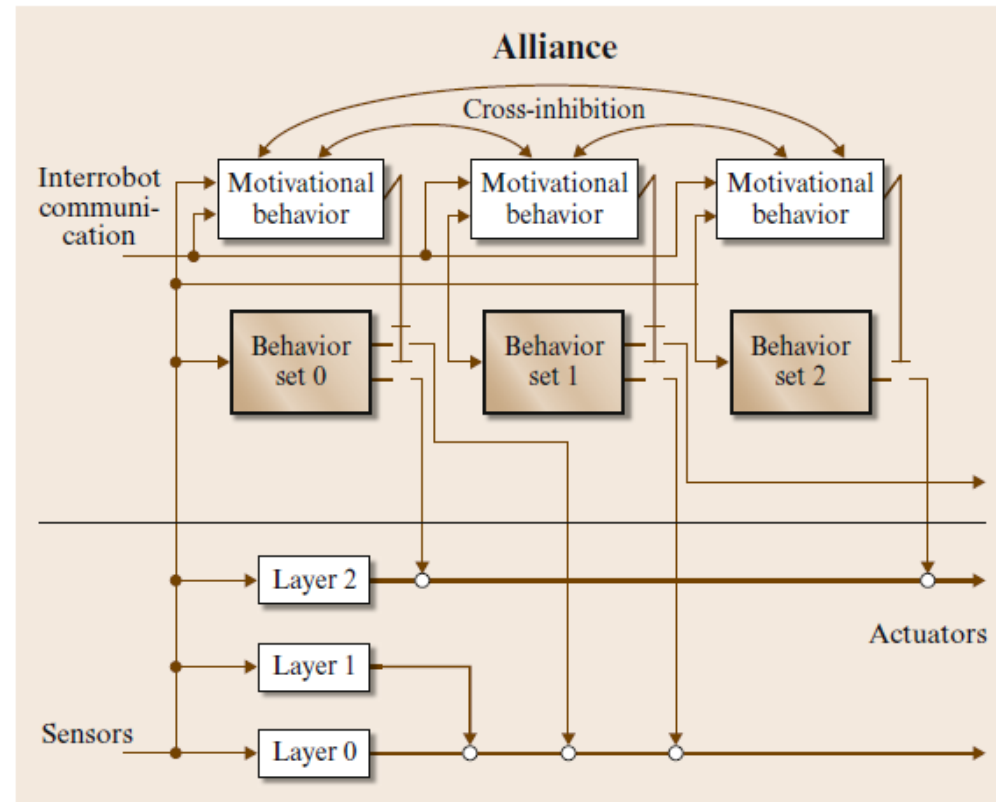


Organizational Paradigms

- **MRS Architecture Examples: ALLIANCE**

- **Behavior sets:** group low-level behaviors together for the execution of a particular task.

- **The motivations:** consist of levels of impatience and acquiescence that can raise and lower a robot's interest in activating a behavior set corresponding to a task that must be accomplished.



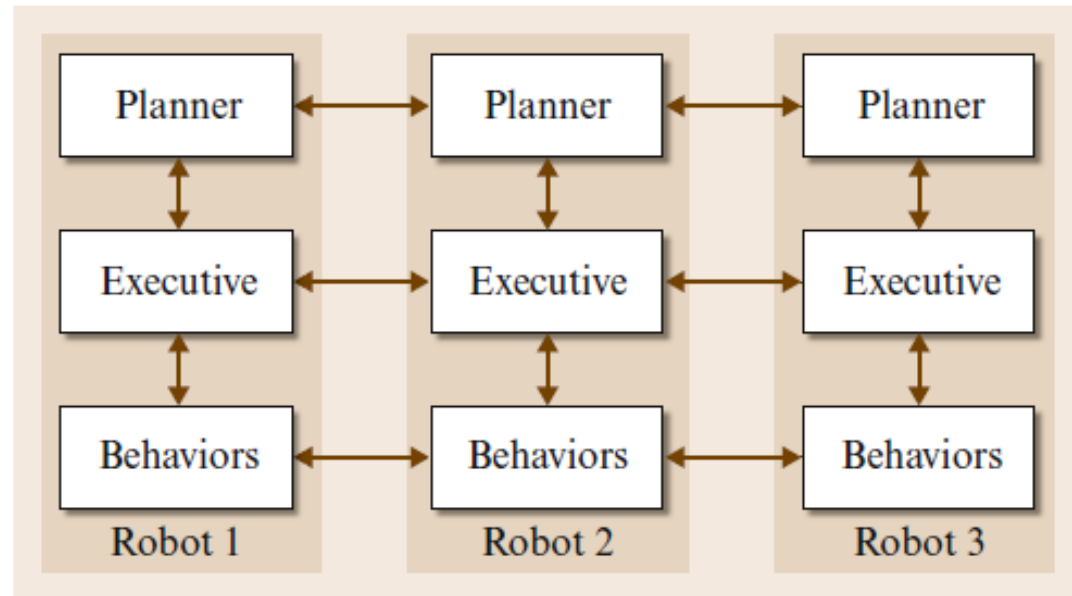
Organizational Paradigms

- **MRS Architecture Examples: DIRA**

Similar to the Nerd Herd and ALLIANCE approaches, the **Distributed Robot Architecture (DRA)** approach allows autonomy in individual robots.

However, unlike the previous approaches, DIRA also facilitates explicit coordination among robots.

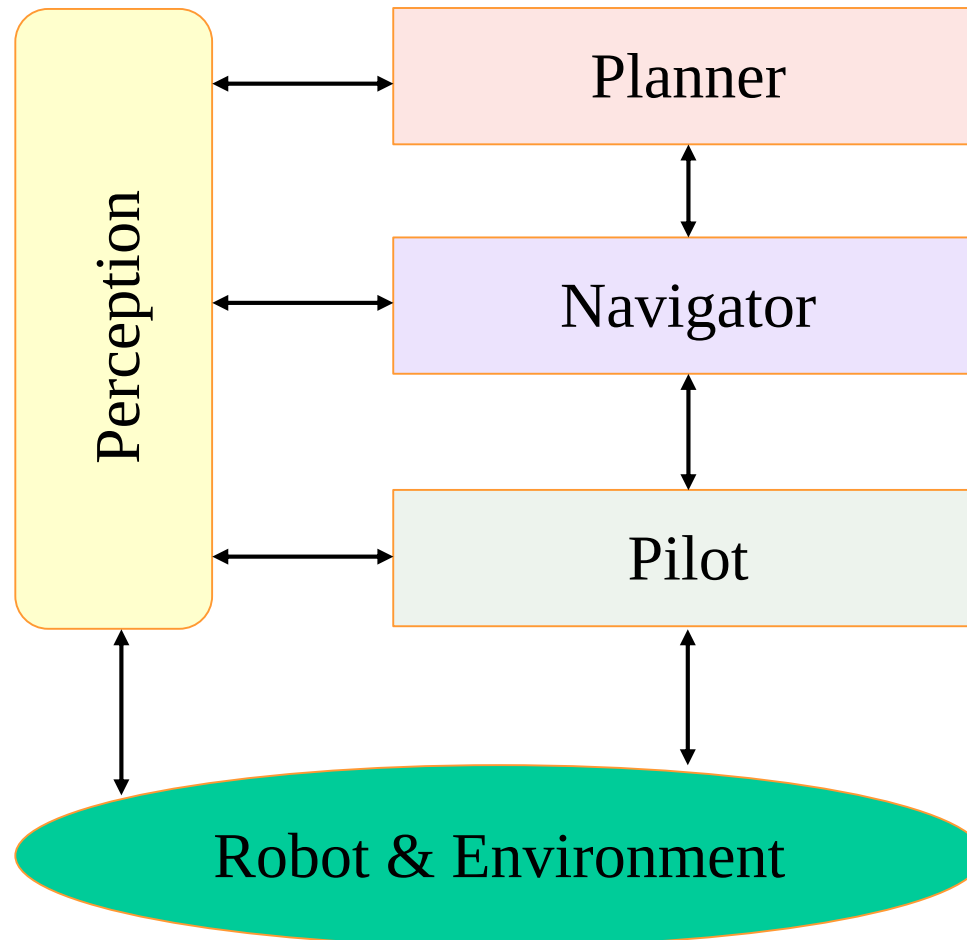
This approach is based on **layered architectures** that are popular for single-robot systems.



Organizational Paradigms

- **MRS Architecture Examples: DIRA**

- ◇ **Layered Architecture**



Organizational Paradigms

- **MRS Architecture Examples: DIRA**

- ◊ **Layered Architecture**

- **Planner:** The planner calculates in advance the **paths that the robot must follow and the main actions** to be performed in order to proceed with a certain mission.
- **Navigator:** The navigator **localizes the robot and elaborates medium term commands** taking into account local and current events and circumstances (e.g., mobile obstacles) that are not considered by the planner.
- **Pilot:** The pilot **controls** the robot actuators.
- **Environment Perception:** The perception module controls the sensors and process the **sensory information**.

Organizational Paradigms

- **Other MRS Architecture**

- ◇ CHARON [1]
- ◇ CAMPOUT [2]
- ◇ CLARAty [3]
- ◇ AuRA [4]
- ◇ CEBOT [5]
- ◇ ...

[1] Fierro, Rafael, Das, Aveek, John, Spletzer, Esposito, Joel, Kumar, Vijay, Ostrowski, James P., Pappas, George, Taylor, Camillo J., Hur, Yerang, Alur, Rajeev, Lee, Insup, Grudic, Greg, and Southall, Ben (2002), A Framework and Architecture for Multi-Robot Coordination, The International Journal of Robotics Research, 21(10-11), 977-995.

[2] Huntsberger, Terry, Pirjanian, Paolo, Trebi-Ollennu, Ashitey, Nayar, Hari Das, Aghazarian, Hrand, Ganino, Anthony J., Garrett, Mike, Joshi, Sanjay S., and Schenker, Paul S. (2003), CAMPOUT: A Control Architecture for Tightly Coupled Coordination of Multirobot Systems for Planetary Surface Exploration, IEEE Transactions on Systems, Man, and Cybernetics -Part A: Systems and Humans, 33(5).

[3] IEEE (2001), The CLARAty Architecture for Robotic Autonomy, IEEE, Big Sky, MT.

[4] Arkin, R.C. and Balch, T. (1997), AuRA: Principles and Practice in Review, Journal of Experimental & Theoretical Artificial Intelligence, 9(2/3), 175-188.

[5] Fukuda, T. and Nakagawa, S. (1987), A dynamically reconfigurable robotic system (concept of a system and optimal configurations), In Proceedings International Conference on Industrial Electronics, Control, and Instrumentation, pp. 588-595.

Outline

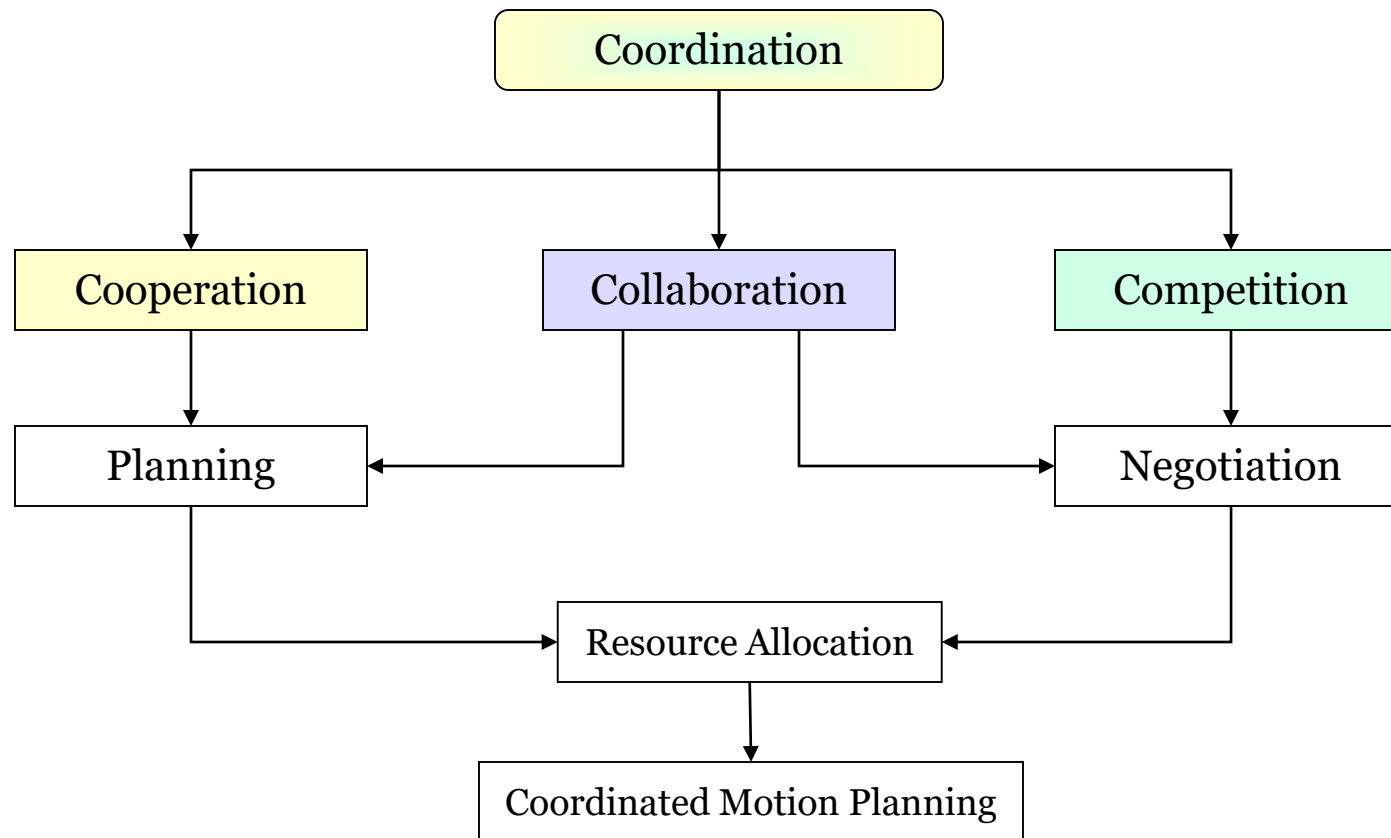
- Seminar Description
- Introduction to Multirobot Systems
- Benchmark Problems of Multirobot Systems
- Organizational paradigms of Multirobot Systems
- **Challenging Problems of Multirobot Systems**
- Summary

Challenging Problems of MRS

- **Coordination**
- **Algorithm Design**
- **Implementation and Test**
- **Analysis and Modelling**

Challenging Problems of MRS

- **Coordination**



Challenging Problems of MRS

- **Algorithm Design**

Swarm roboticists face the problem of designing both the physical morphology and behaviours of the individual robots such that when those robots interact with each other and their environment, the desired overall collective behaviours will emerge. At present there are no principled approaches to the **design of low-level behaviours** for a given **desired collective behaviour** [1].

“collective behavior is not simply the sum of each participant’s behavior, as others emerge at the society level” [2].

[1] E. Sahin and A. Winfield, “Special issue on swarm robotics,” *Swarm Intelligence*, 2: 69–72, Springer Science, 2008.

[2] Pasteels et al. *From Individual to Collective Behavior in Social Insects*. Pages 155-175, 1987.

Challenging Problems of MRS

- **Algorithm Design (cont'd)**

Main decisional abilities:

- Mission planning,
- Task allocation and
- Coordinated task achievement
 - Management the task allocation,
 - Scheduling,
 - Cooperation/collaboration between the entities,
 - Conflict avoidance, etc.

Challenging Problems of MRS

- **Implementation and Test**

To build and rigorously test a swarm of robots in the laboratory requires a **considerable experimental infrastructure**.

Real-robot experiments thus typically proceed **hand-in-hand with simulation** and good tools are essential [1].

[1] E. Sahin and A. Winfield, “Special issue on swarm robotics,” *Swarm Intelligence*, 2: 69–72, Springer Science, 2008.

Challenging Problems of MRS

- **Implementation and Test**

- ◇ **Advanced Robotics Interface for Applications (ARIA):** Robotic Sensing and Control Libraries.



- ◇ **Open Robot Control Software (OROCOS):** open-source real time control architecture for different machines.



- ◇ **Microsoft Robotics Studio:** is a Windows-based environment for robot control and simulation.



- ◇ **Player/Stage/Gazebo:** PSG is open source software that used and developed by an international community of researchers from over 30 universities/companies.

Challenging Problems of MRS

- **Implementation and Test: Player**
 - ◇ **Server** running on a robot providing **access to hardware and sensors** to remote programs.
 - ◇ Player supports multiple hardware platforms such **as iRobots rFlex, including IR, sonar, PTZ control, cameras, bumper and gripper devices.**
 - ◇ The interface to access those sensors is the same regardless of the underlying hardware platform

Challenging Problems of MRS

- **Implementation and Test: Player**

- ◇ Built in **camera server** to display pictures remote;
- ◇ Built in **blobfinder** from the camera images using the **CMVision** algorithm;
- ◇ Built in support for **Festival speech synthesis**;
- ◇ support for **sphinx speech recognition**;
- ◇ Different algorithms for **localization** on a known map;
- ◇ logfile writer/reader;
- ◇ and even more...

Challenging Problems of MRS

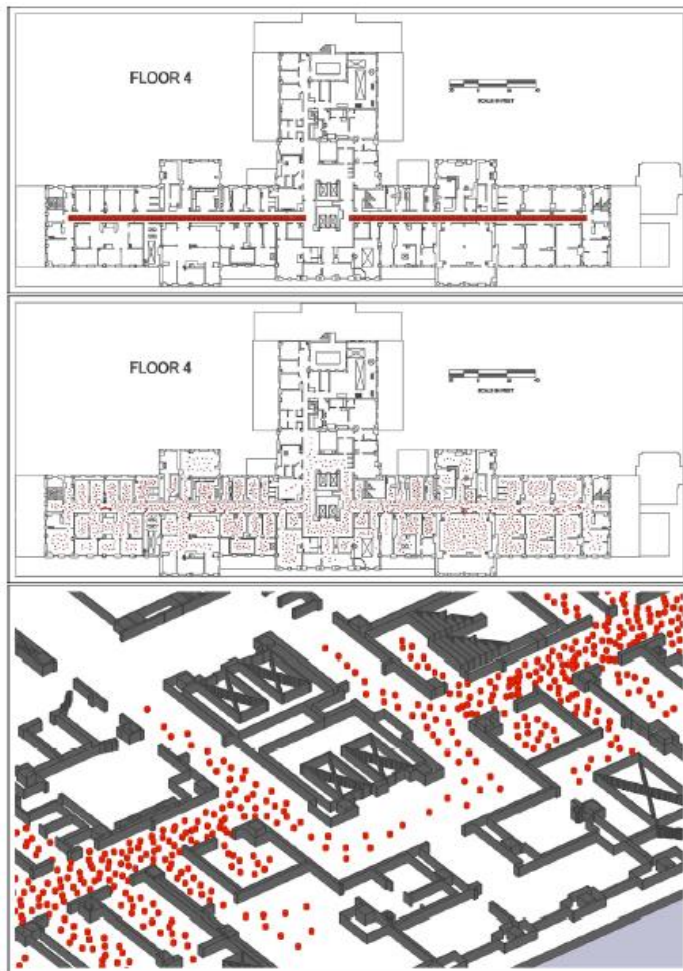
- **Implementation and Test: Stage**

- ◇ Stage is a **scalable** multiple robot simulator;
- ◇ It simulates a population of mobile robots moving in and sensing a **two-dimensional** bitmapped environment, controlled through Player.
- ◇ Stage provides **virtual Player** robots which interact with simulated rather than physical devices.
- ◇ Various **sensor models** are provided, including sonar, scanning laser rangefinder, pan-tilt-zoom camera with color blob detection and odometry.

Challenging Problems of MRS

- **Implementation and Test: Stage**

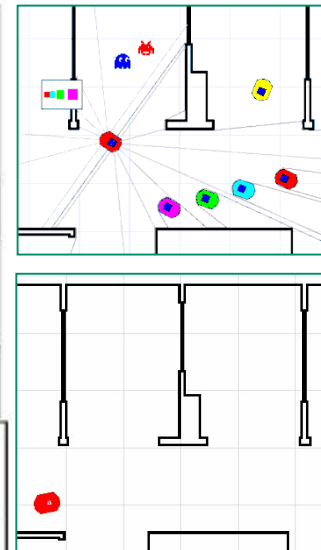
Stage 3.0 [Richard Vaughan, “Massively multi-robot simulation in stage,” Swarm Intelligence 2: 189–208, 2008.]



2,000 minibots



100 Pioneer Robots



Challenging Problems of MRS

- Implementation and Test: Stage



Group Formation, PAMI Lab



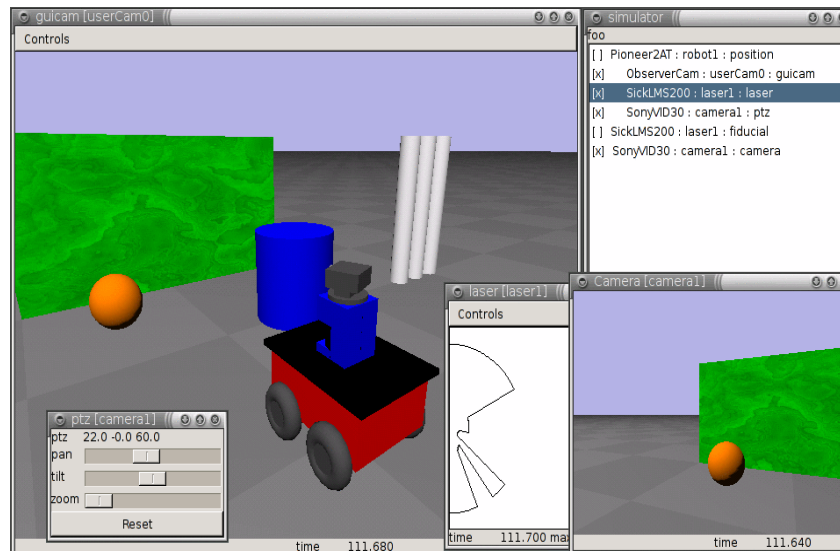
Yun-Qian Miao, Alaa Khamis, Mohamed Kamel, "Applying Anti-Flocking Model in Mobile Surveillance Systems," Special Session on Distributed Surveillance Systems, International Conference on Autonomous and Intelligent Systems (AIS 2010), June 21-23, 2010, Pova de Varzim, Portugal

Challenging Problems of MRS

- **Implementation and Test: Gazebo**

High quality/ low quantity **3D simulator** like Stage.

Gazebo is a 3D, dynamic, multi-robot simulator. Whereas Stage is intended to simulate the behavior of very large populations of robots with moderate fidelity, Gazebo simulates the behavior of small populations of robots (less than 10) with high fidelity.



Challenging Problems of MRS

- **Analysis and Modelling**

A robotic swarm is typically a **stochastic, non-linear system** and constructing **mathematical models** for both validation and parameter **optimization** is challenging.

Such models would surely be an essential part of constructing a safety argument for real-world applications [1].

[1] E. Sahin and A. Winfield, “Special issue on swarm robotics,” *Swarm Intelligence*, 2: 69–72, Springer Science, 2008.

Outline

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Summary

- Multirobot systems (MRS) are a group of robots that are designed aiming to perform some collective behavior.
- The MRS is gaining great interest because of the following reasons:
 - ◇ Some tasks may be quite complex for a single robot to do or even it might be impossible.
 - ◇ Increasing the performance when multiple robots are involved in one goal to achieve.
 - ◇ Having small, simple robots will be easier and cheaper to implement than having only one single powerful robot.
 - ◇ Increasing the system reliability.